

# Air:bit 2

## The micro:bit drone



# Teachers! Read this

The Air:bit 2 is a great way to engage the students through cross curriculum teaching and learning. Here is some useful information.

## **Time frame**

Building takes 2-3 hours

Coding the remote controller takes 3-4 hours or you can download the readymade code.

Flying takes about 30 minutes (flying and adjusting before battery is empty)

We recommend charging the batteries while coding. Charging with Air:bit V2 takes approx. 1 hour.

## **Micro:bits**

We recommend two micro:bits for a drone/transmitter pair. The drone should have micro:bit V2 on board. See the next page.

## **Radio channels**

Each student should select their own radio channel for transmitter and receiver. (0-255). If you use ready made code, make sure to edit the variable called radioGroup in makecode before downloading to micro:bit (both transmitter and receiver)

## **Flying**

Use an open space close to where you build the drones. A soft carpet is better than a hard surface in case of crashes. A high ceiling is not needed (in case of crash, high falls are worse)

Read more about flying at the end of the document or see the video at [www.makekit.no/docs](http://www.makekit.no/docs)

## **Safety**

The propellers are small but spins fast and can make small cuts in the skin. Students should be cautious when the propellers and battery are both connected. If you leave the propellers unmounted until the end, you can test run the motors first.

## **Repairs**

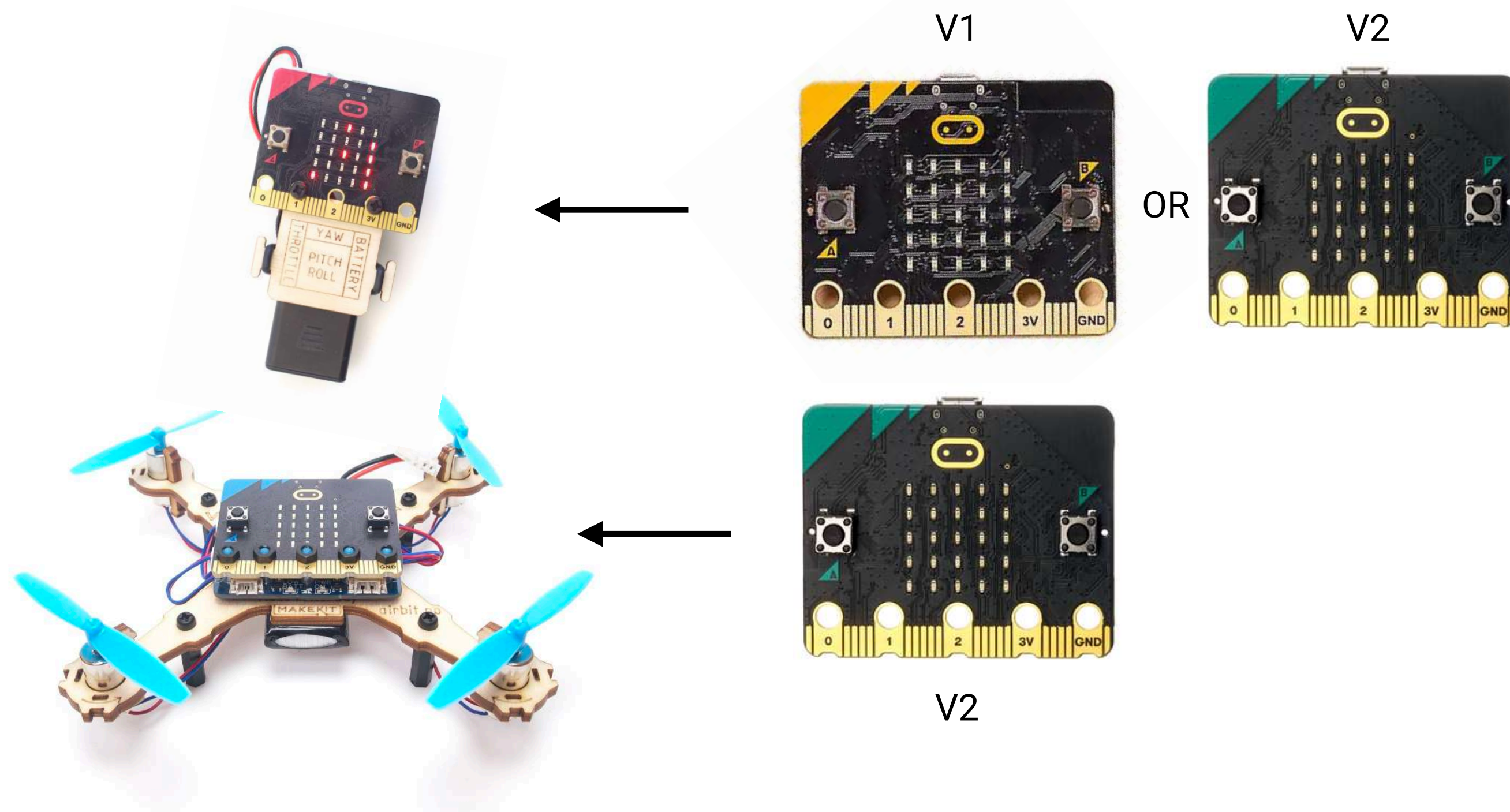
All parts can be replaced. The most wear goes to propellers and motors. Keep some spares at hand or order more from your distributors or directly from Makekit ([sales@makekit.no](mailto:sales@makekit.no))

## **More activities**

Did you know you can 3d-print a frame for Air:bit? Or make a smart glove remote? Check out more projects at [www.makekit.no/docs](http://www.makekit.no/docs)



# Micro:bit V1 or V2?



The remote control works great on both micro:bit V1 and V2.  
Air:bit 2 relies on the micro:bit for high speed calculations and stabilisation. Therefore, we highly recommend using micro:bit V2 on the drone.

**If you experience trouble getting the  
air:bit to fly, please see page 61.**





# Safety warning

## **Fast rotation propellers can hurt humans and animals**

### **To prevent injury, do:**

- Mount the propellers at after all building and coding is done. Test that you can start and stop motors (shake to stop)
- Keep a distance when arming and flying the drone.
- Children under 8 years and animals must be kept at a safe distance, at least 5 meters away even indoors.
- Use propeller protector if possible
- Follow local FAA regulations



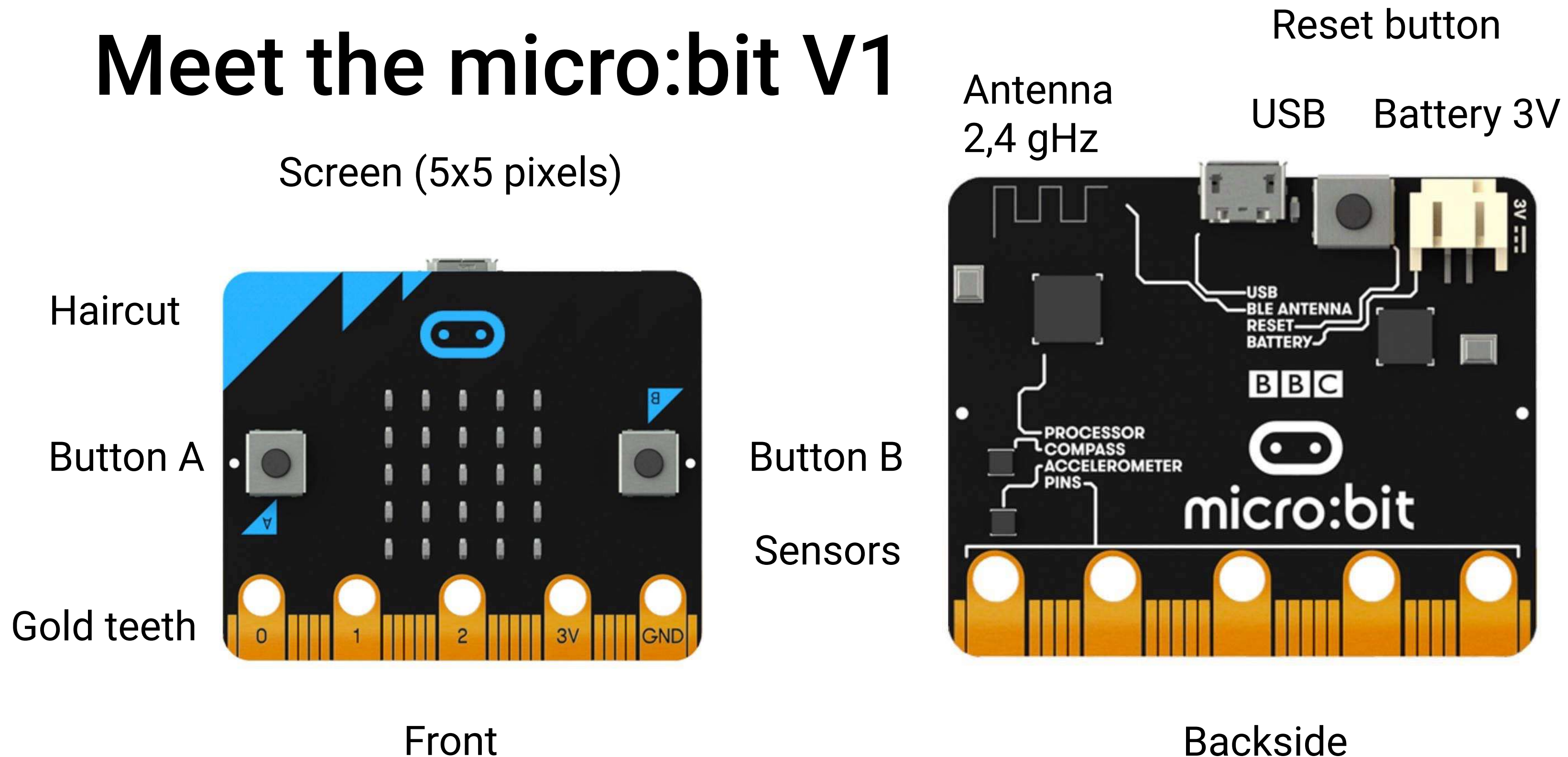
## **Lithium batteries can release smoke or cause a fire**

### **To prevent damage, do:**

- Don't charge the batteries unattended
- Don't use a damaged or punctured battery
- Do not short circuit the battery
- Avoid temperatures below -10 and +50 degrees celcius.
- Don't use batteries that are colder than 15 degrees celcius
- Always have a plan for what to do in case of a fire: If you are indoor, open a window and get the battery outside to prevent smoke or fire.
- Do not open or modify the battery in any way.
- For optimal performance, store the battery at around 50% capacity and between 10 and 20 degrees celcius
- Follow airport regulations for carrying lithium batteries on airplanes. (Usually hand luggage only)



# Meet the micro:bit V1

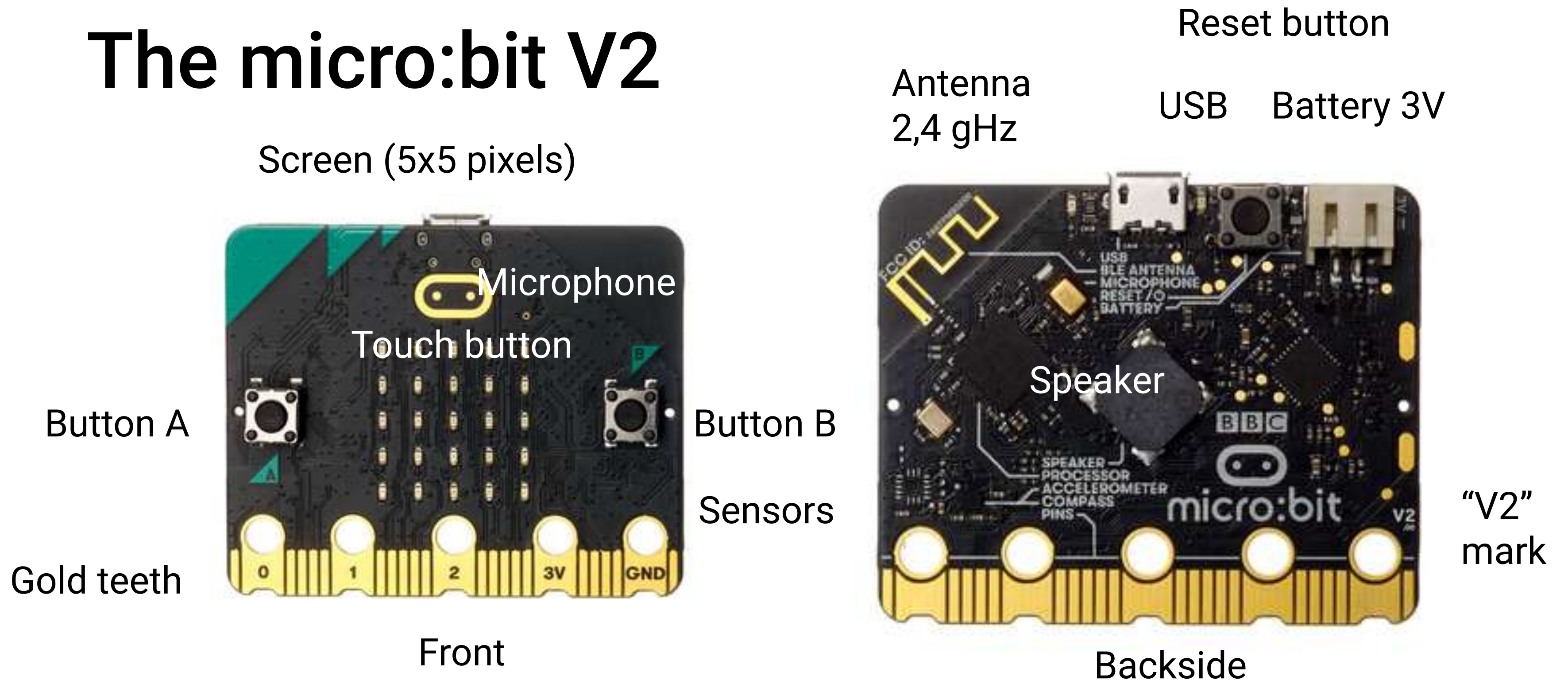


micro:bit is a small computer with prosessor, sensors, display and radio. It has connection pins for external components like LEDs, speakers or various sensors.

You can learn more at: <https://tech.microbit.org/hardware/>



# The micro:bit V2



The micro:bit V2 is very similar to V1 but with extra features, like: Microphone, speaker, touch button, faster processor and more memory. The "gold teeth" looks different compared to V1, for better grip with crocodile clips.

**Air:bit can work on V1 and V2, but V2 gives more processing power for advanced users.**



# Air:bit 2

The micro:bit drone

micro:bit with  
Microsoft  
make:code

Screen showing flight  
directions and battery  
status

Propeller  
protection mount

Air:bit control  
board with charger

10x20mm  
coreless  
dc-motors 3.7v  
(replacable)

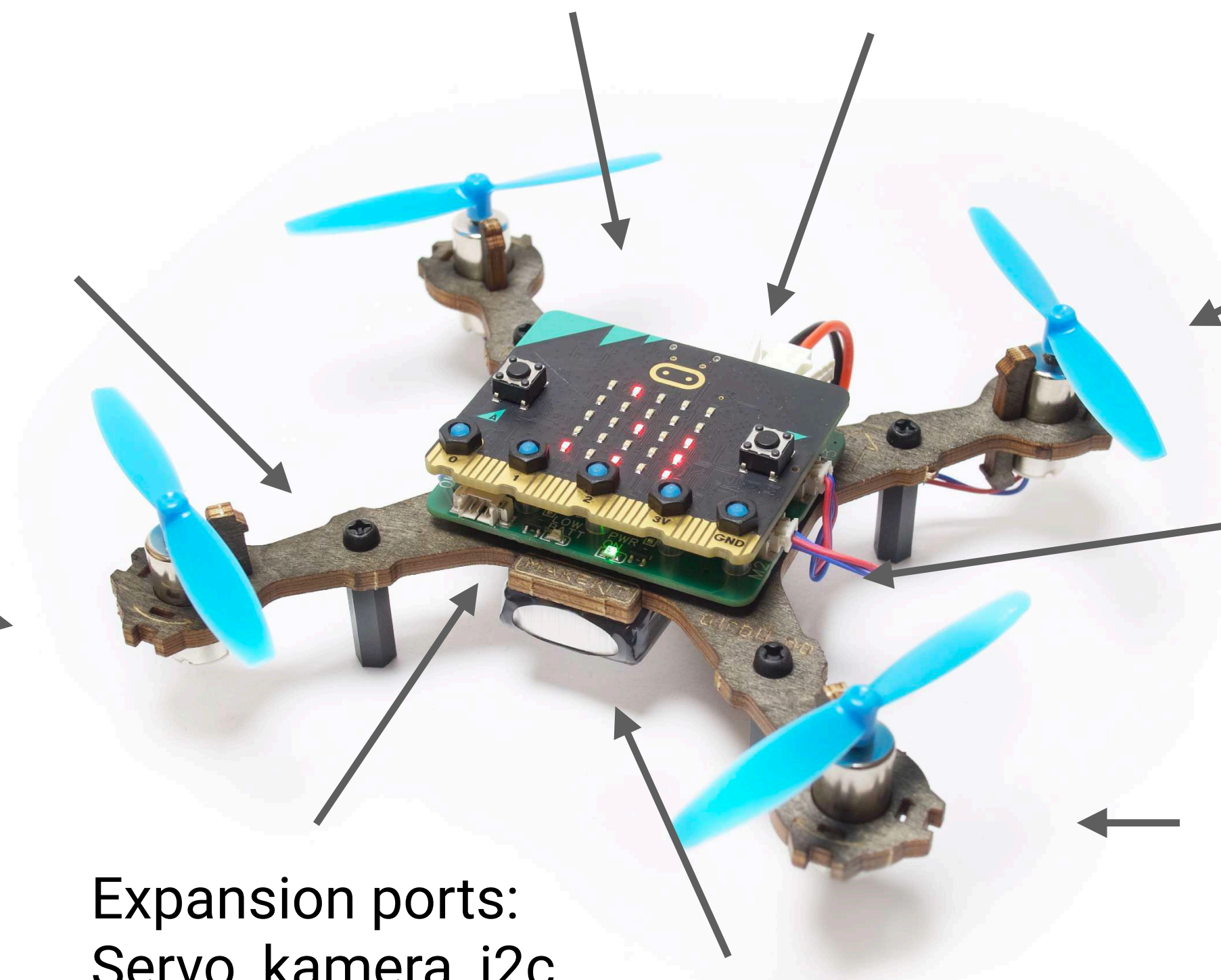
1 cell 3,7v  
800 mAh  
LiPo-battery

Expansion ports:  
Servo, kamera, i2c,  
serial (jst picoblade)

Propeller:  
Gemfan 65mm  
Polycarbonate

6 layers  
“unbreakable”  
birch plywood

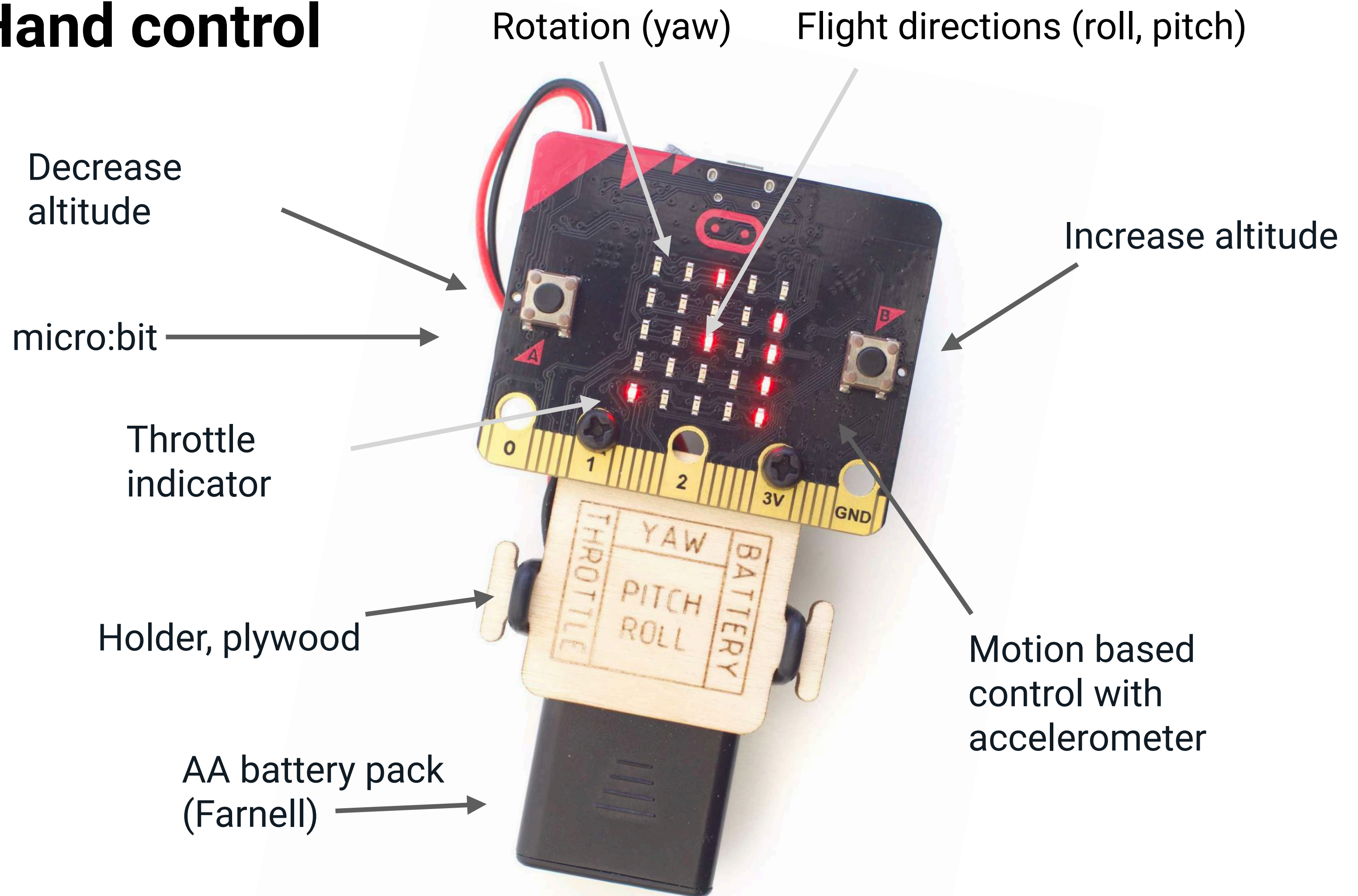
Nylons screws/  
spacers M3



More info: Visit [airbit.no](http://airbit.no)



# Hand control





# Control board

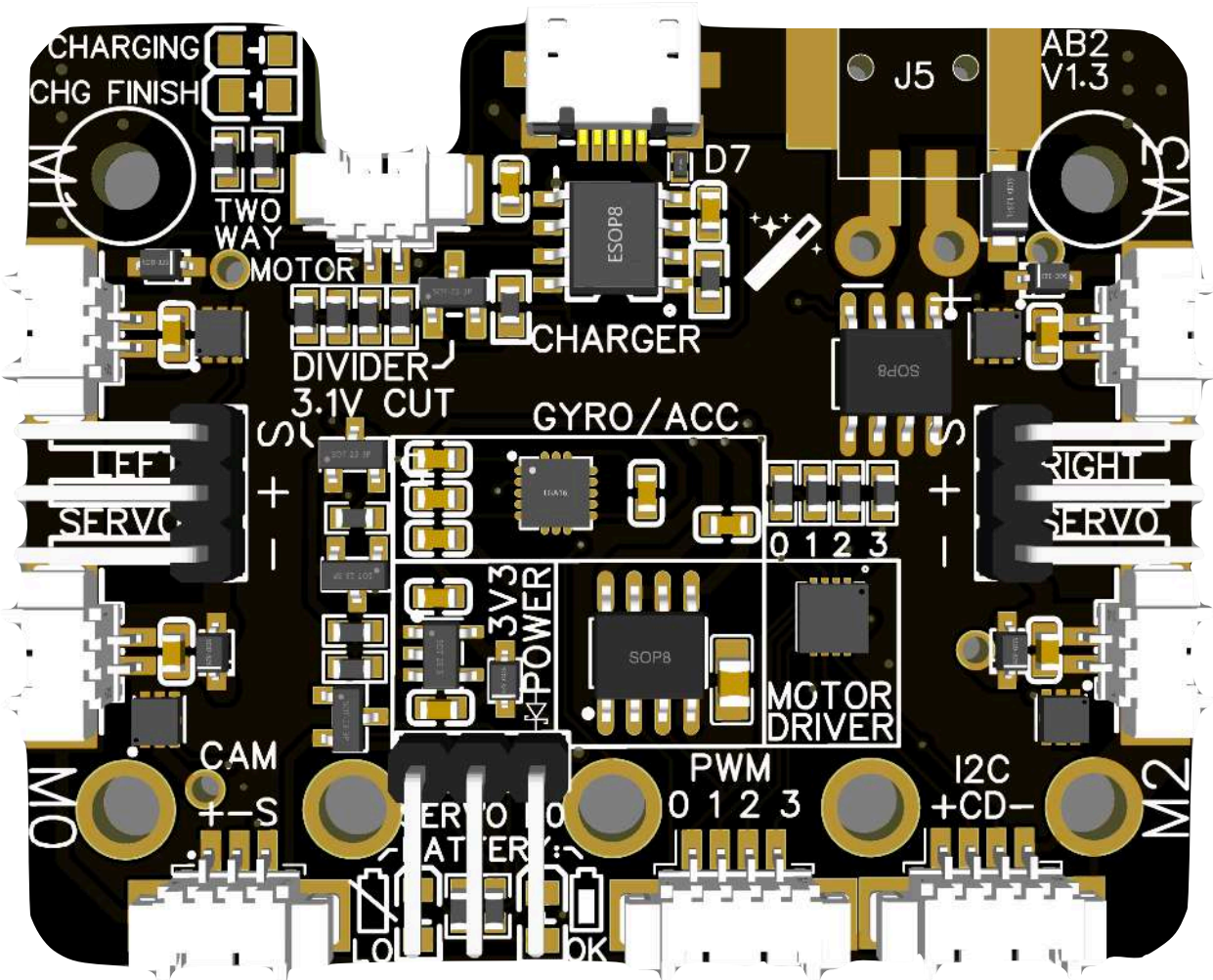
By MakeKit

USB power  
(charging)

Red: Charging  
Green: Finished

Battery plug

Front-left motor  
Servo connector  
Back-left motor



Front-right motor  
Servo connector  
Back-right motor

Micro  
servo/  
camera

i2c (sensors)

# Parts

Note: CW means Clockwise rotation, CCW means counter clock wise



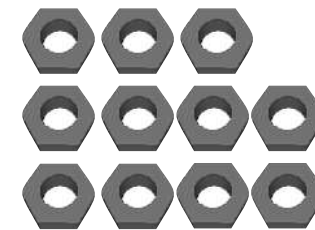
6 nylon screw m3x8



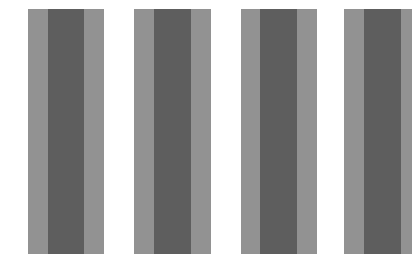
5 nylon screw m3x12



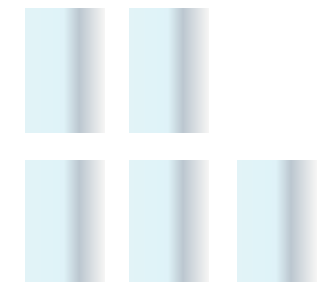
2 nylon screw countersunk



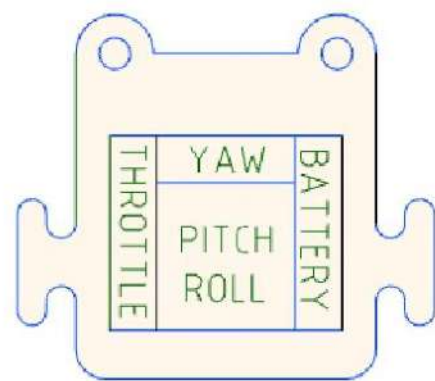
11 nylon nuts



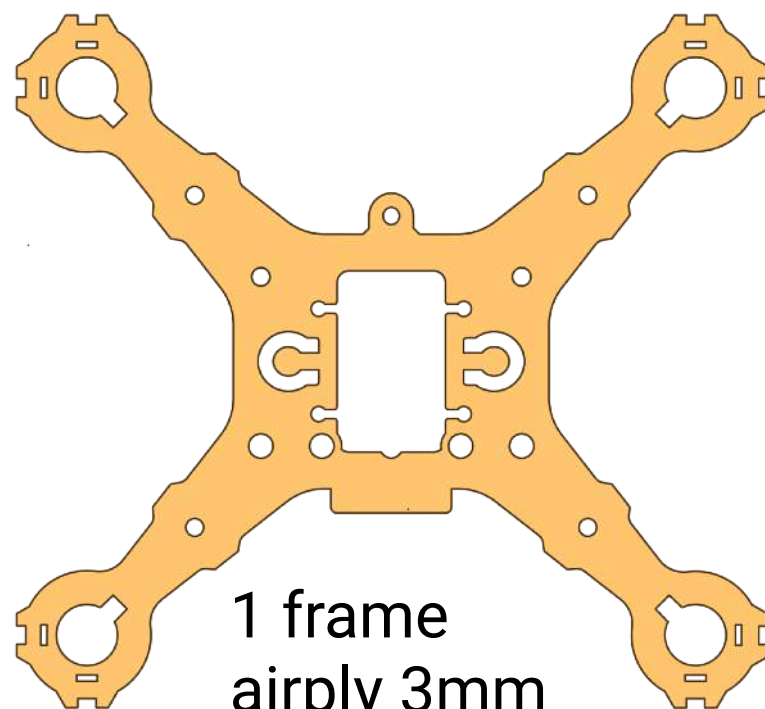
4 spacers m3x20



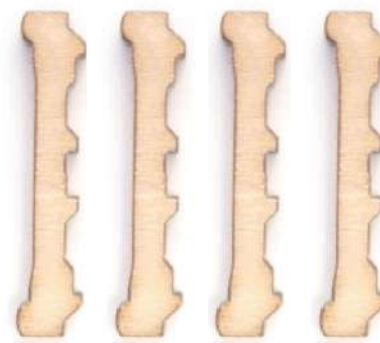
5 aluminium Spacer rings



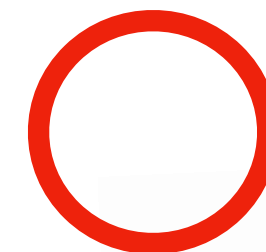
remote - holder



1 frame airply 3mm



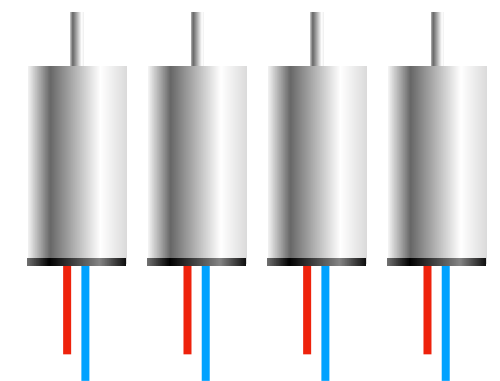
4 wedges



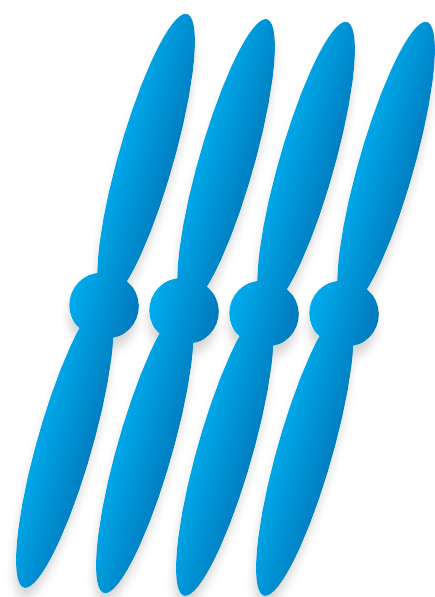
Red silicone ring



Rubber Band



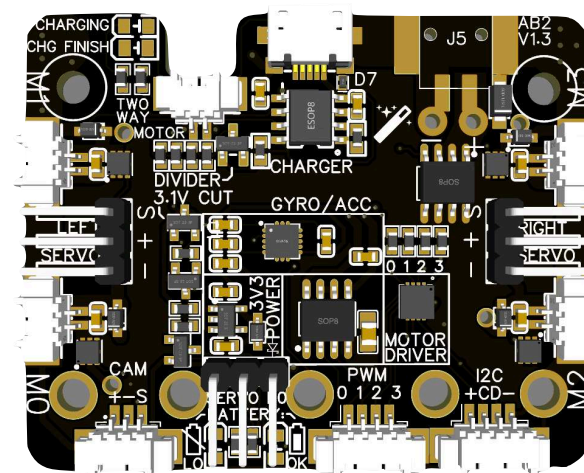
Motors: 4 CW/CCW



4 propeller (+4 spares)  
4 CW, 4 CCW



1 Lithium (LiPo)-battery



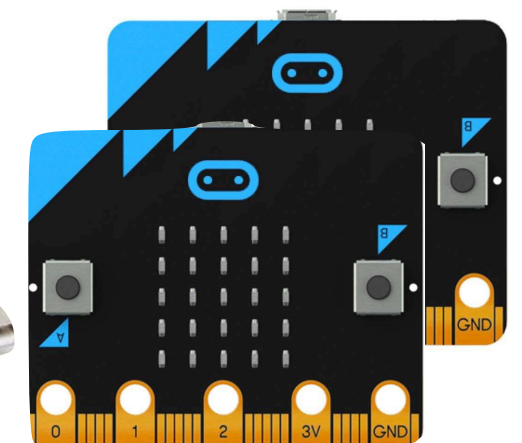
Control board



Micro:bit battery box\*



1 micro usb cable\*



2 micro:bits\*



# Tools



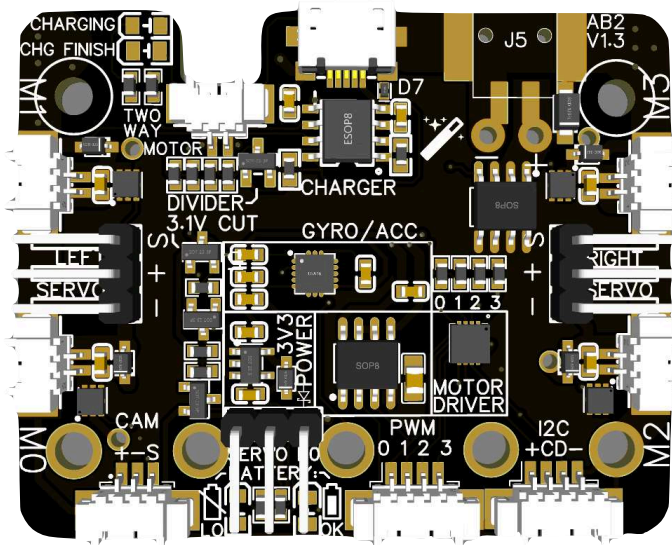
Small philips  
screwdriver



Socket wrenches  
5.5mm (included)

# Charging

## Parts:



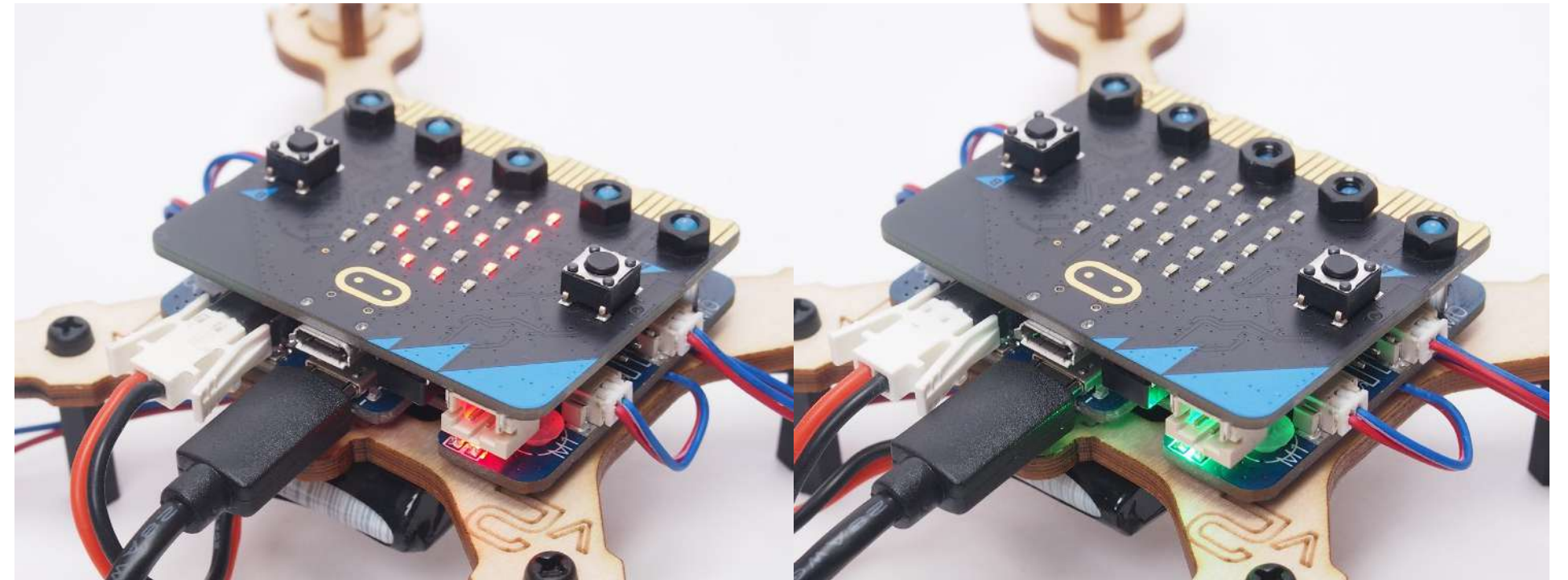
Air:bit control board



Lithium (LiPo)-  
battery



Micro USB-  
Cable



1. The battery can be charge with or without the micro:bit connected.
2. Plug the big white battery plug into the grey connector
3. Connect the micro USB into the Airbit control board (not the microbit)
4. Connect the other end into a USB charge outlet
5. Red light indicates charging. Green light indicates charging finished. It takes about 1 hour to charge.
6. To prevent battery drain, always unplug battery when not in use!

**Fire hazard:**  
**Never charge a Lithium battery unattended!**





# Assembly

# Assemble the remote

Tools: Philips Screwdriver

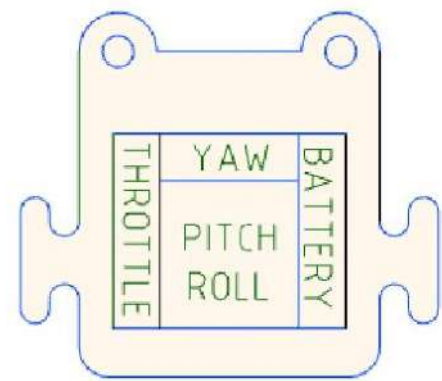
## Parts:



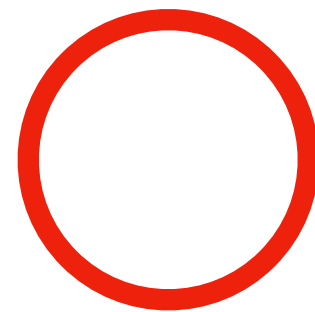
2 nylon screws  
m3x8



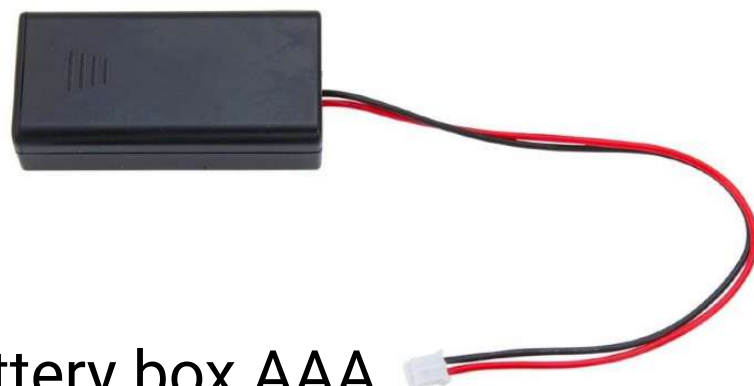
2 nylon  
nuts m3



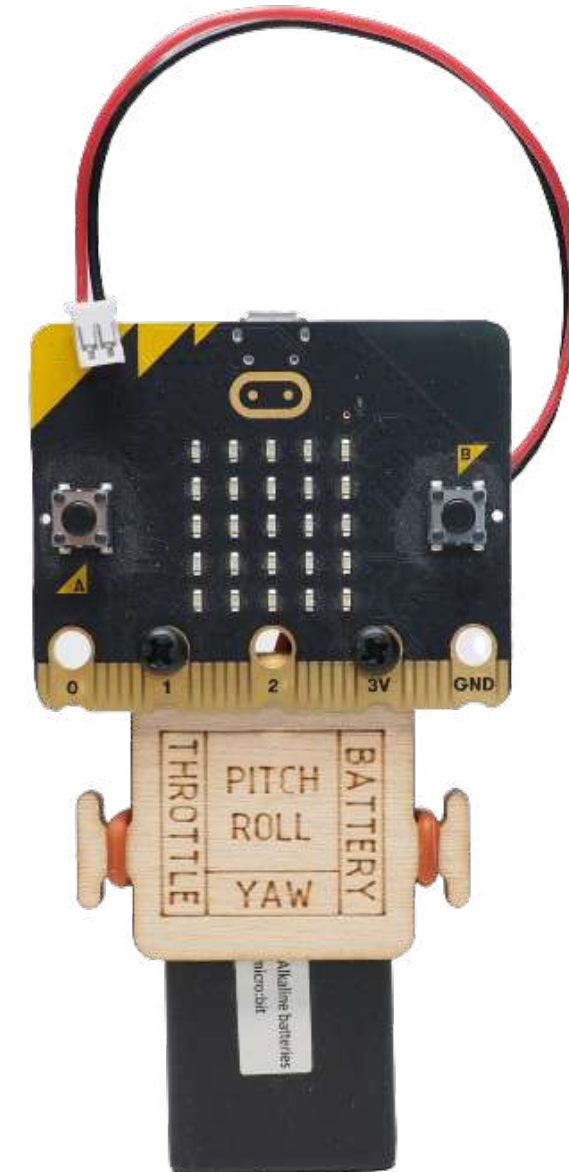
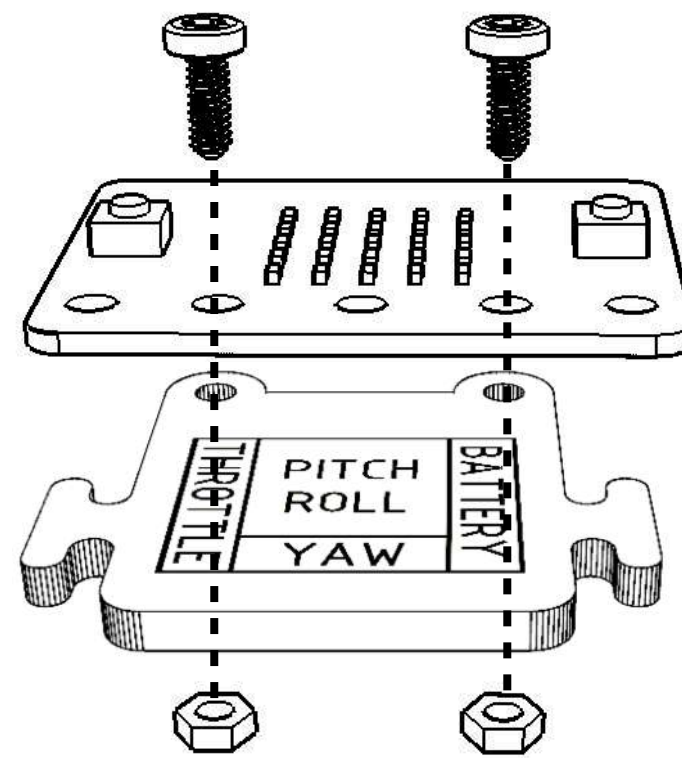
Remote holder



Red silicone ring



Battery box AAA



- Place the micro:bit with screen facing up on top of the holder and mount it with the screws somewhat tightened
- Mount battery box with the silicone ring
- You can use different battery boxes



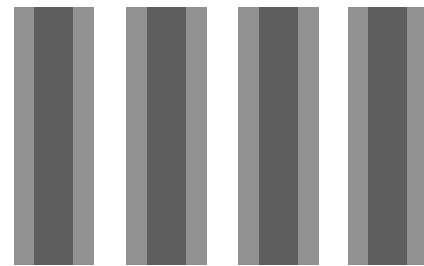
# Mounting the frame

**Tools:** Philips Screwdriver

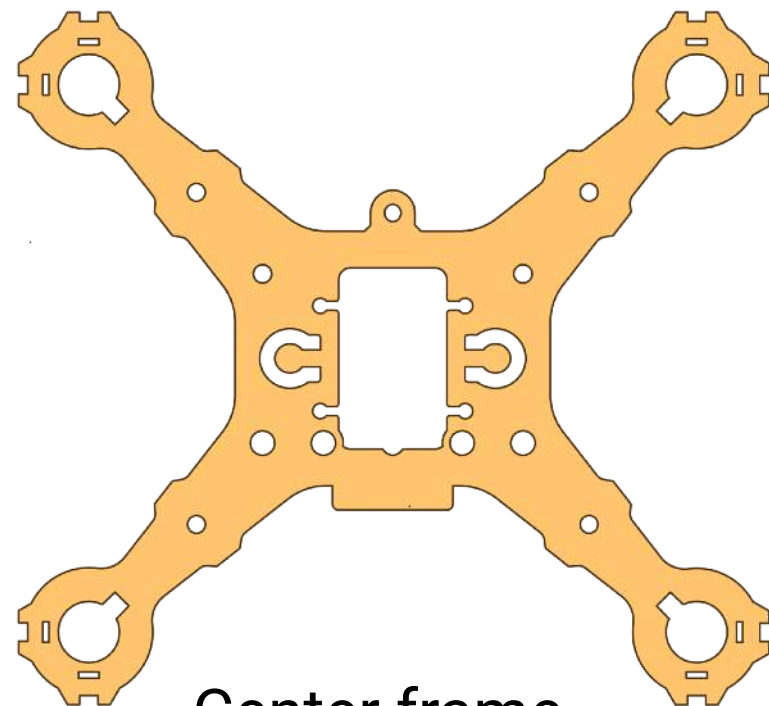
## Parts:



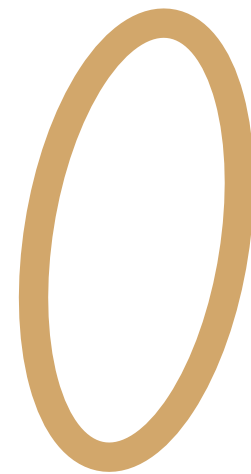
4 stk nylon screws m3x8



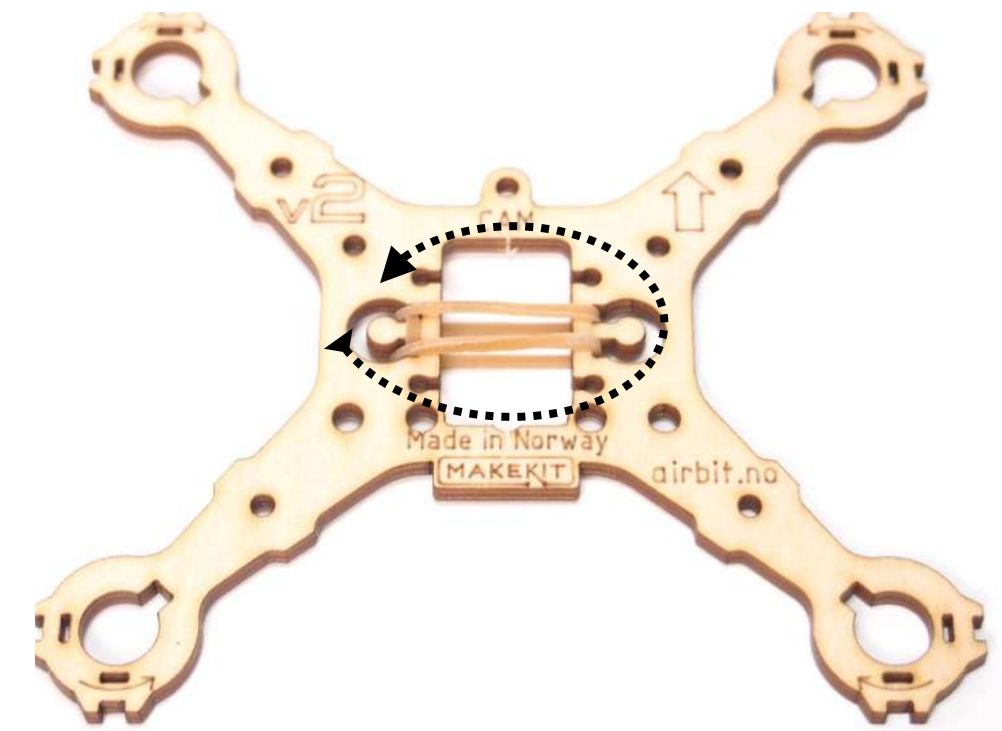
4 spacers  
m3x20



Center frame



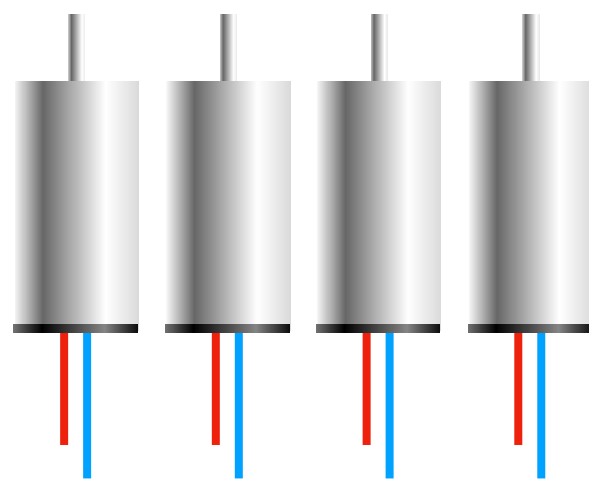
Rubber band



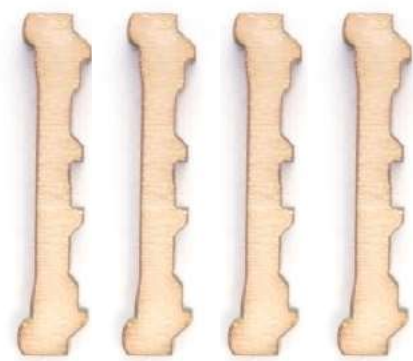
- Hook the rubber band from one knob, under the frame, to the next hook, over the frame, and back.
- Mount the spacers with the screws. Notice, the text should face up while the legs points down.
- The rubber band will be visible on both sides of the frame

# Mounting the motors

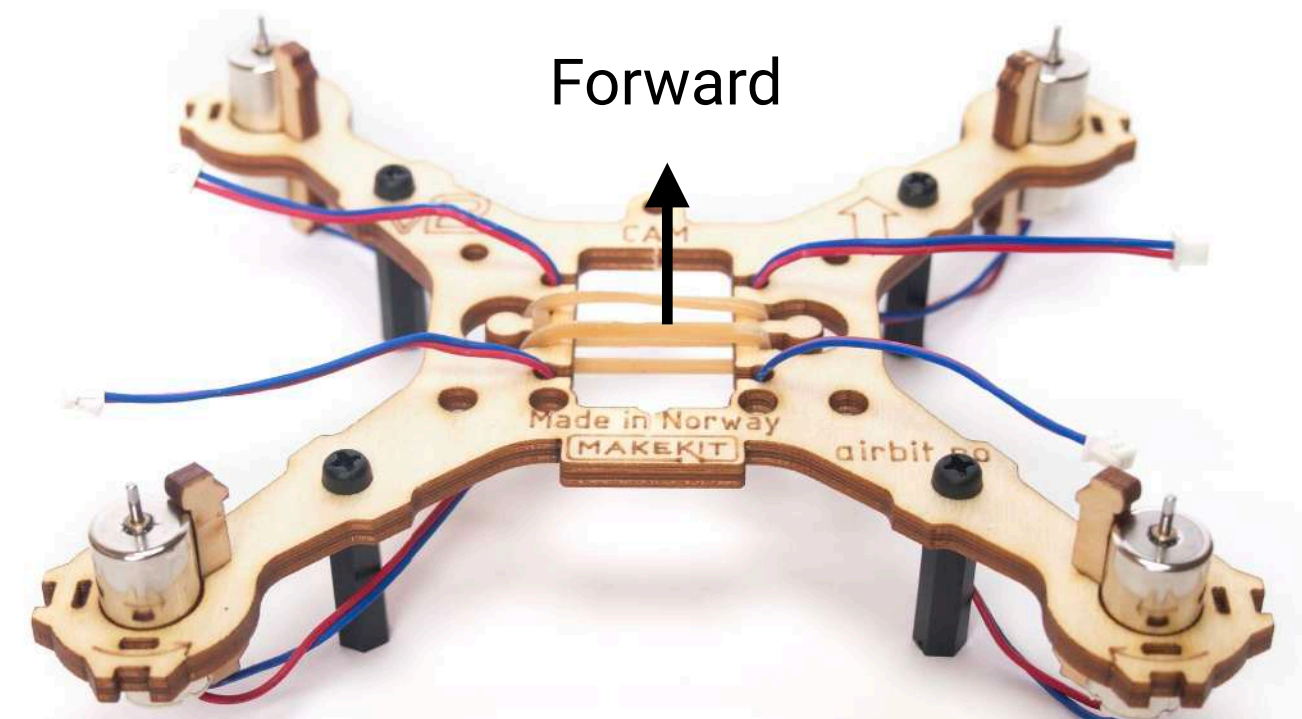
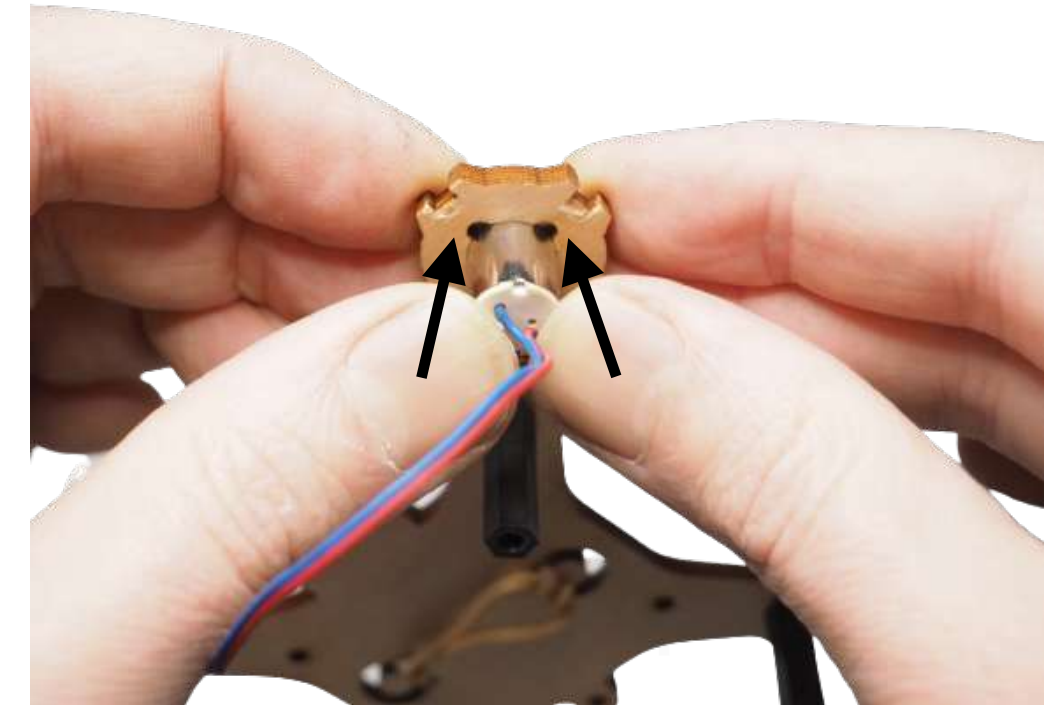
## Parts:



4 motors



4 wedges



- First insert a wedge and wiggle it in place into the frame,
- Then push the motor upwards to click into place. (All 4 motors). Use both hands and both thumbs to push the motor forward.
- Then pull the motor cables through each notch to tidy them up a bit. Lay the cables to the sides.



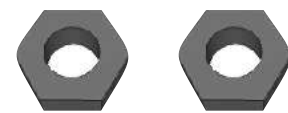
# Countersunk screws

**Tools:** Philips Screwdriver, socket wrench

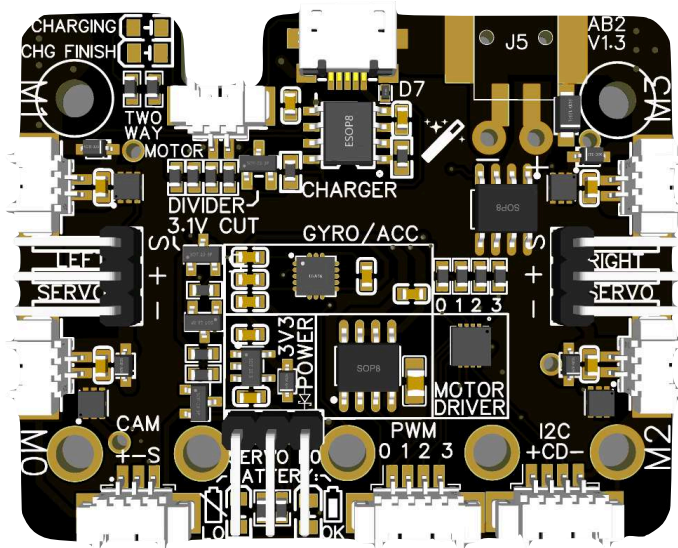
## Parts:



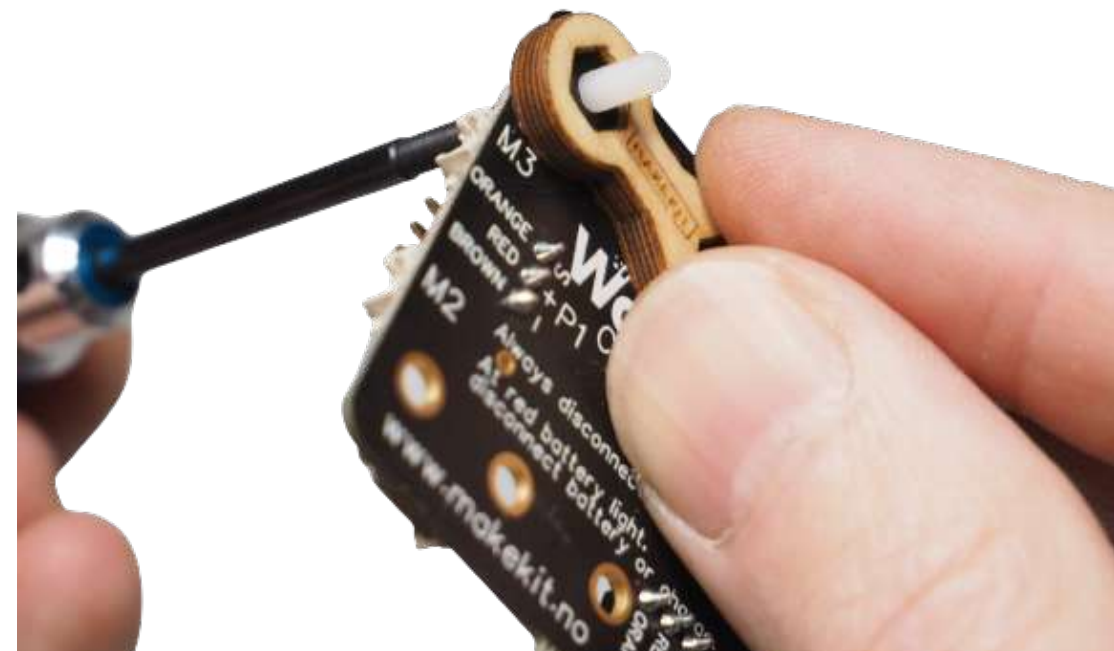
2 x  
countersunk  
screws



2 x  
Nylon nuts



Control board



- Pull the screws through to the back side of the board (where the big letters are)
- Attach the nuts with a screwdriver and a socket wrench

# Control board

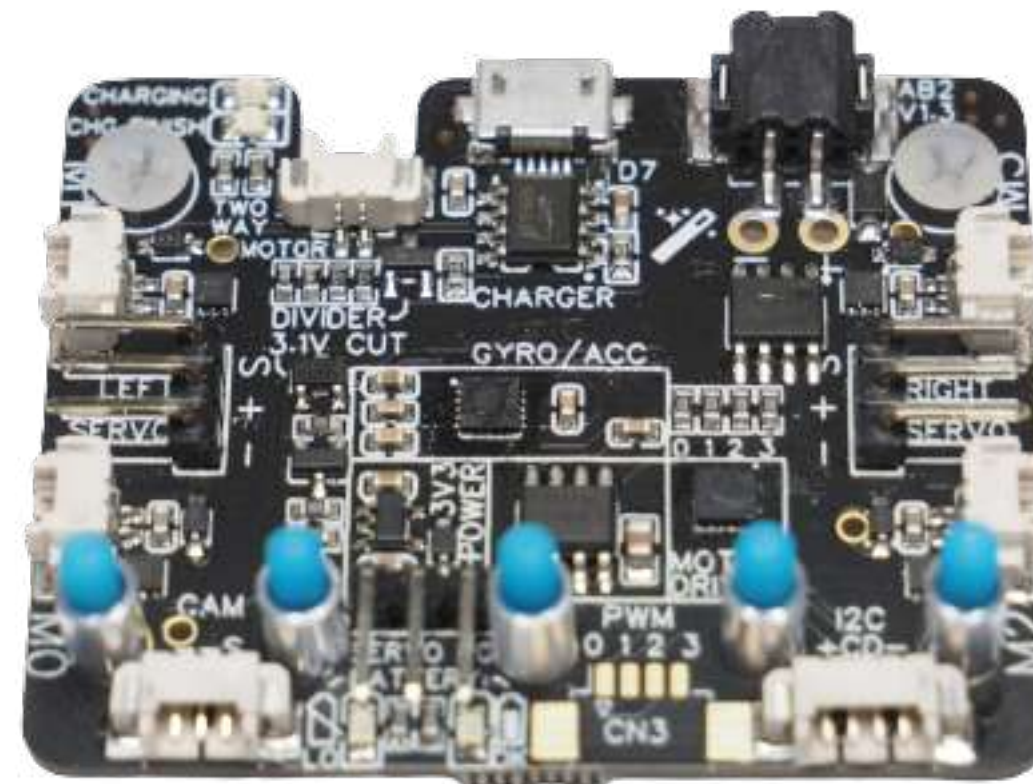
## Parts:



5 nylon screws m3x12 (blue)



5 aluminium  
spacer rings



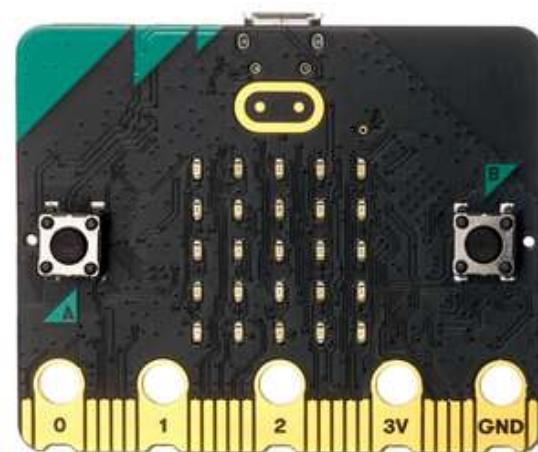
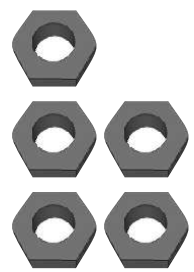
- From the backside of the board, insert the 5 screws.
- Hold the screws with your index finger while you flip the control board.
- Insert the 5 spacer rings.



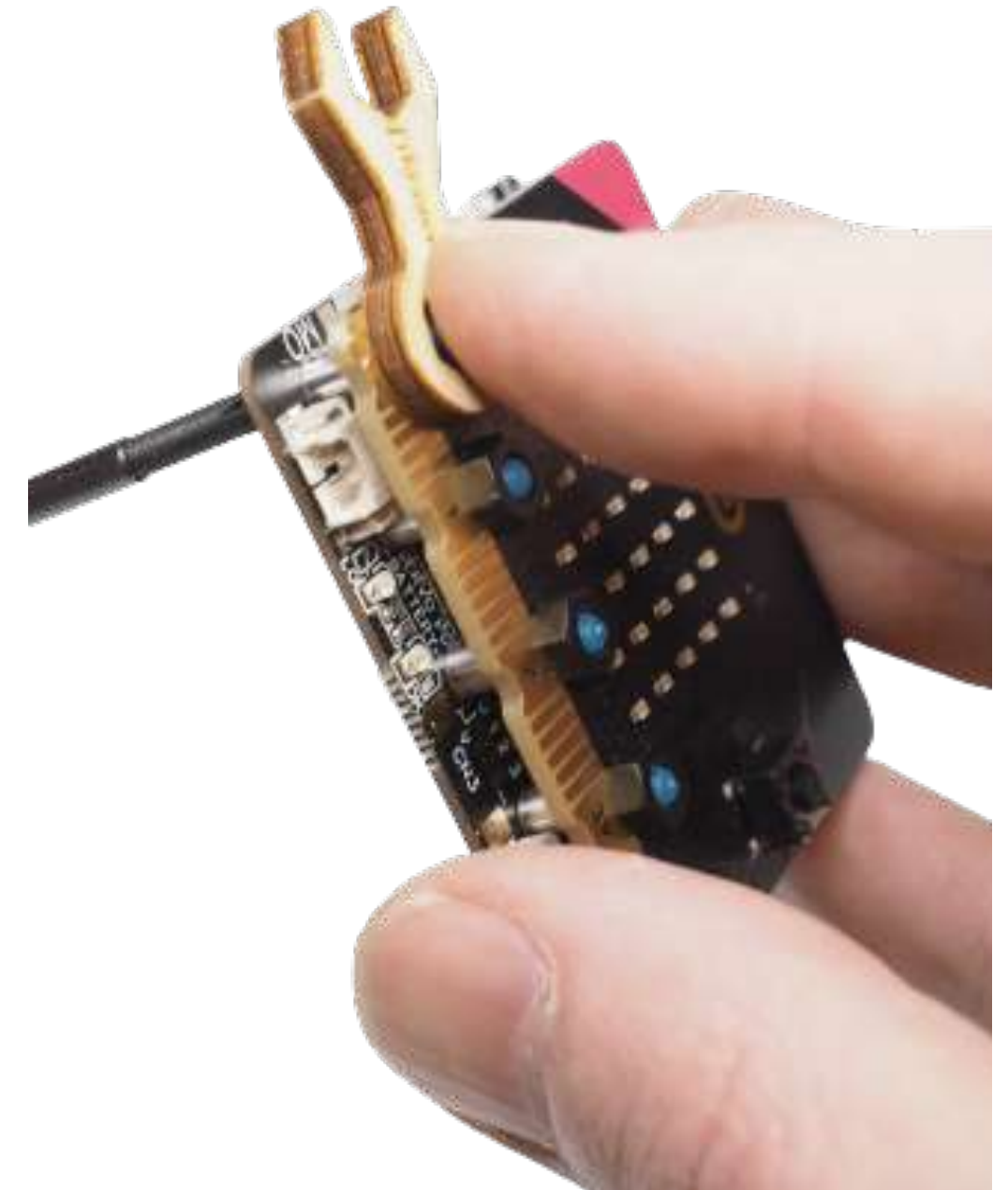
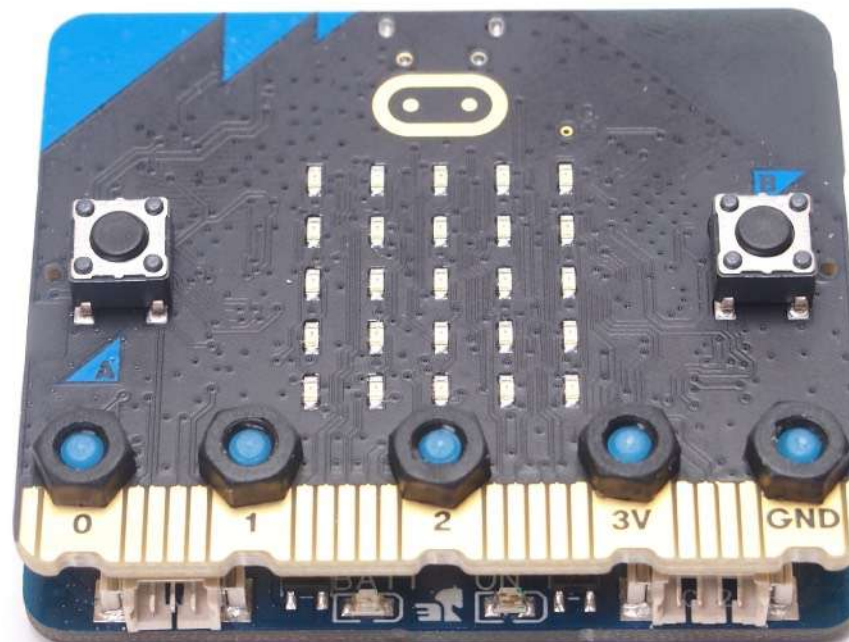
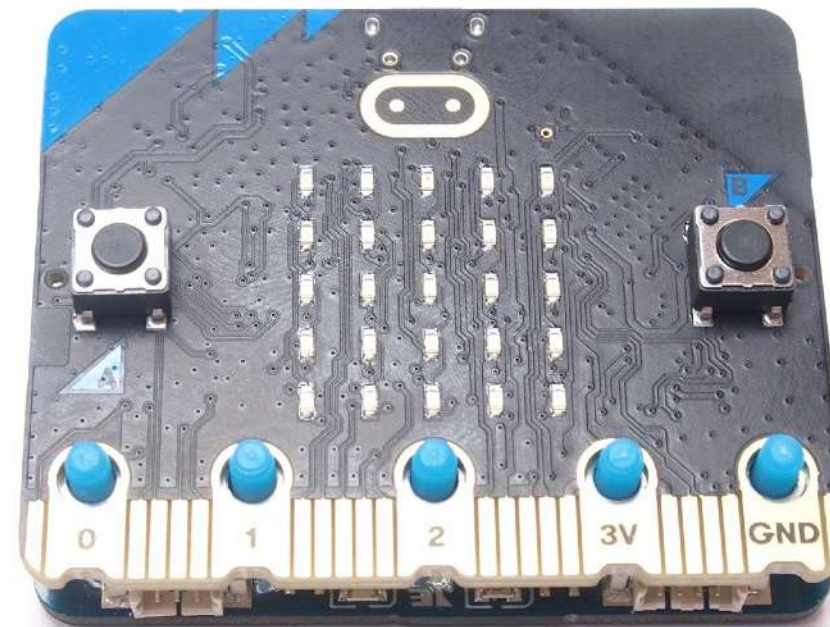
# micro:bit

**Tools:** Philips Screwdriver, wrench  
5.5mm

**Parts:**



micro:bit

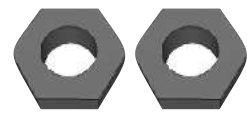


- Place the micro:bit on the five screws
- Carefully add the five nuts and mount them loosely before lifting the unit
- Add some tension using a screwdriver and the socket wrench.
- **The screws must be slightly tightened for the drone to work as they conduct electric signal between the boards.**

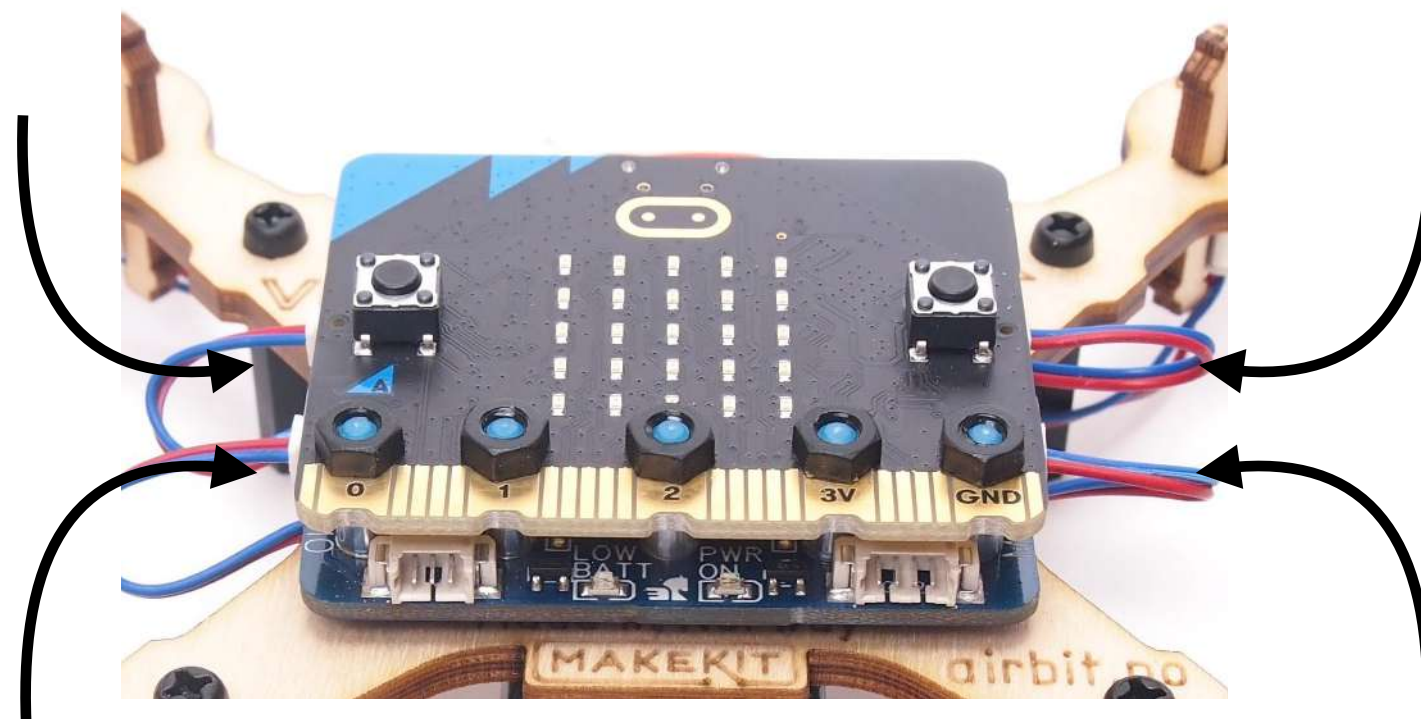
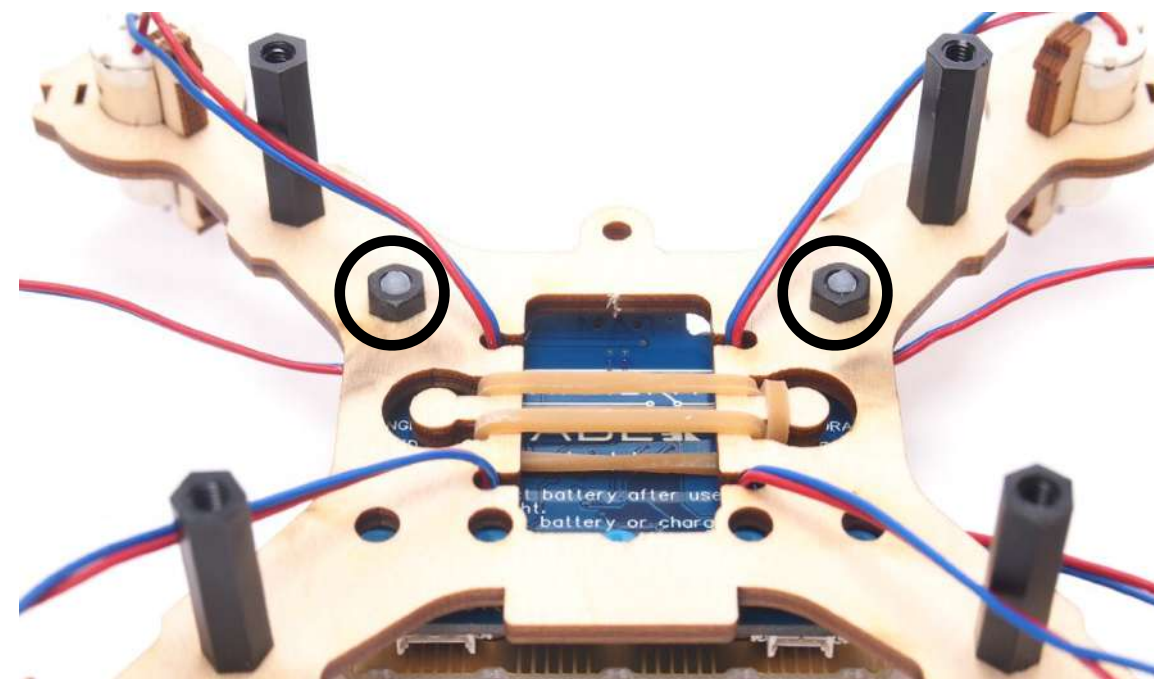
# Assembling the drone

**Tools:** Wrench 5.5mm

**Parts:**



2x nylon nuts



- Flip the drone upside down
- Attach the control board with two nuts underneath the drone. Slightly tighten with the wrench.
- Plug each motor in the nearest connector.



# Battery

## Parts:

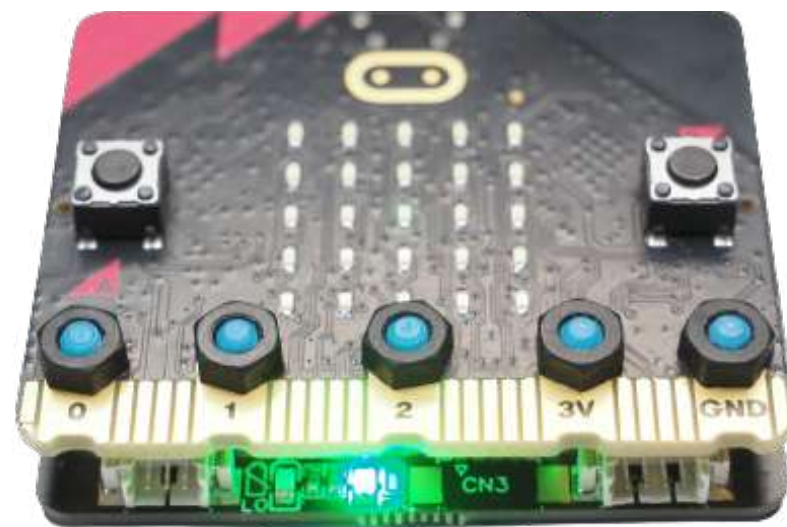


1 Lithium (LiPo)-battery

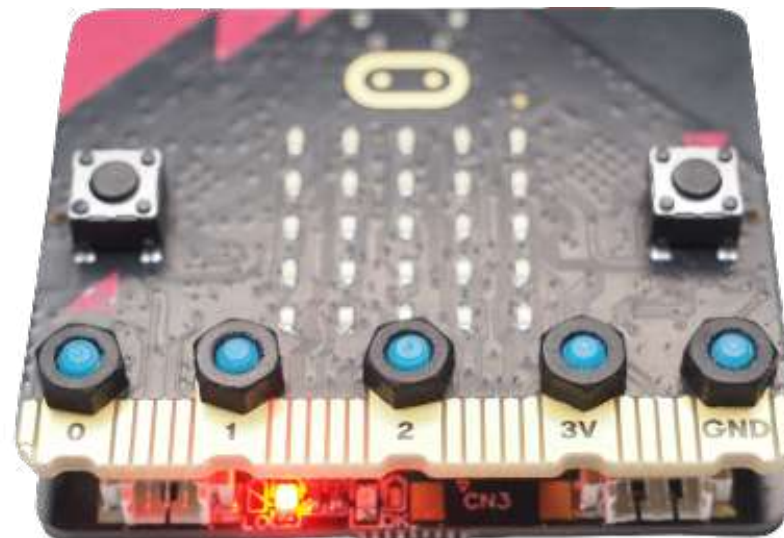


- Turn the drone upside down
- Attach the battery under the rubber band, in the very center of the drone.
- To power the drone, connect the battery to the grey connector.
- Disconnect the battery whenever you are not using or charging the drone. The battery can suffer over discharge if connected for extensive time.

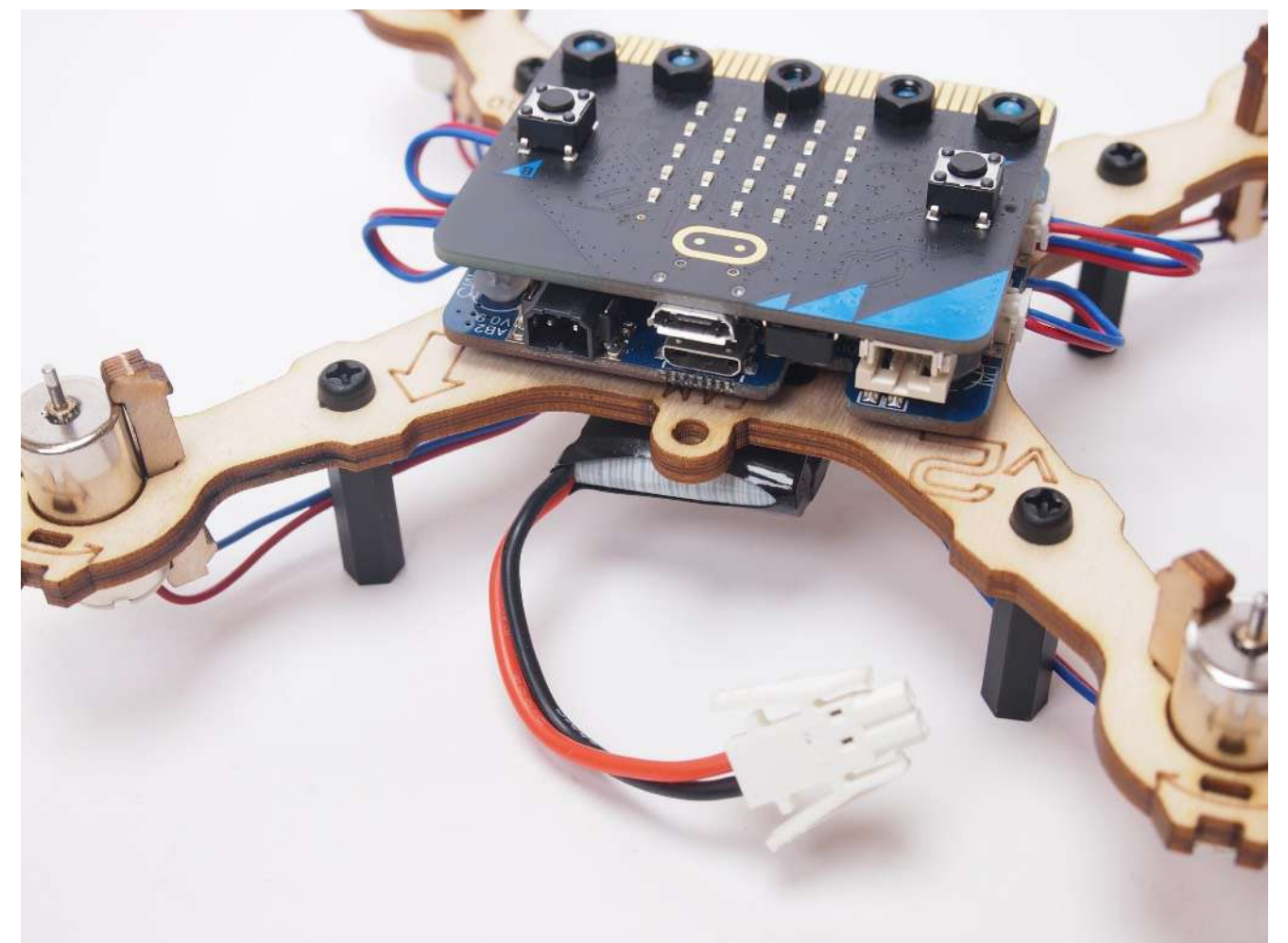
# Battery indicator and care



Normal



Low battery

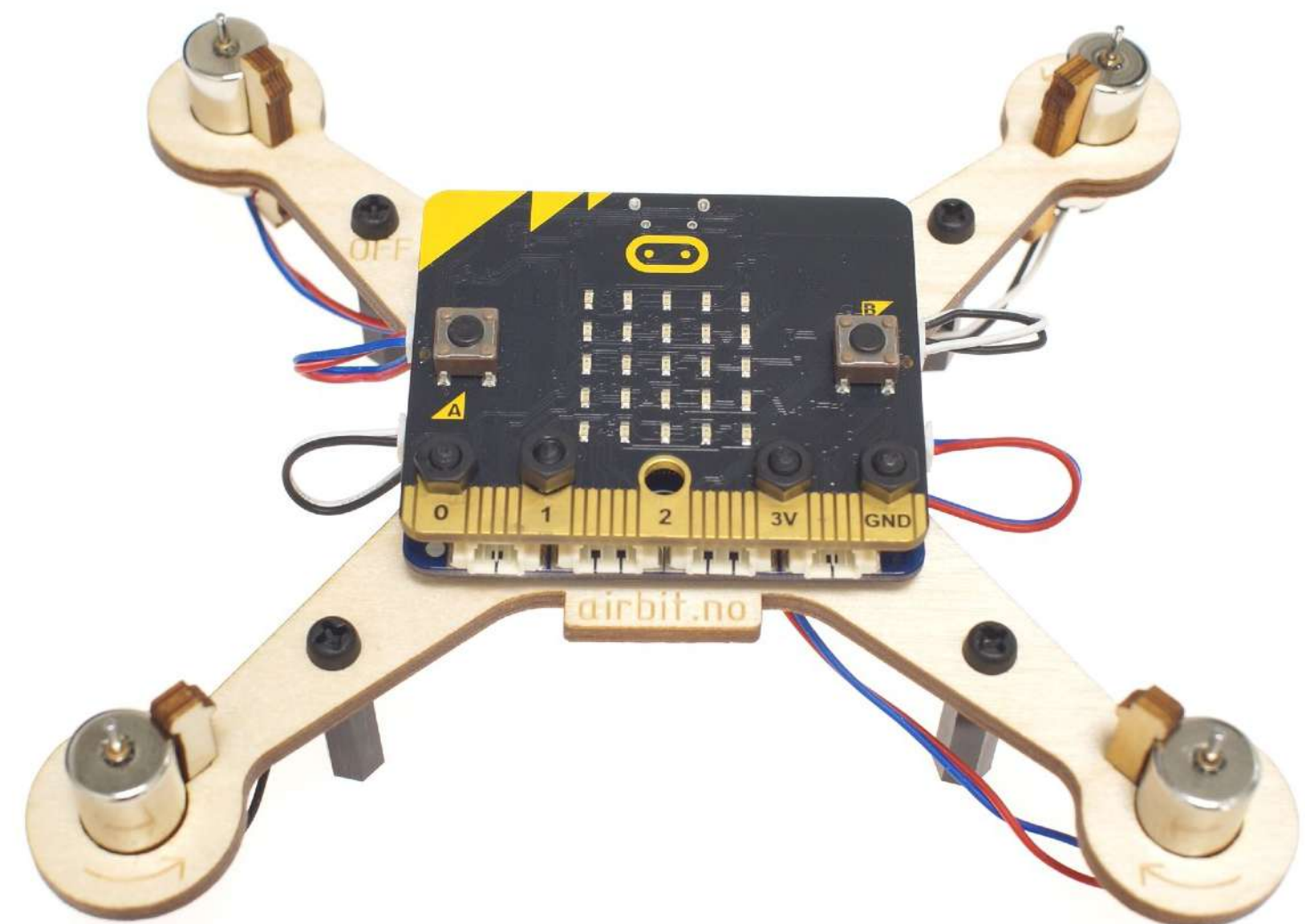
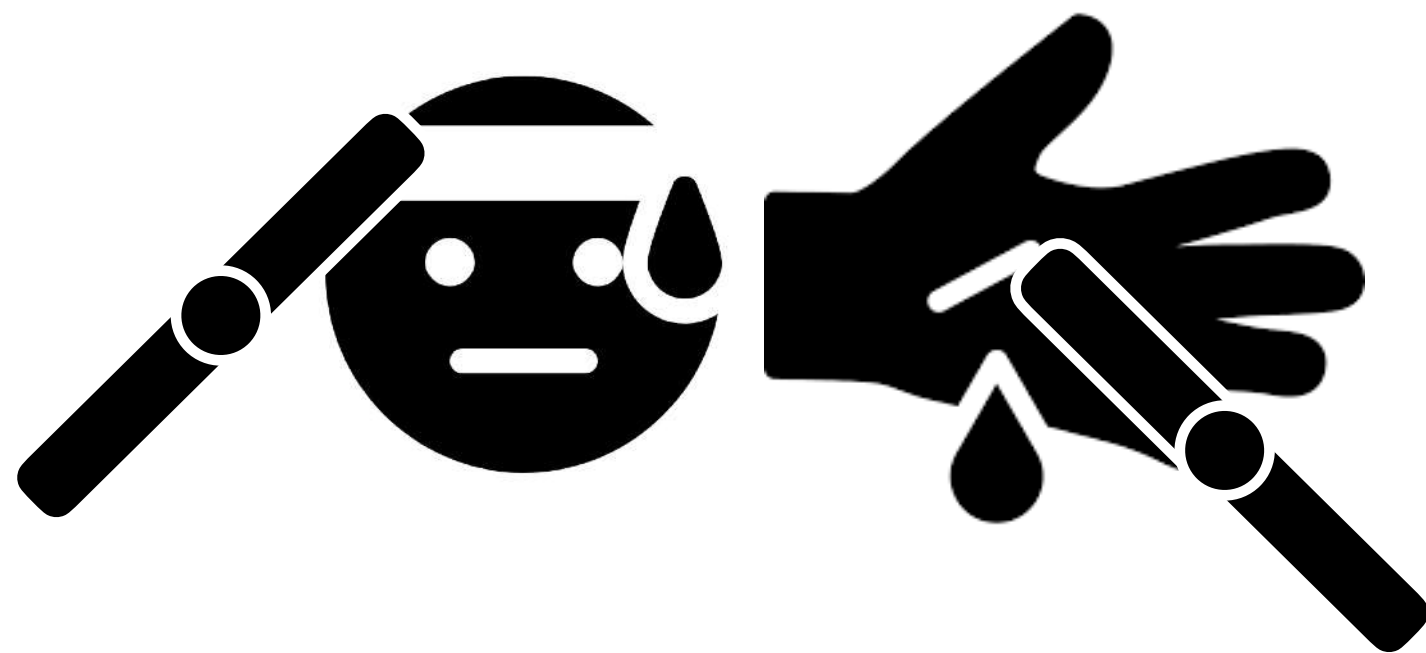


The control board has a battery protection function.  
When the **green** light is lit, battery has sufficient power for flight.  
A **red** light indicates that the battery needs to charge.  
Do not attempt to fly the drone after this light has been seen.  
Disconnect the battery until you are ready to charge again.

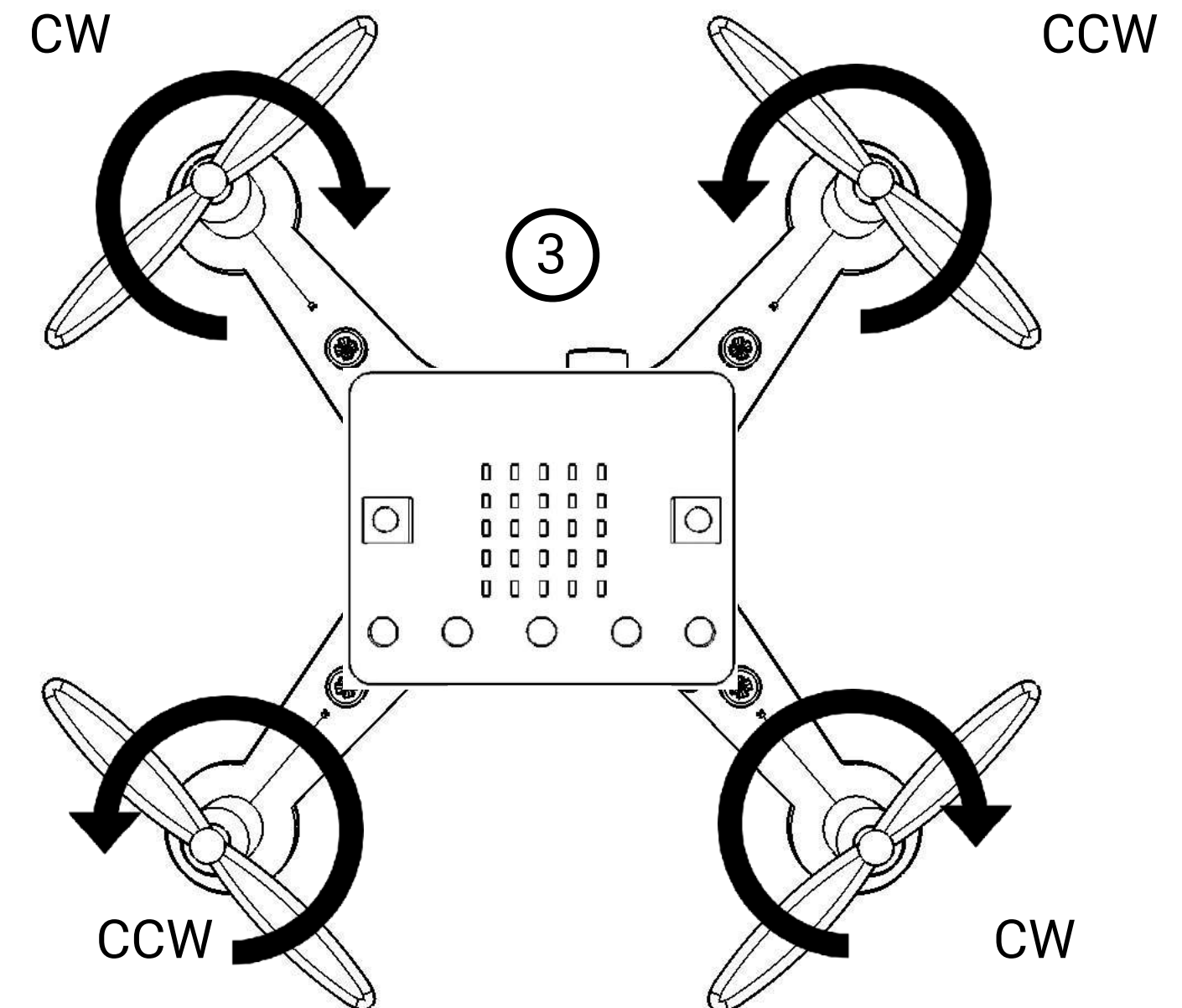
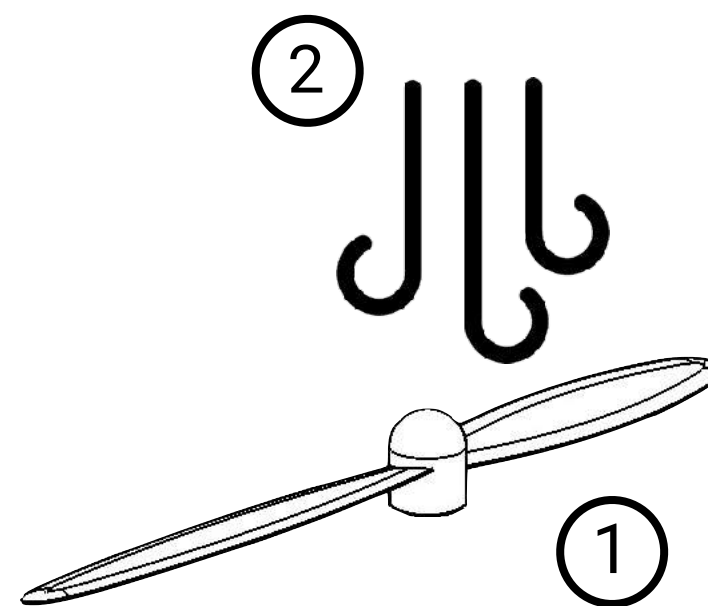
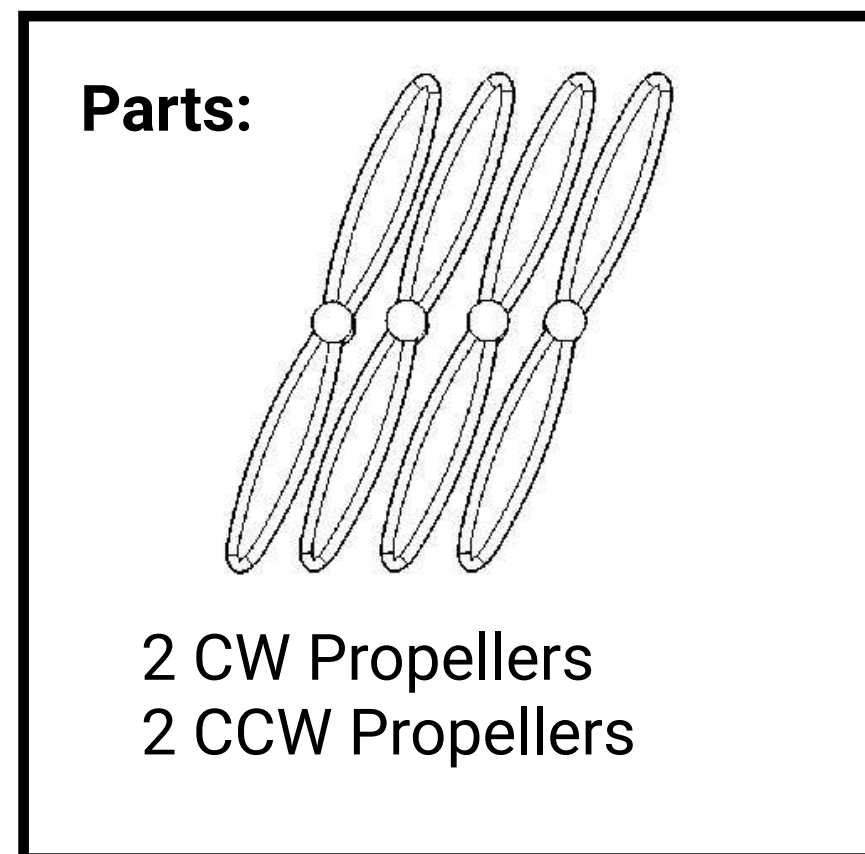


# For your safety

Mount the propellers after testing the drone. Can you start and stop the motors in a controlled manner?



# Propellers



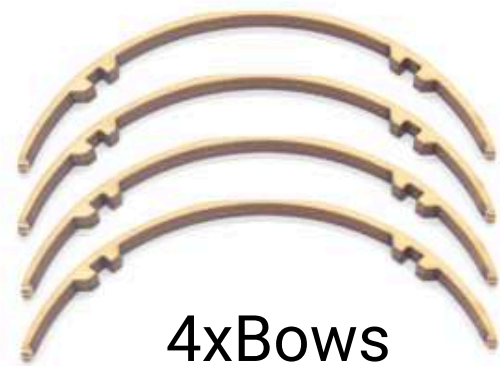
- Make sure the battery is disconnected
- Identify your propellers between CW (clockwise) and CCW (counter clock wise):
  - Take a random propeller. Place it on a table, able to spin (1)
  - Blow gently straight down above (2)
  - If it rotates with the clock, its a CW propeller
  - If it rotates against the clock, its a CCW propeller
- Place all four propeller on the correct motors
- The letters "CW" or "CCW" can also be read on the propeller near the center top.
- If you need to remove a propeller, use some pliers and eject the center of the prop. Do not bend the propeller blades.



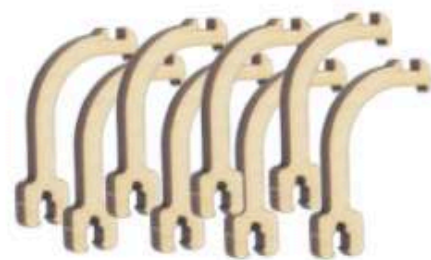


# Propeller protection

## Parts:



4xBows



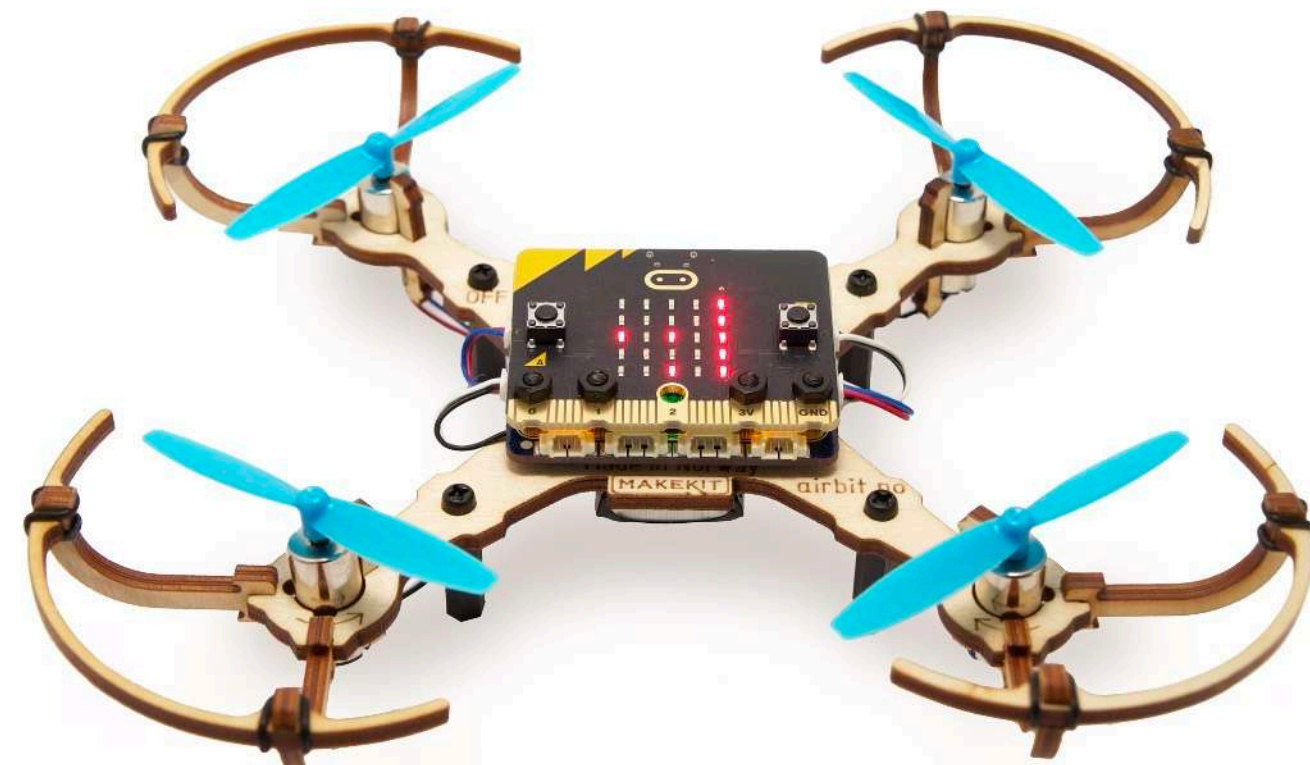
8xArms



8xSmall rubber rings

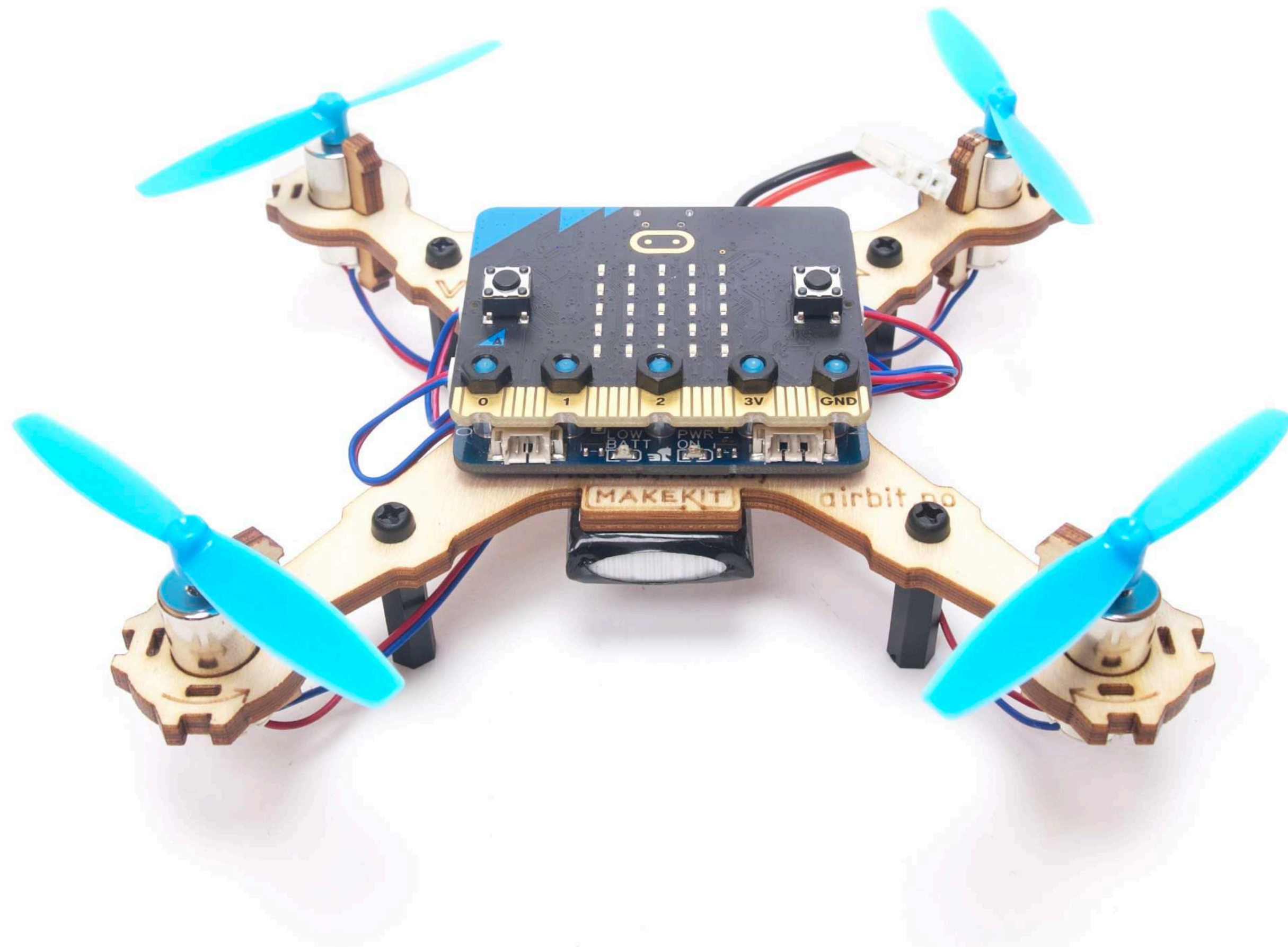


Connect the bows and the arms with the rubber rings



On each protector, insert both arms onto the main frame. Adjust so they are straight and do not obstruct the propellers when they rotate.

# Ready for coding!





Feb. 16, 1926.

G. DE BOTHEZAT

1,573,228

HELICOPTER

Filed June 27, 1923

3 Sheets-Sheet 1

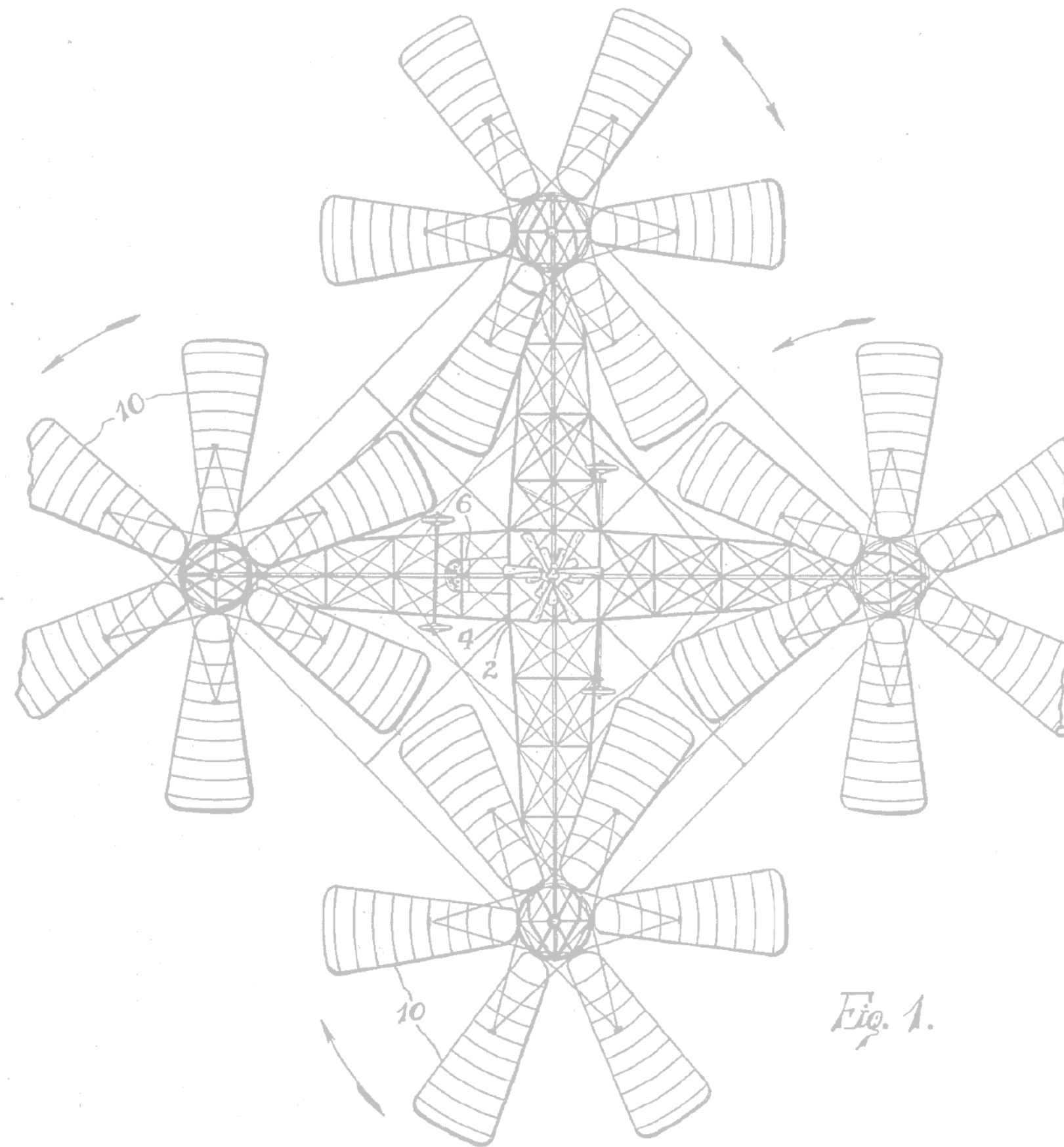


Fig. 1.

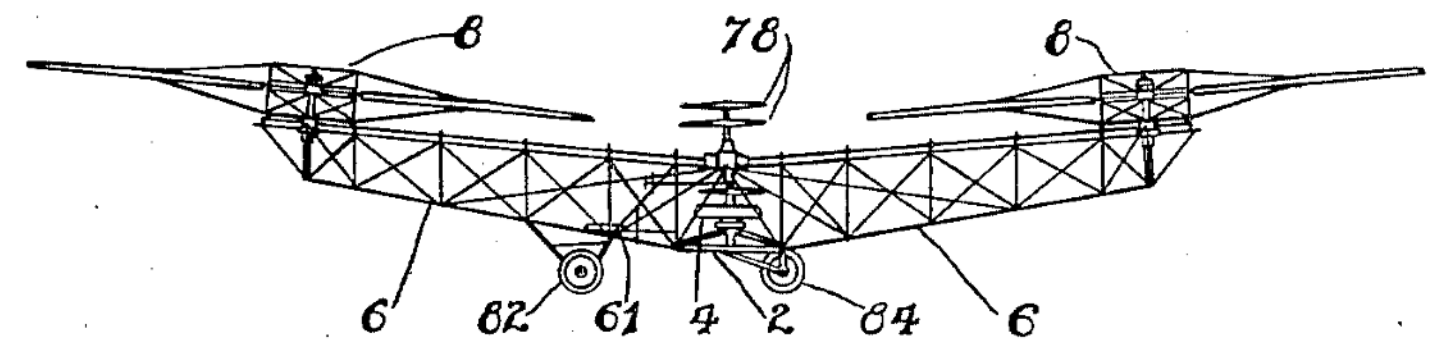
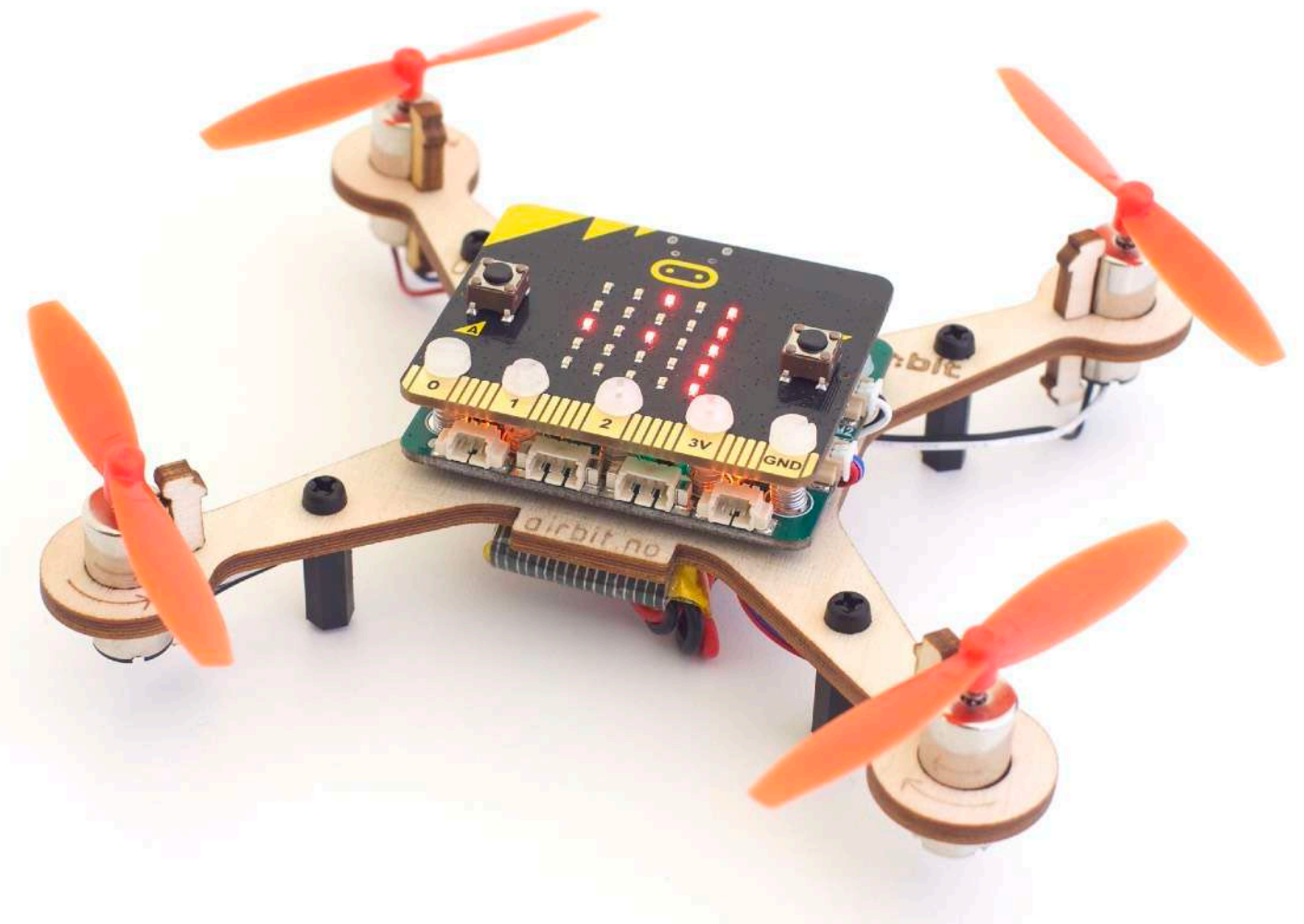
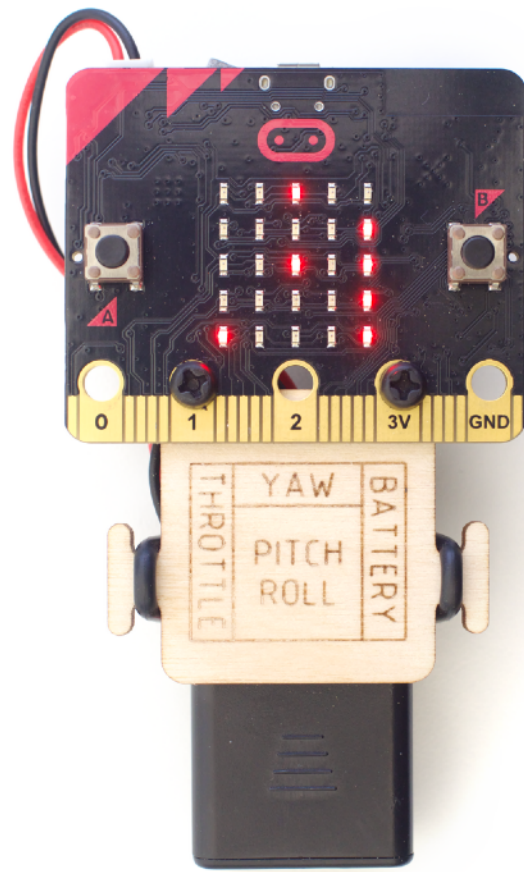


Fig. 2.

INVENTOR

BY George de Bothezat

Robert H. Young ATTORNEY

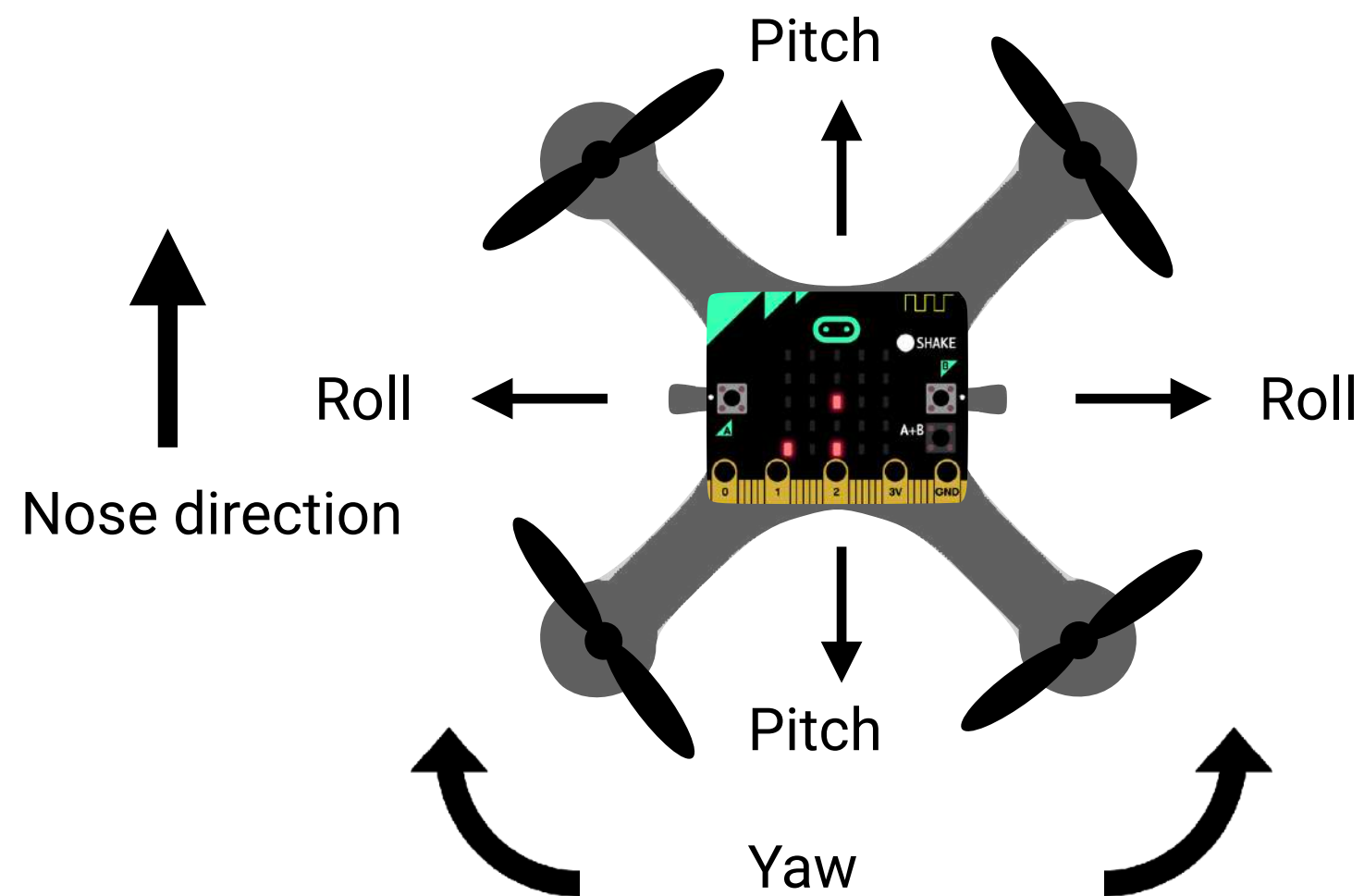
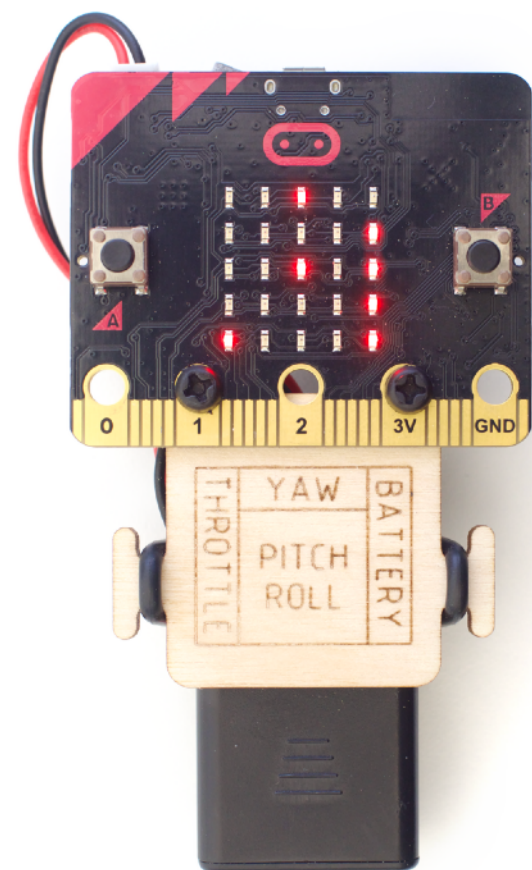


# PARTY

Five values to control the drone



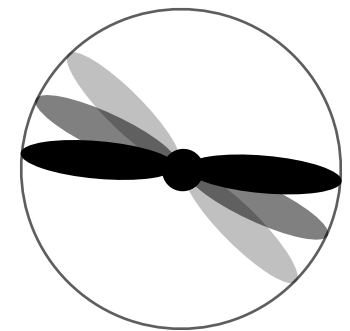
# Flight directions



# PARTY

pitch, arm, roll, throttle, yaw

Arm:  
Start propeller

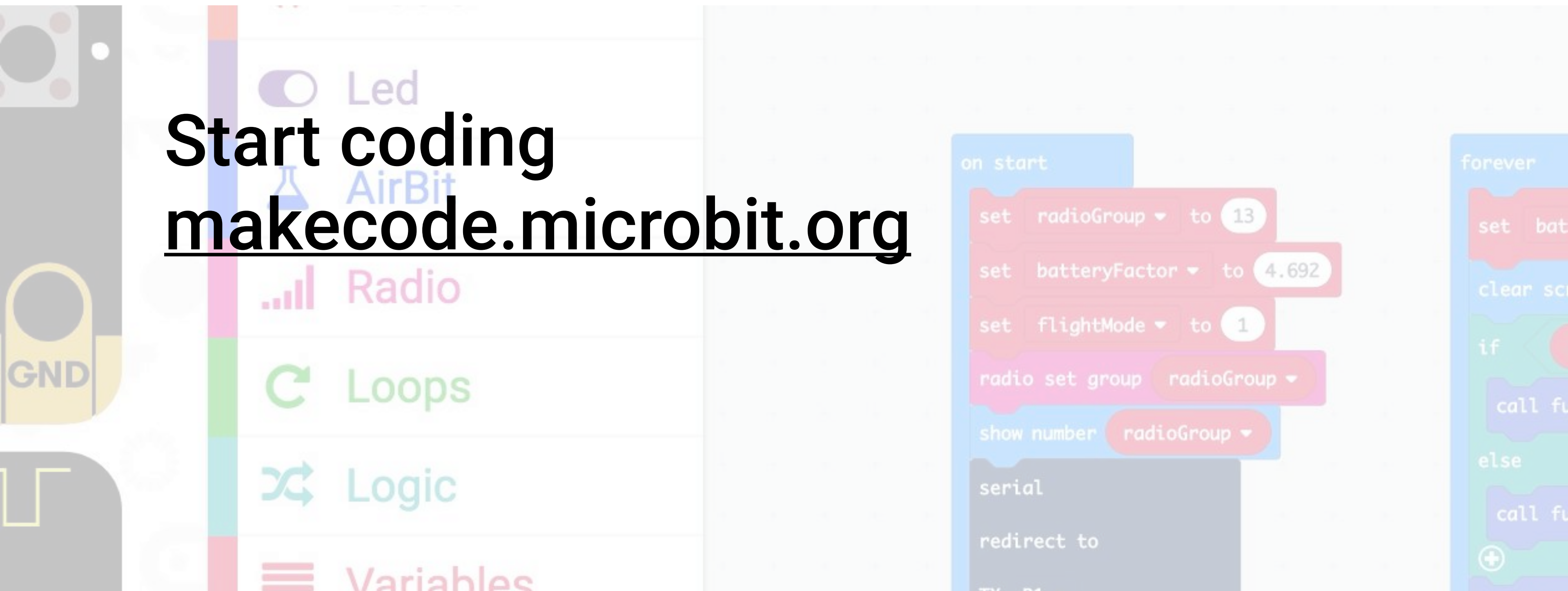


Control the drone using 5 parameters:

| Variable: | Minimum | Neutral | Maximum |
|-----------|---------|---------|---------|
| Arm       | 0       |         | 1       |
| Throttle  | 0       | 50      | 100     |
| Pitch     | -45     | 0       | 45      |
| Roll      | -45     | 0       | 45      |
| Yaw       | -30     | 0       | 30      |

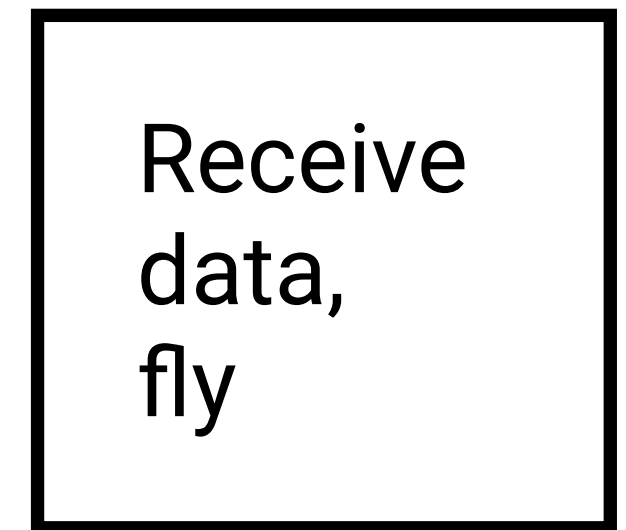
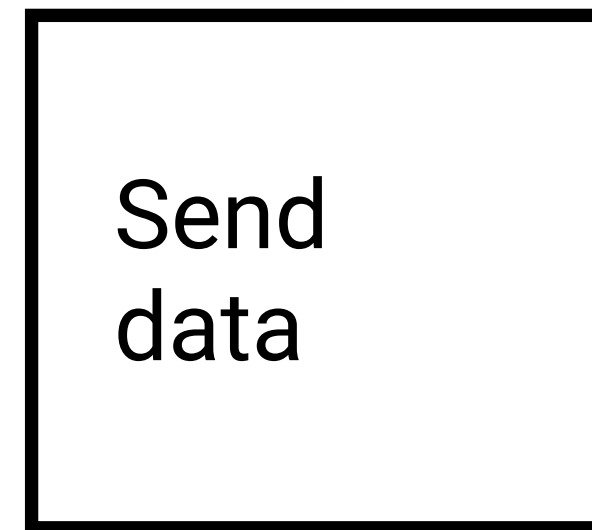
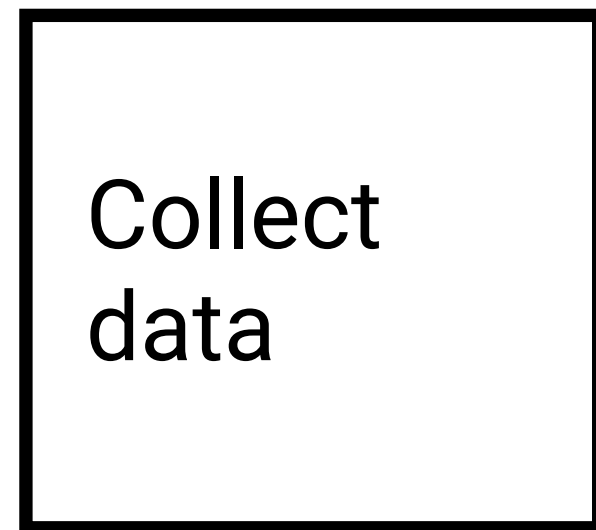
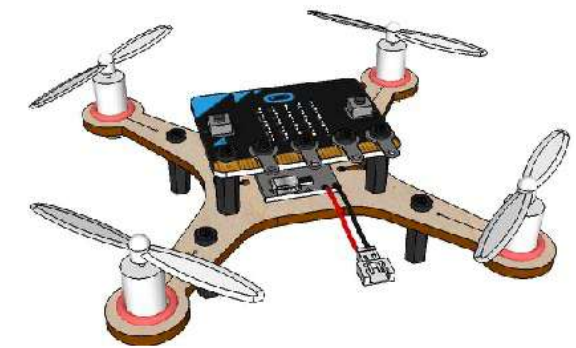
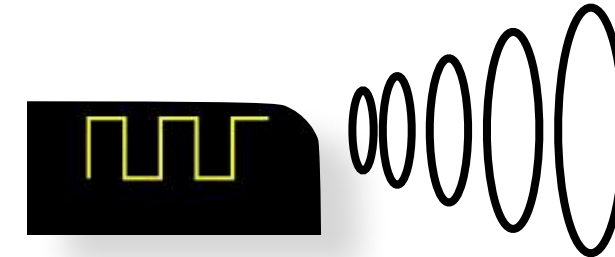
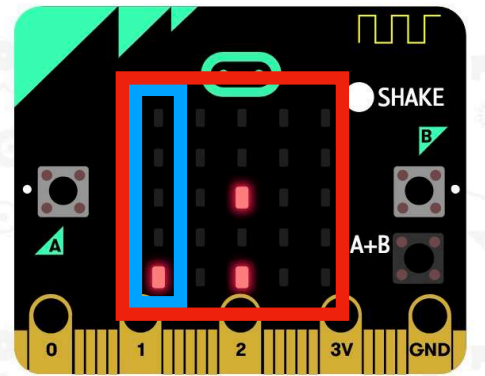
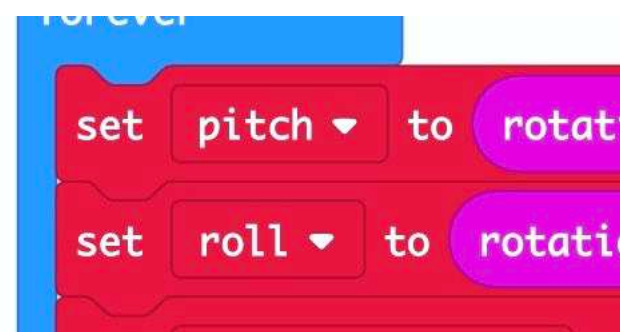


**Start coding**  
**[makecode.microbit.org](https://makecode.microbit.org)**





# Four steps



Collecting data for  
Pitch, Arm, Roll,  
Throttle, Yaw

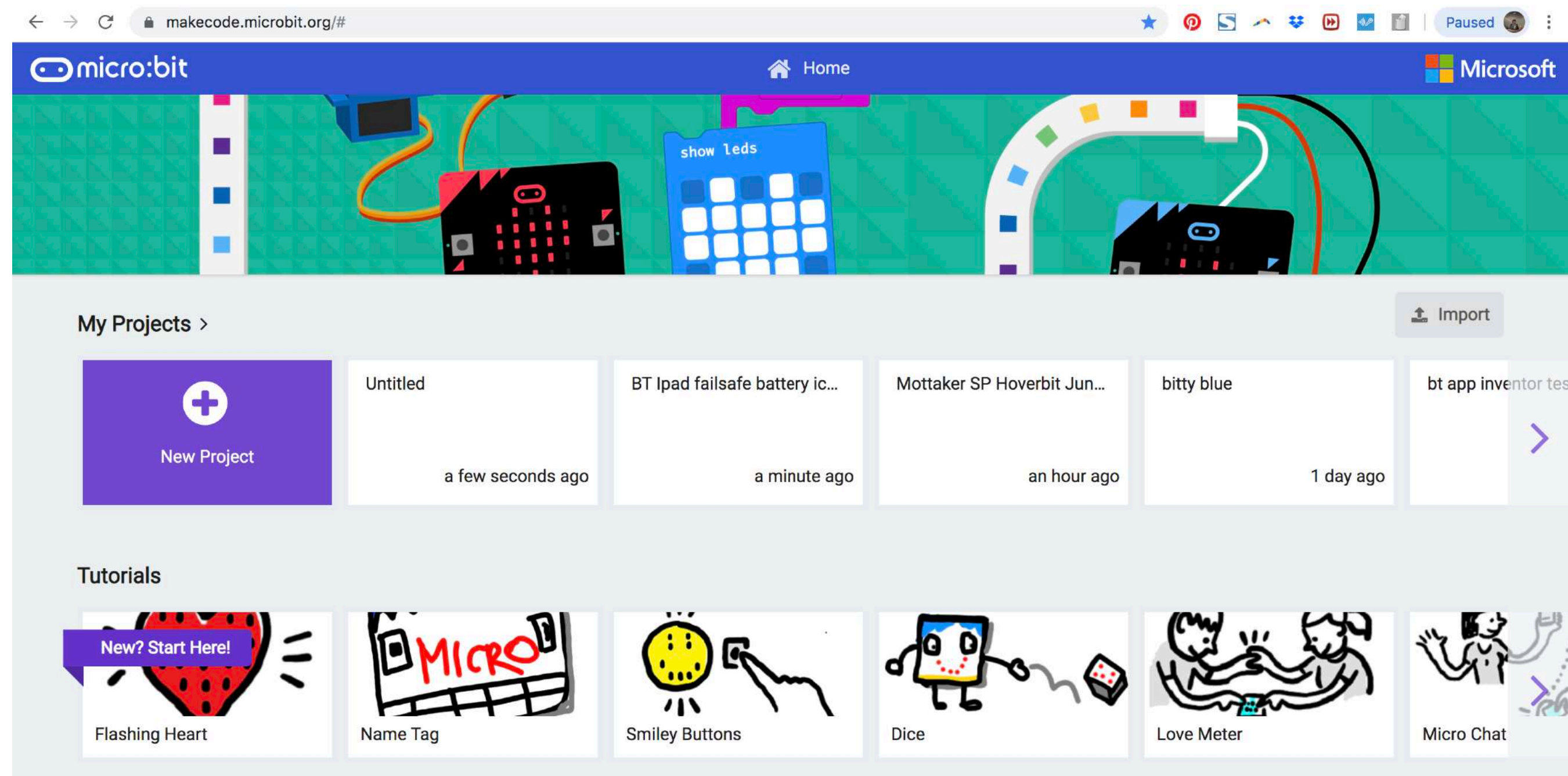
Convert the values to  
coordinates for the  
display

Send the value over  
the radio

Receive the values  
and turn the into  
control signal for the  
Air:bit control board

# Start at [makecode.microbit.org](https://makecode.microbit.org)

Chrome is recommended for better connection with the micro:bit



Select “New project”.

PS. If you’re new to micro:bit you should try one of the tutorials above first.



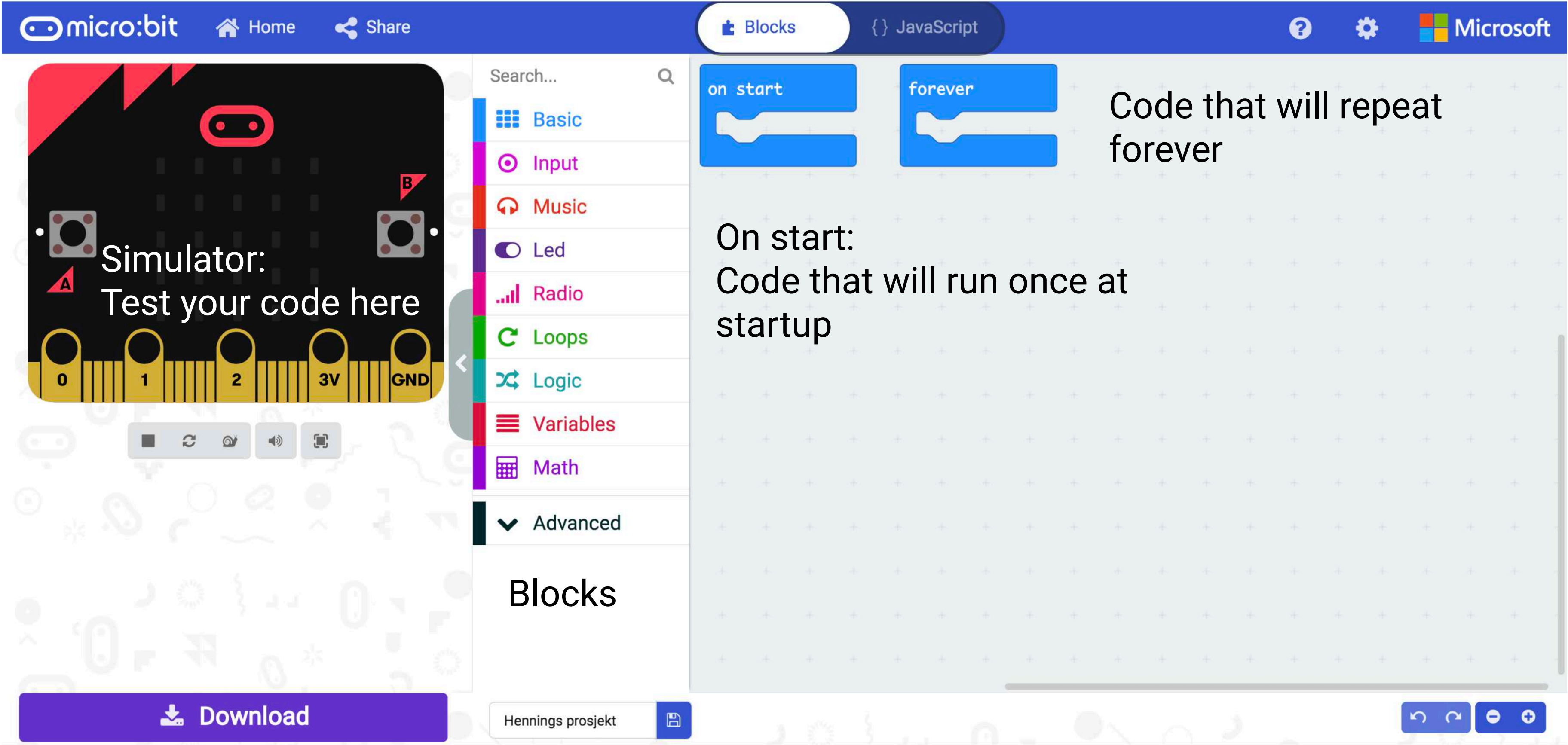
# The editor

Main menu

Share your code

Block mode   Javascript mode

Settings

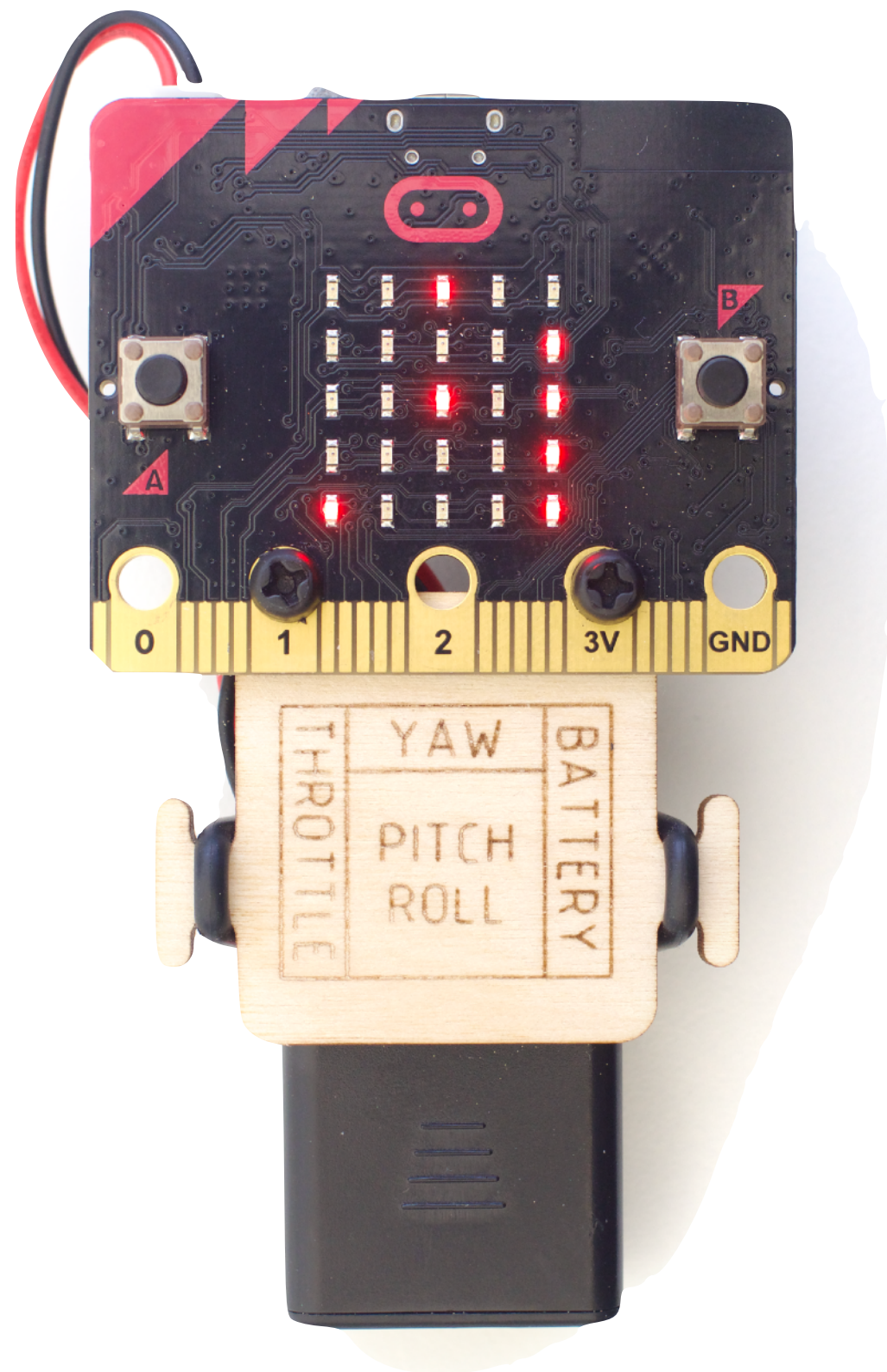


Download to micro:bit

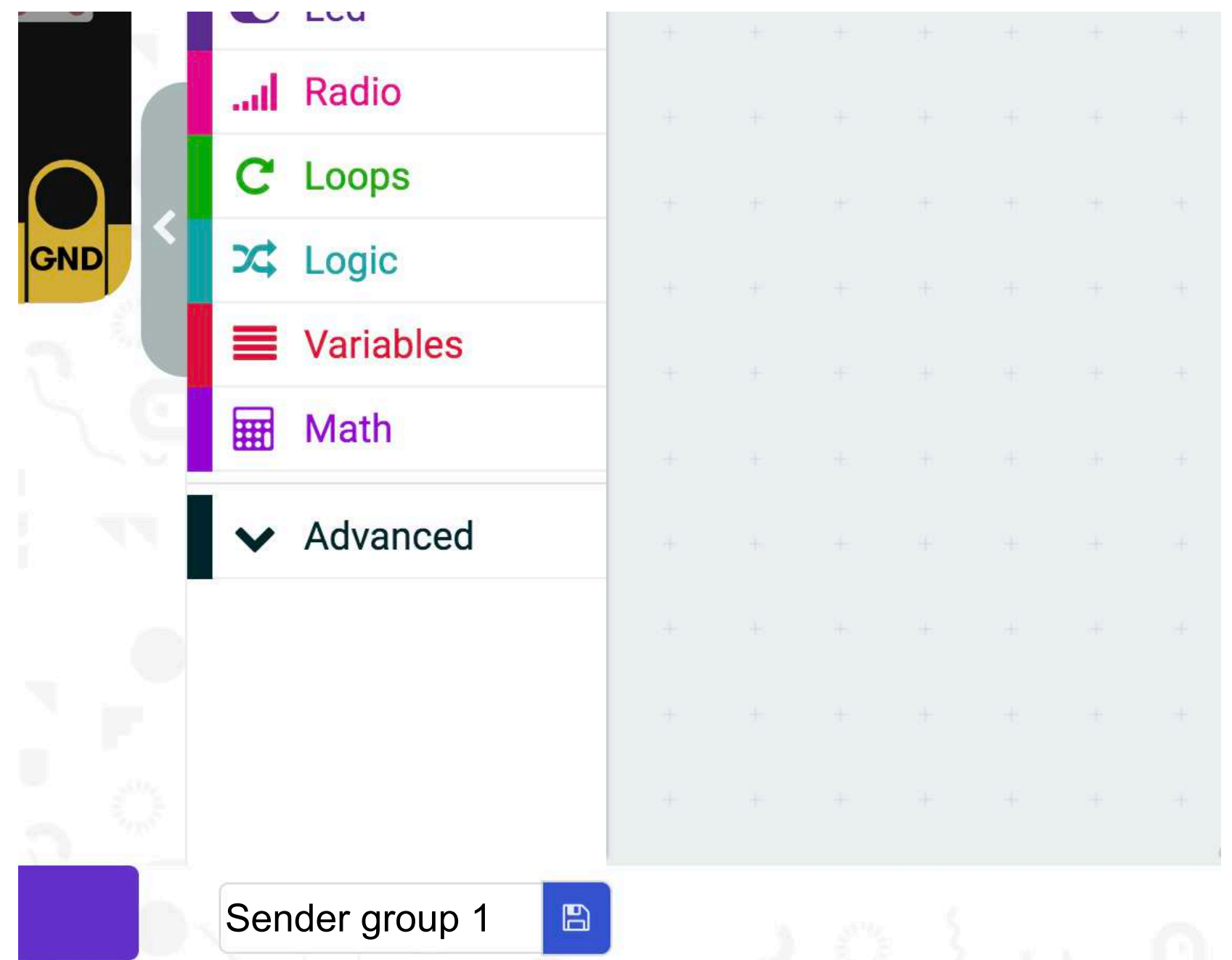
Save a backup  
Name your project

Undo   zoom

# Get the variables and code the remote



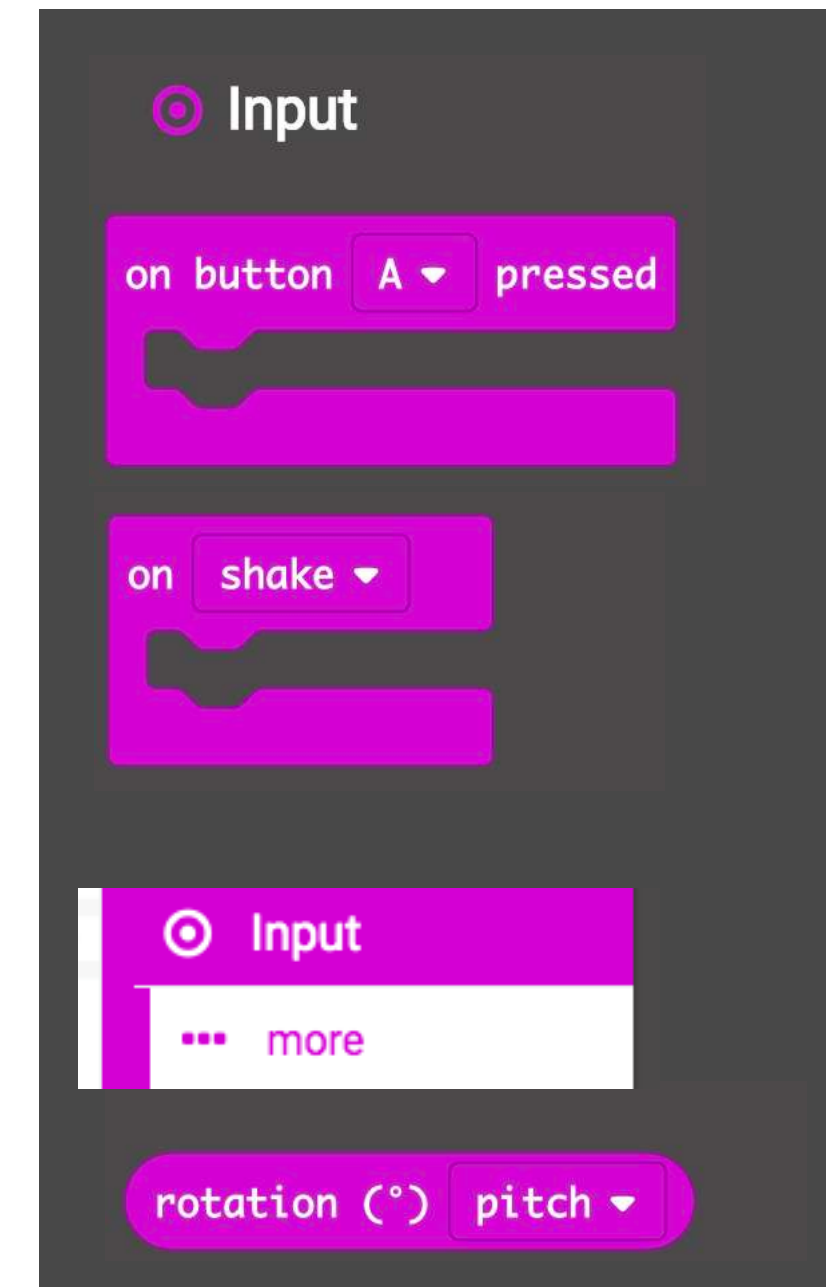
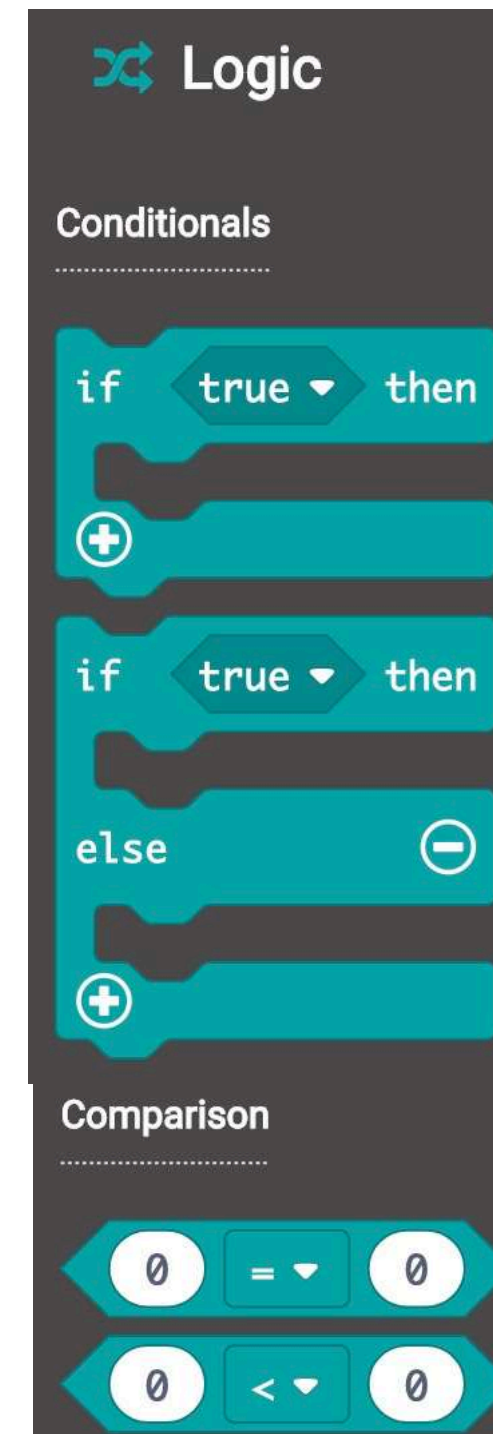
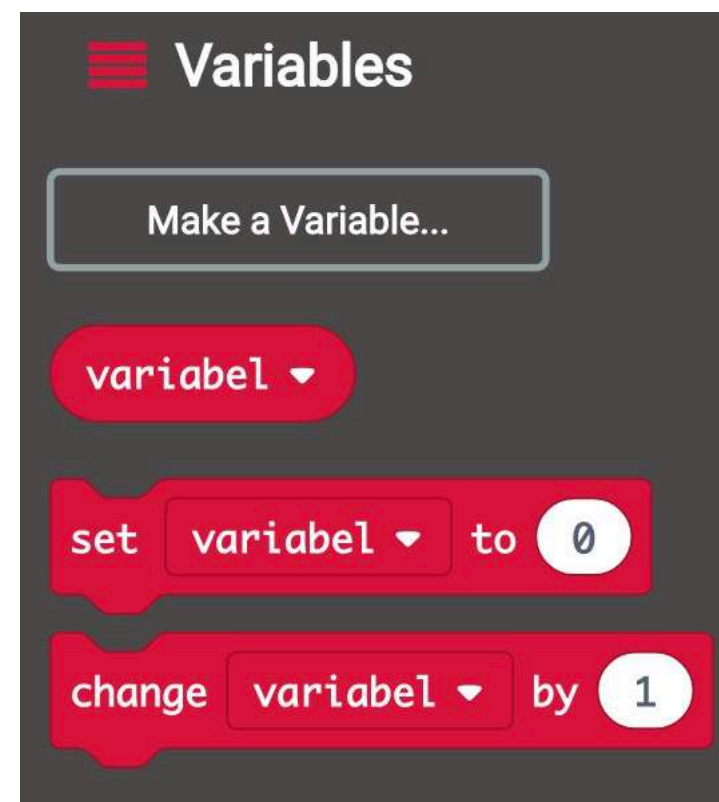
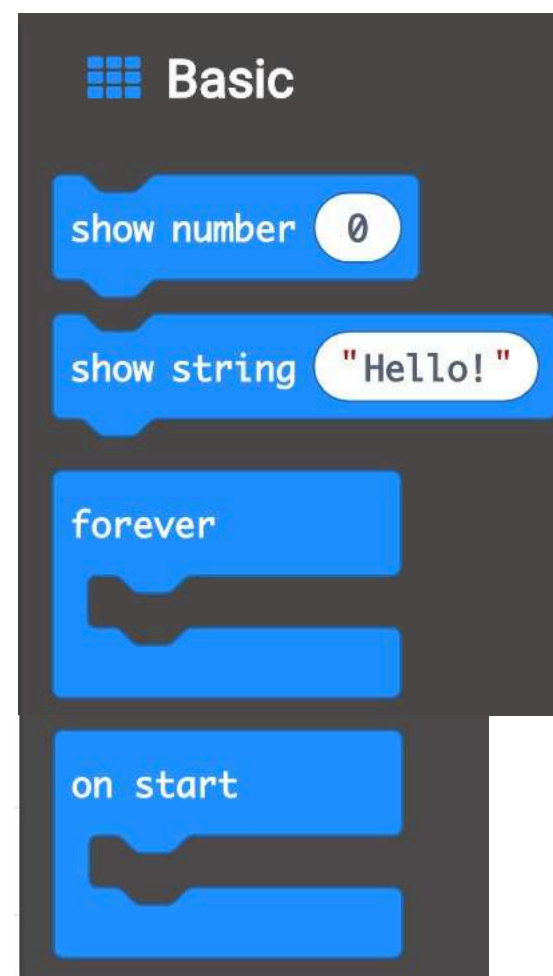
Start by giving your project a name like “Sender group x”. This is your unique radio channel. If you are flying alone you can use channel 7



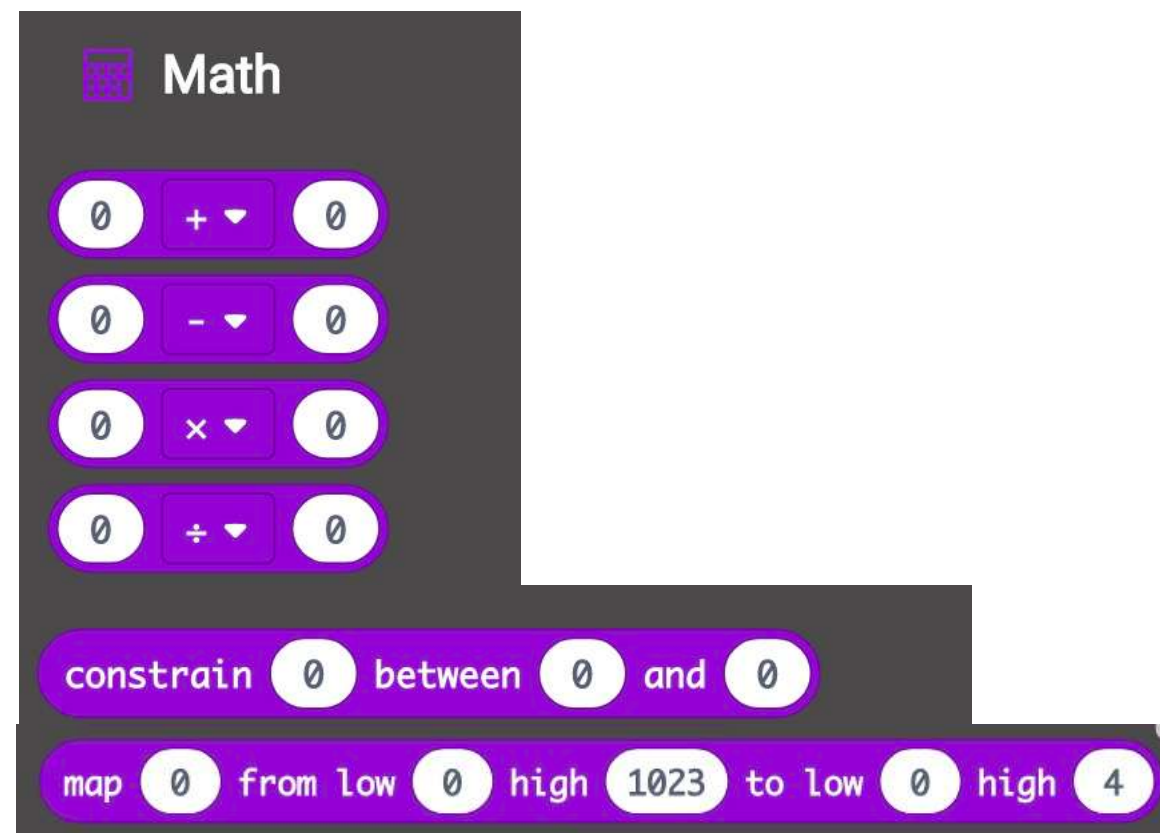


# Code blocks

We will be using these blocks



# More blocks





# PARTY

**pitch**, arm, **roll**, **throttle**, yaw

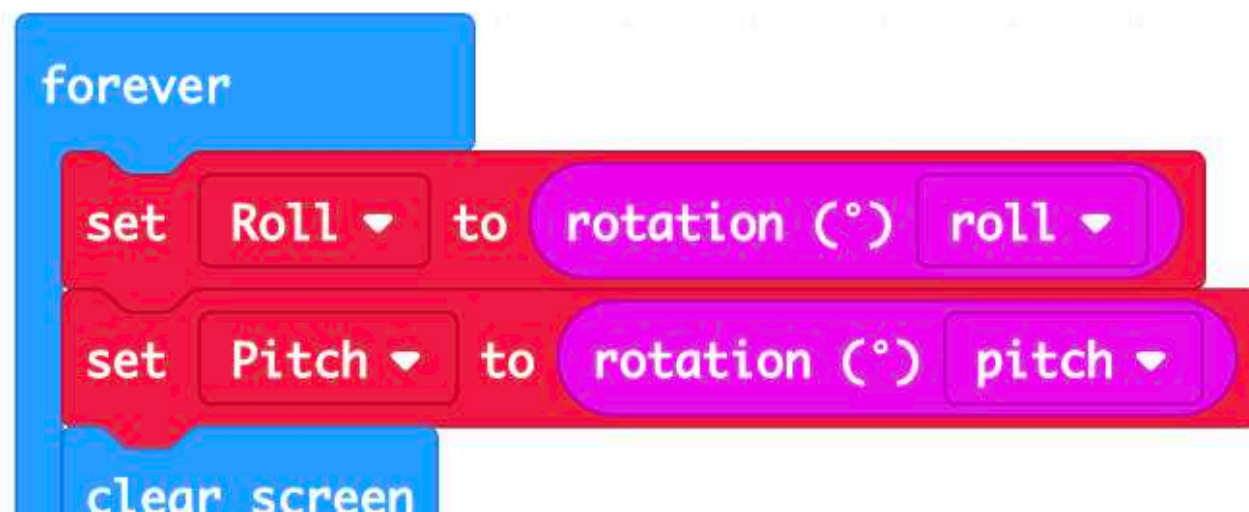
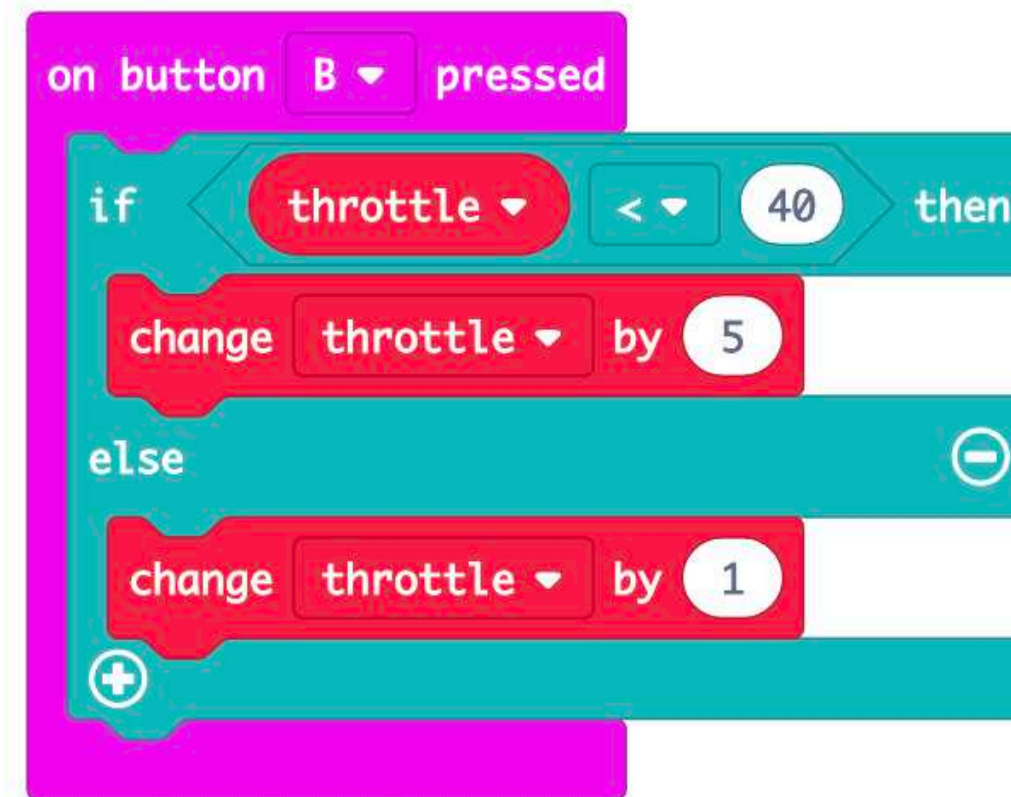
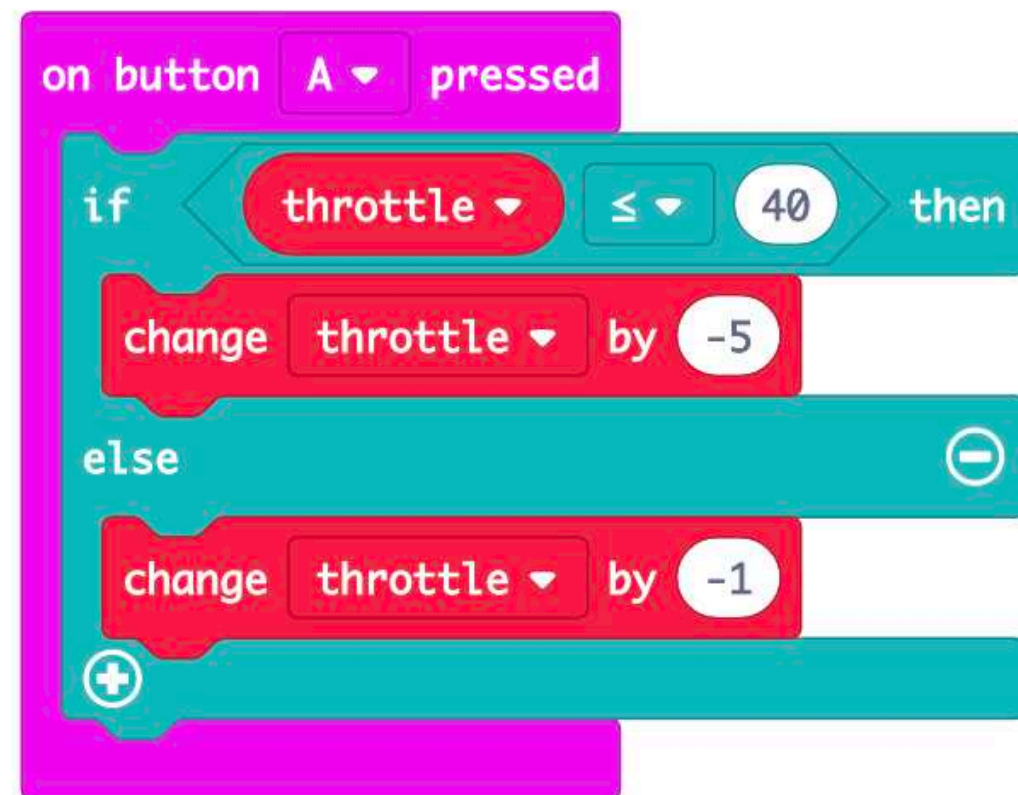
## Throttle

- Make a variable called Throttle
- Use the buttons A and B to change the throttle. A should reduce it with 5, B should increase it with 5
- Use Show Number (in the forever loop) to verify that throttle is changing correctly
- When above 40, throttle should only change by 1 for better precision when drone is hovering.

## Roll og Pitch

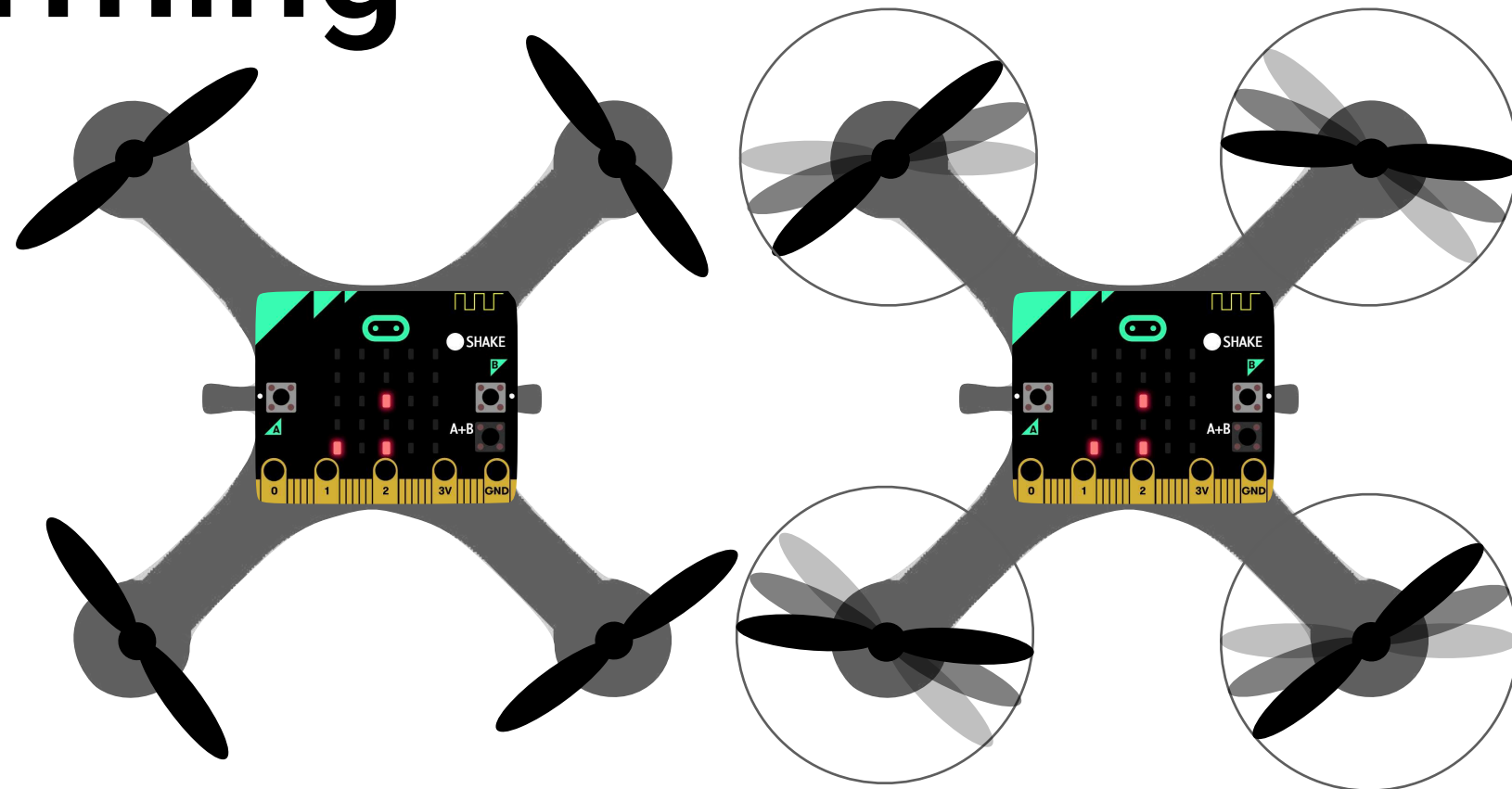
- Make a variable called Roll and one called Pitch
- In the forever loop: Set the variables roll and pitch to the input -> more -> rotation.roll and rotation.pitch

# Solution





# Arming



Disarmed

Armert

## Arming

- Arming is a safety switch and the propellers will only spin when arm is set to 1

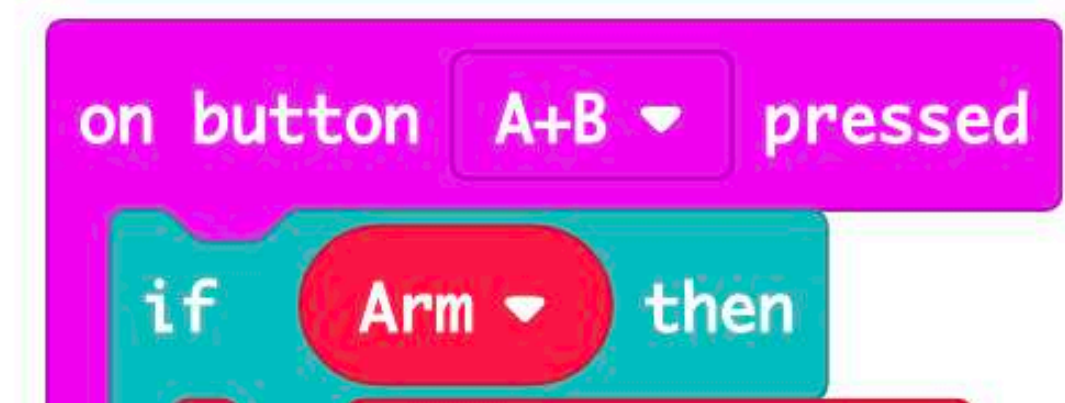
## Task:

- Make a variable called Arm
- When pressing A+B, change the Arm between 0 and 1
- If Arm is 1 make it 0 and visa versa
- Use Show Number: Arm to verify
- When pressing A+B also reset throttle to 0

Bryteren setter Arm til 0 og 1 annenhver gang

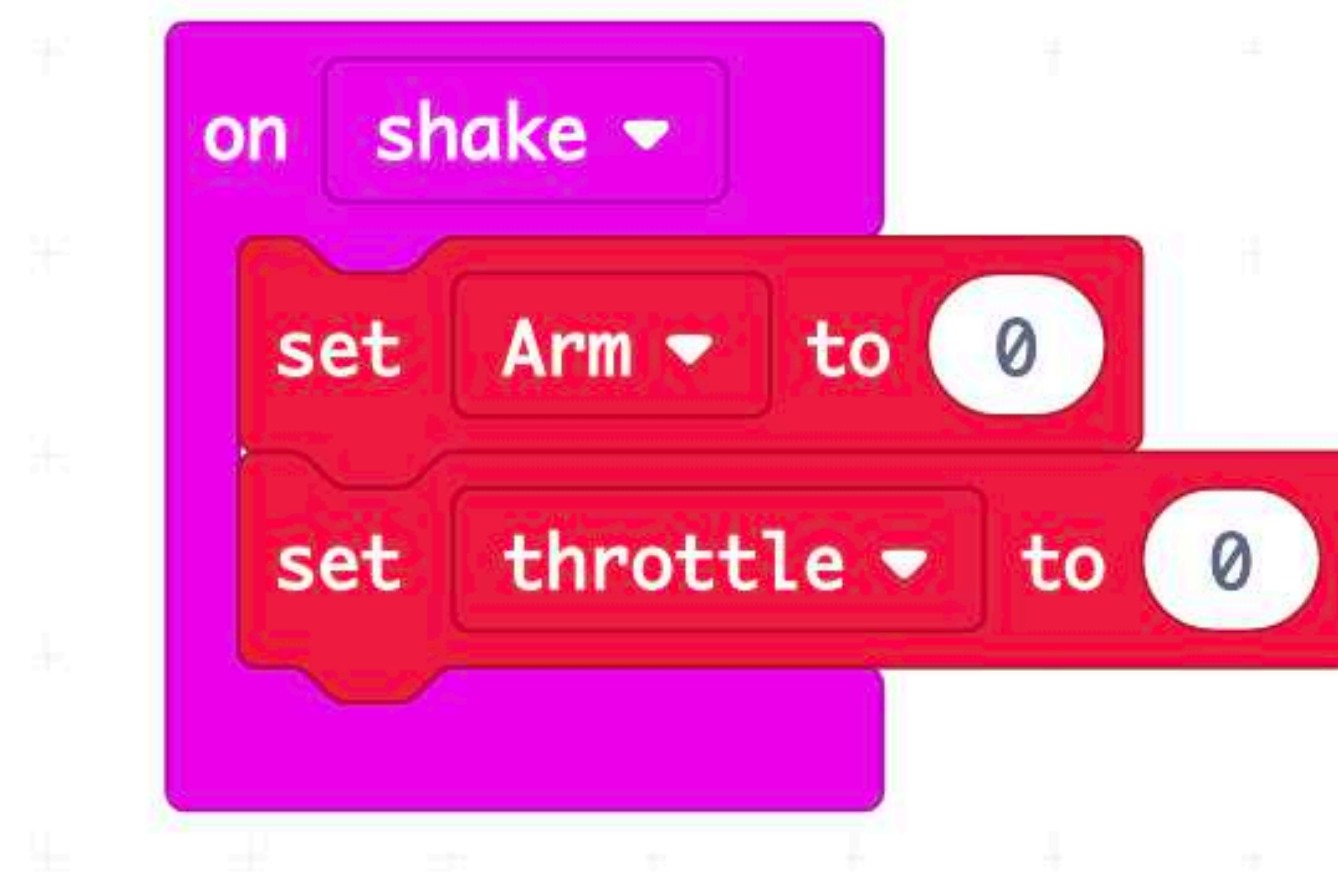
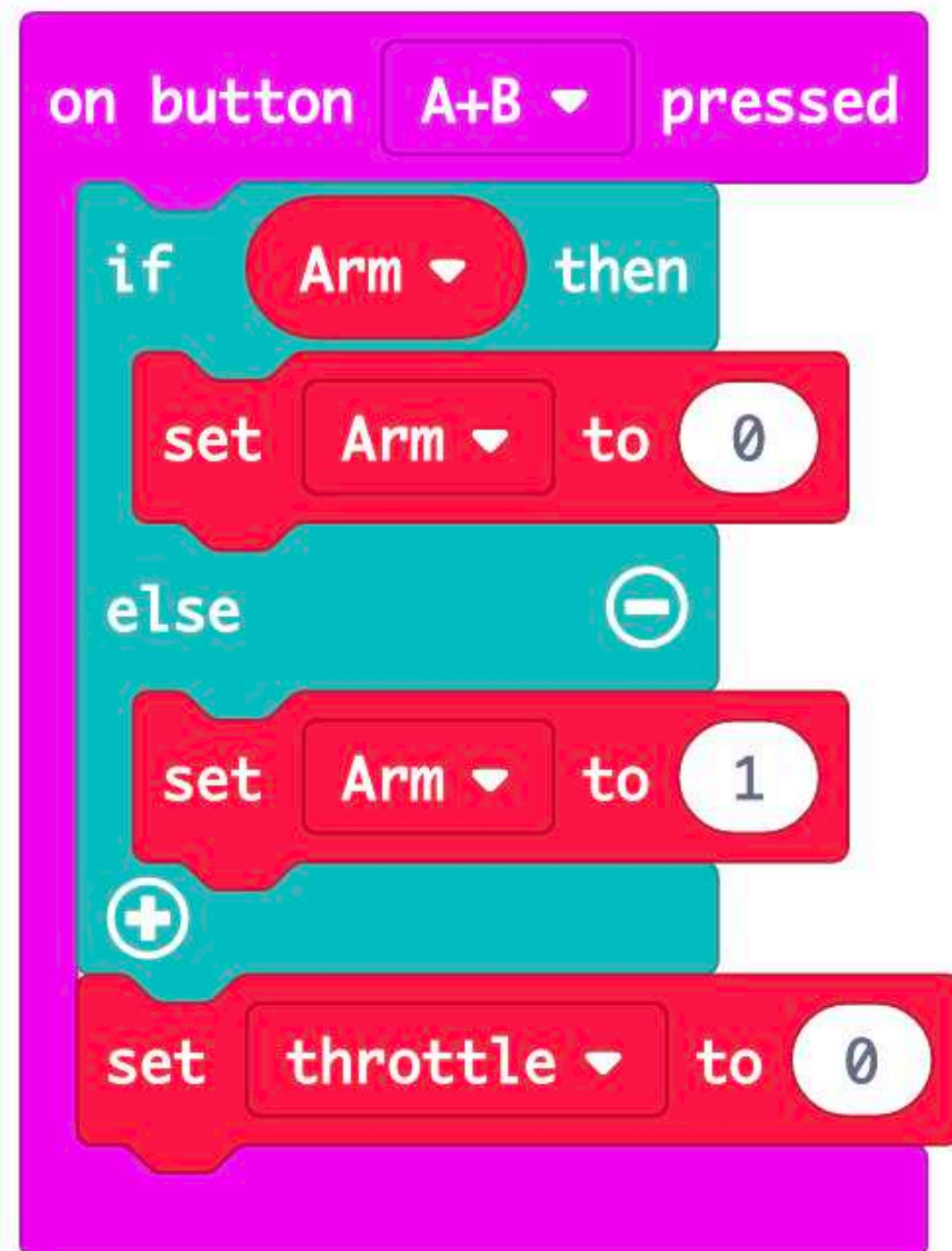


# PARTY



**Tips!** Bruk **If Arm** (betyr "hvis arm ikke er 0")  
I praksis betyr dette "hvis dronen er armert"

# Solution



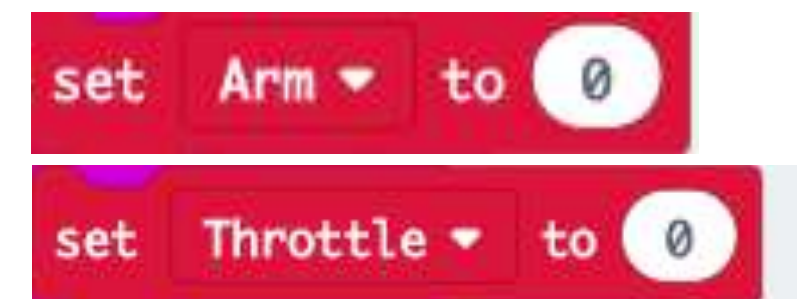
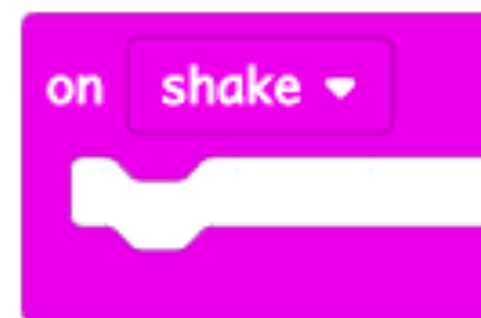
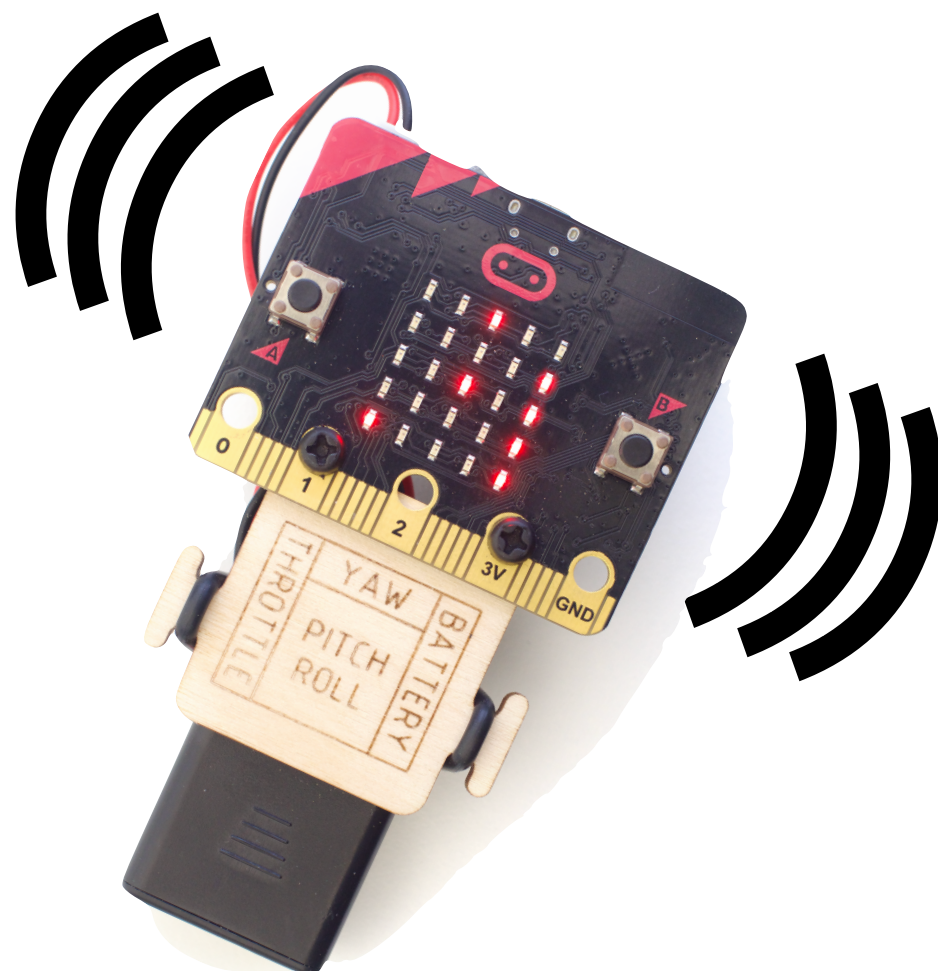


# Panic feature

If we loose control of the drone or crash, a panic button will be handy. It will cut off the motors immediately.

Task:

- Use the On Shake feature to disarm the drone. It will detect a shake from the remote control.
- Think twice before using the feature from above 2 meters altitude or the drone can crash and be damaged.



# Solution

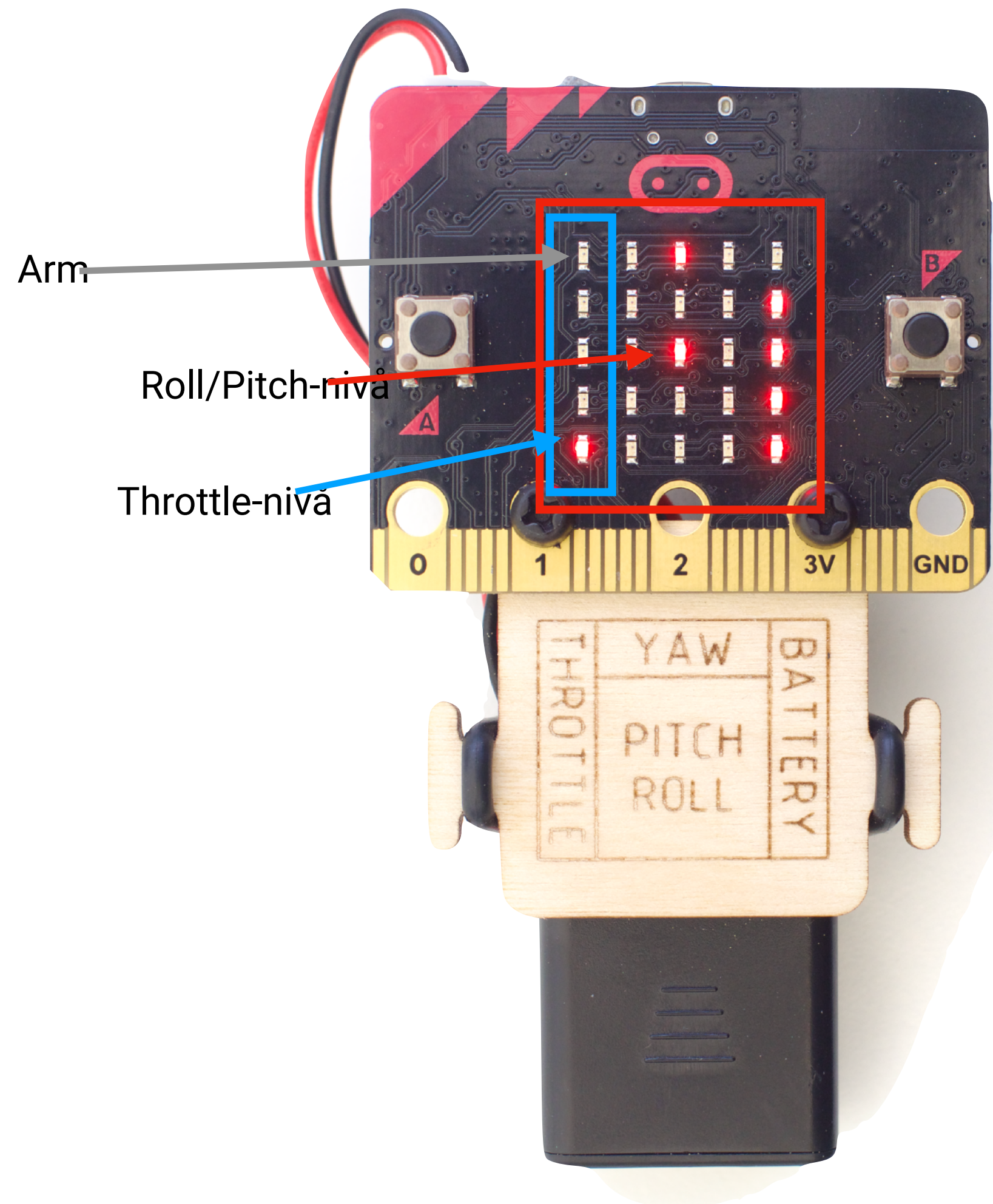




# Visualise the variables

We can use the display to show basic information from our PARTY values.

- Arm-pixel lights when armed
- Throttle-pixel moves upwards when throttle increases (thrust)
- Roll and pitch is the X and Y position of the center dot



# Visualise Arming

## Arming

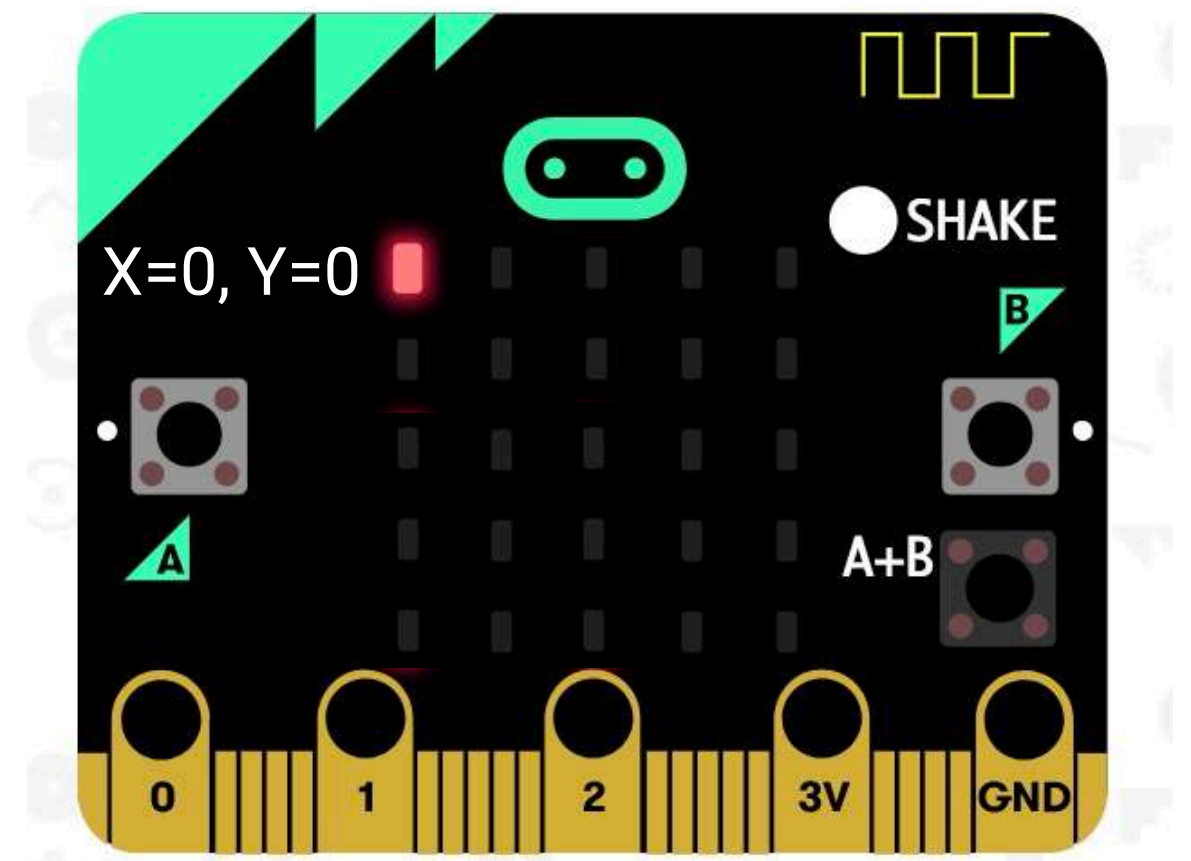
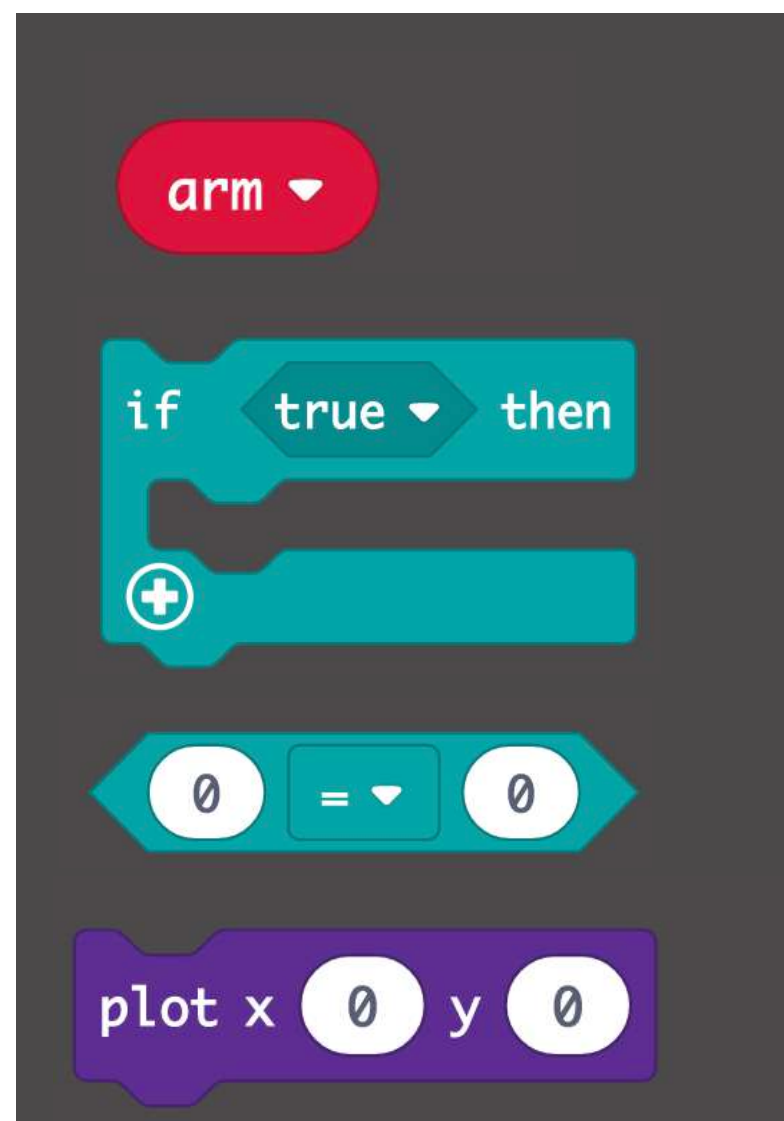
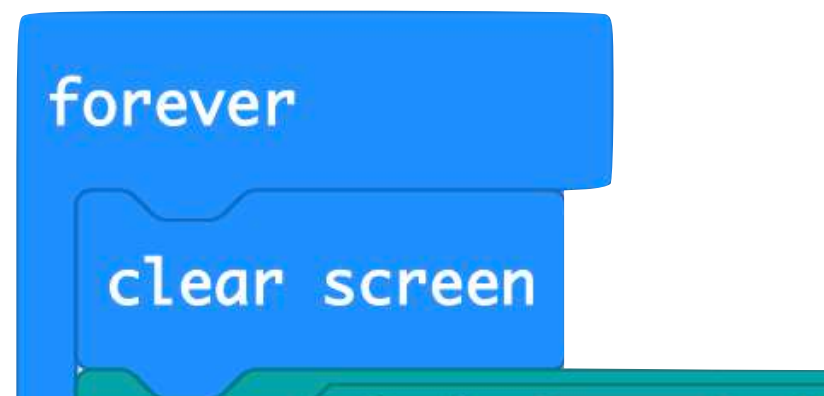
Use the screen to show if the drone is armed.

Find the forever-block.

Insert a clear screen block.

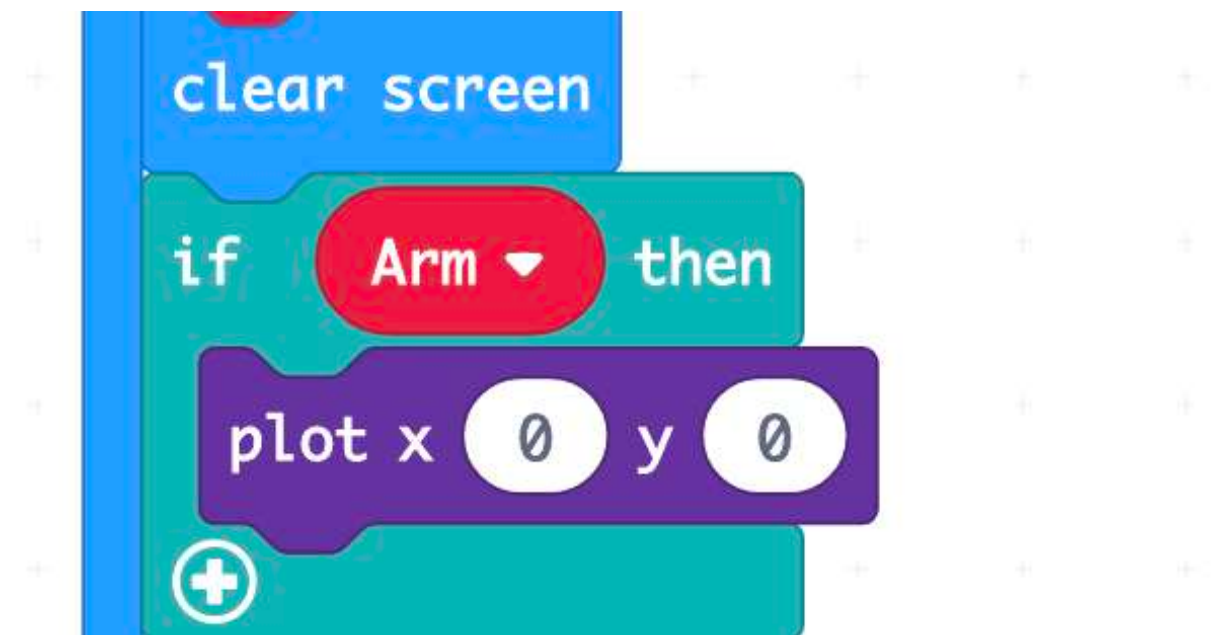
Use the if-block to check if arm is on (armed).

If armed plot at 0,0 (or another place off your choice)



- **Extra challenge:** Make the pixel blink

# Solution

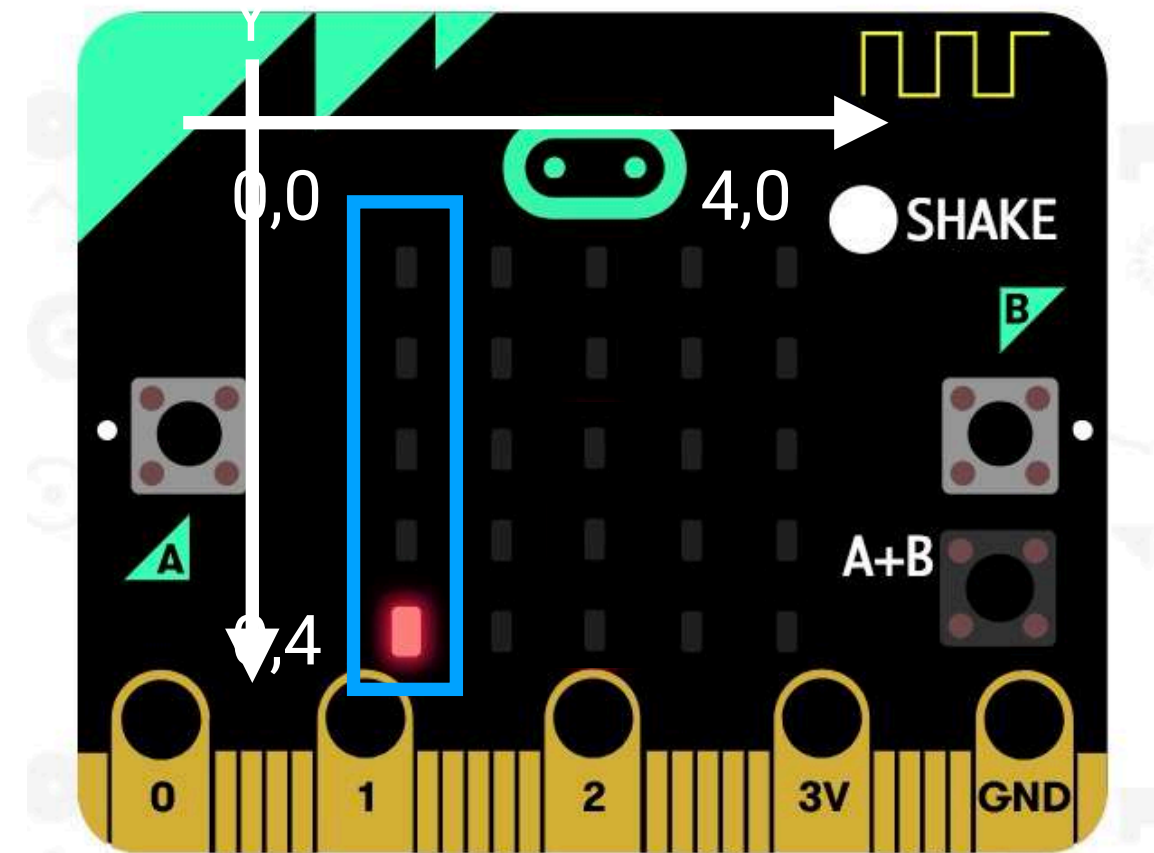




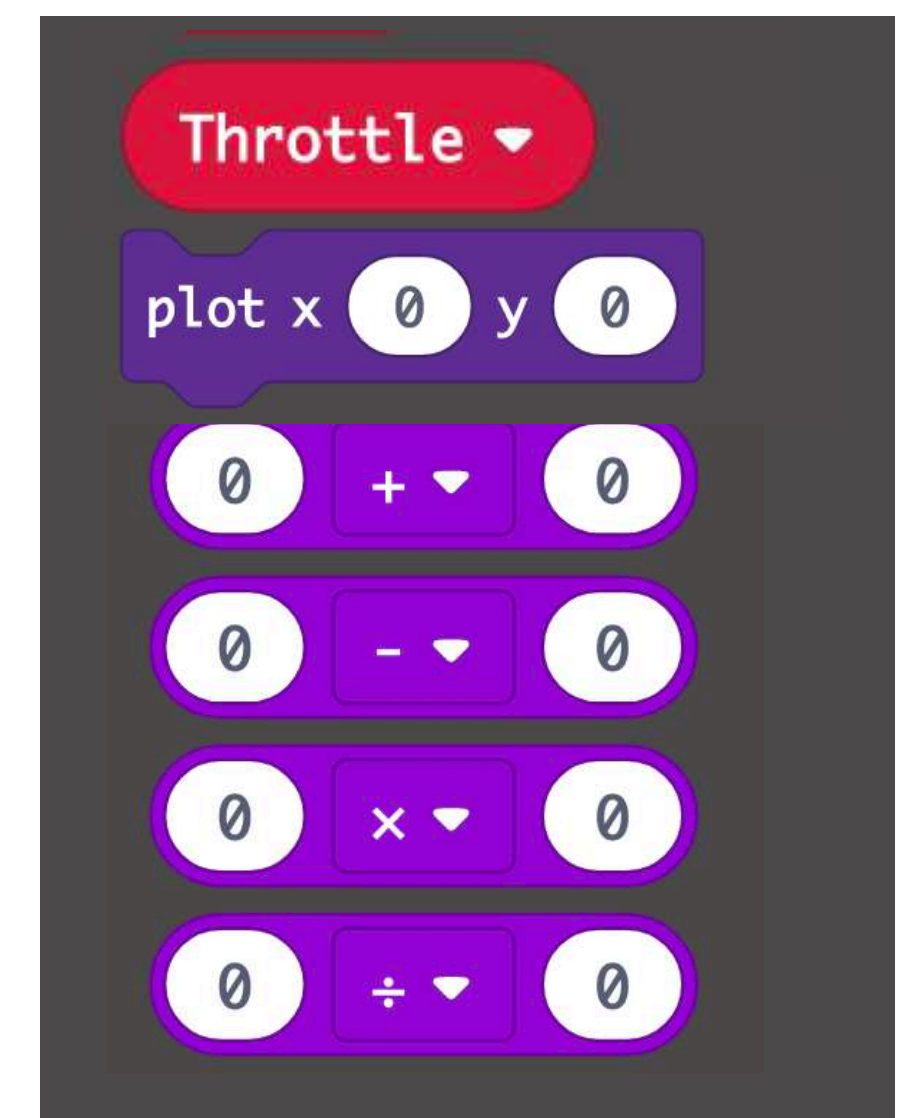
# Visualise values: Throttle

## Throttle

- We can indicate our throttle using the pixels on the left (blue frame)
- We need to translate our numbers according to the table below so that 0 gives 4, 50 gives 2 and 100 gives 0.
- Can you solve the equations?
- Hint:
  - We need to do a division in the first calculation
  - Then we need to do a subtraction in the second.



| Throttle           | 0 | 50 | 100 |
|--------------------|---|----|-----|
| First calculation  | 0 | 2  | 4   |
| Second calculation | 4 | 2  | 0   |



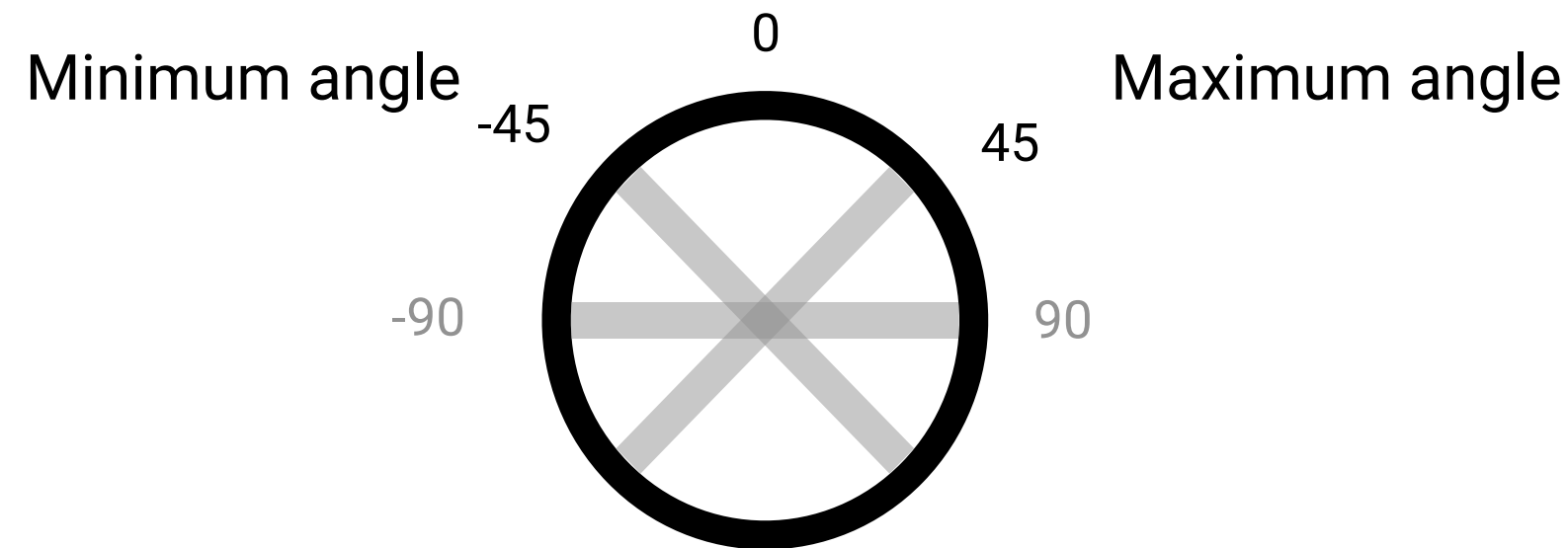
# Solution

Pixel Y = 4 - throttle / 25



Throttle / 25 goes in the inner block

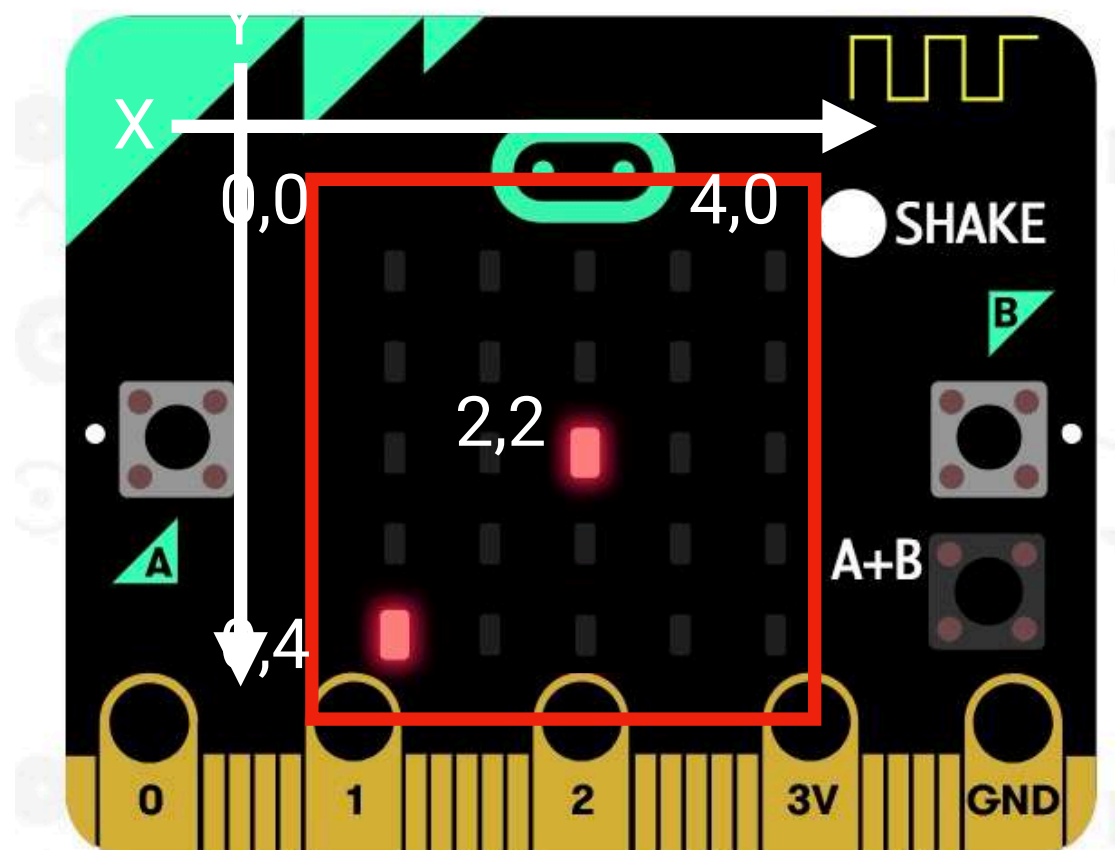
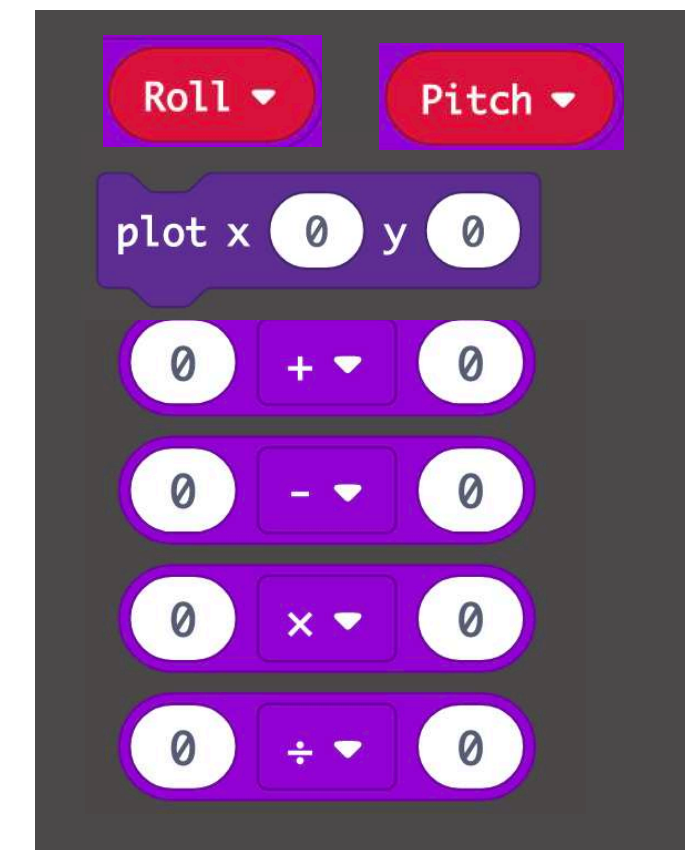
# Visualise values: Roll/Pitch



## Roll / Pitch

- We can indicate our Roll and Pitch angles using center pixel's X and Y position
- We need to translate our numbers according to the table below so that -45 gives 0, 0 gives 2, and 45 gives 4
- Can you solve the equations?
- Hint:
  - We need to do an addition in the first calculation
  - Then we need to do a division in the second.

| Angle              | -45 | 0  | 45 |
|--------------------|-----|----|----|
| First calculation  | 0   | 45 | 90 |
| Second calculation | 0   | 2  | 4  |





# Solution



Roll + 45 in the innermost block      Pitch + 45 i the innermost block



**Tip!** You can also use javascript. It can be easier to see the whole formula. In code, division and multiplication goes before subtraction and addition automatically. You will need parentheses if you want the + or - to be done first.

Be careful: There is not much room for typing errors in javascript.

```
led.plot(0, 4 - throttle / 25) (Division before subtraction)
led.plot((Roll + 45) / 22.5, (Pitch + 45) / 22.5) (Addition before division)
```

# An easier alternative to functions

We can also map the values using this block



Original value

Lowest  
expected  
value

Highest  
expected  
value

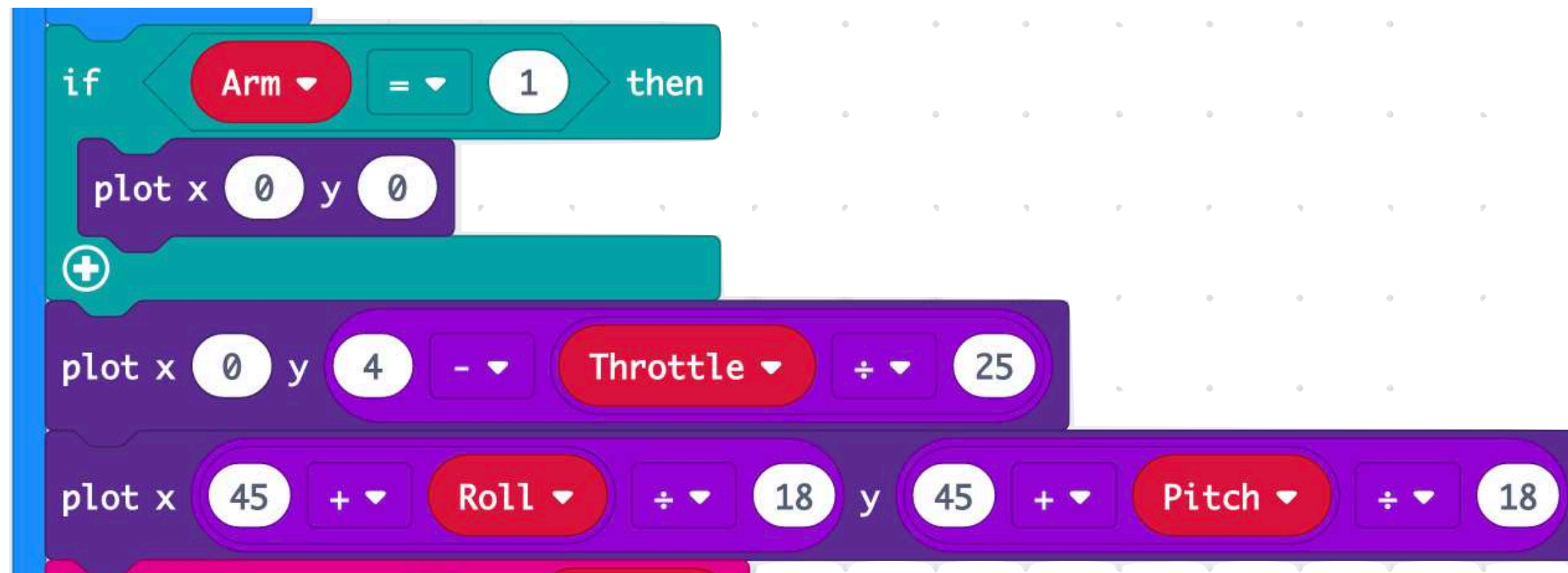
Lowest  
output  
value

Highest  
output  
value

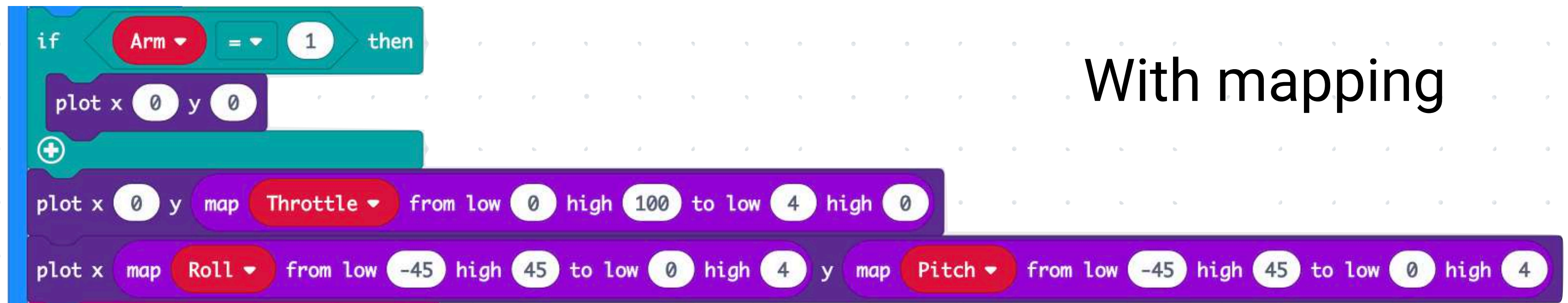
Example:



# Two solutions



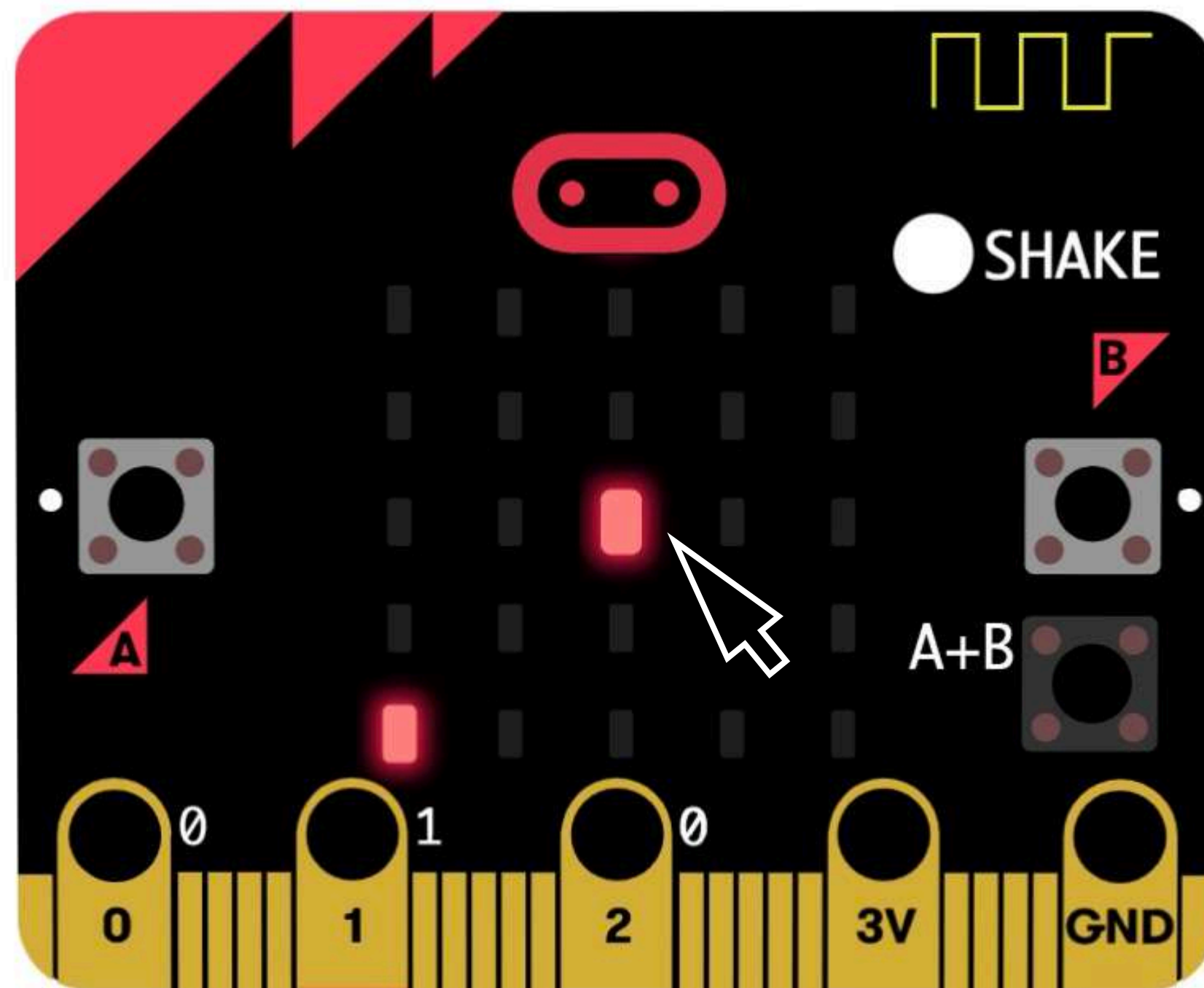
With functions



With mapping



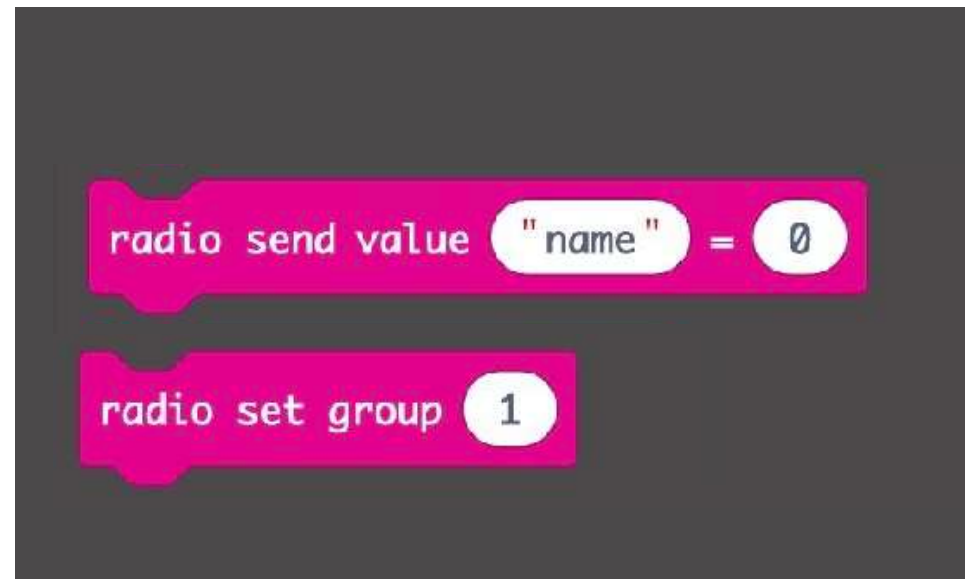
# Does it work?



## Test it in the simulator:

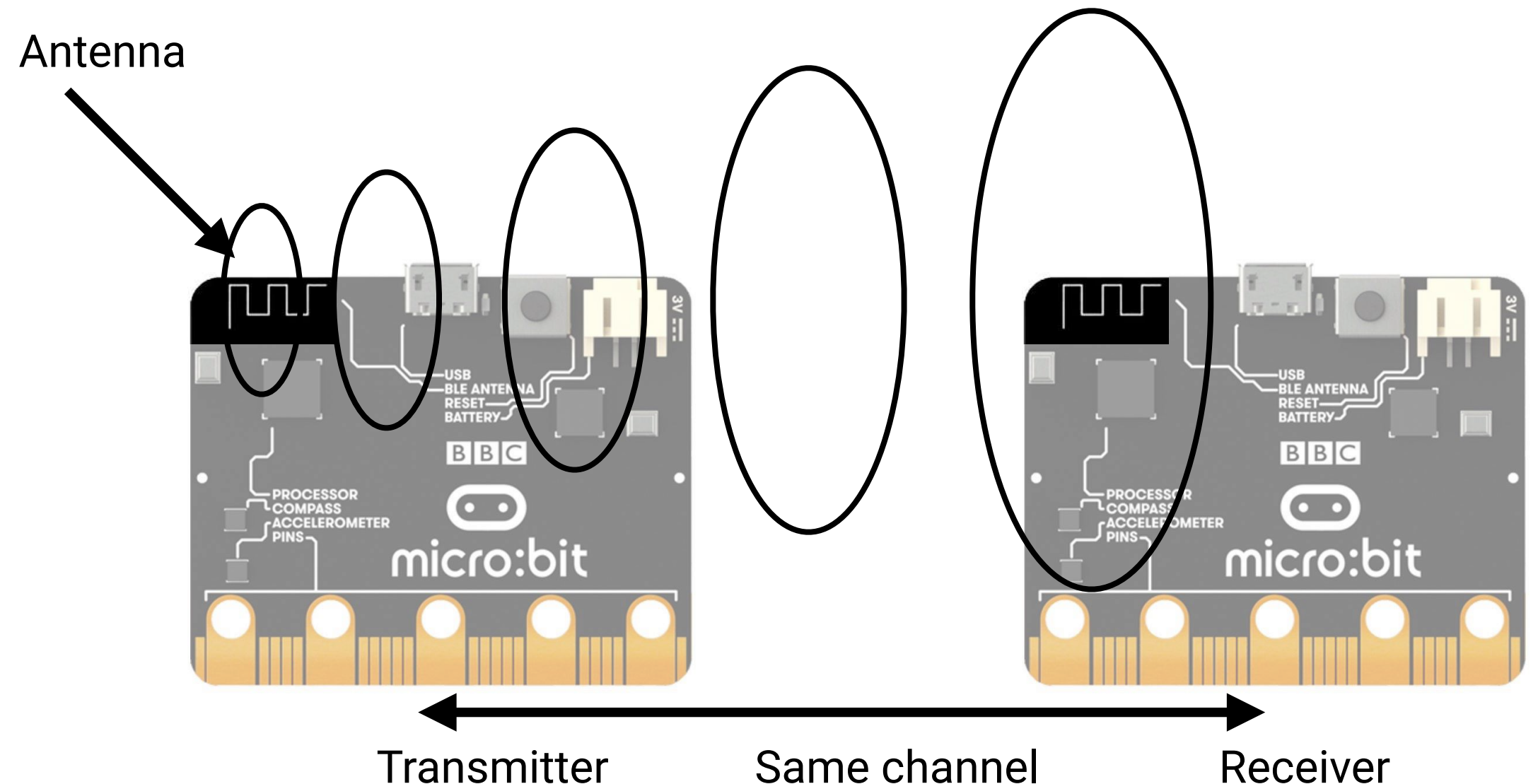
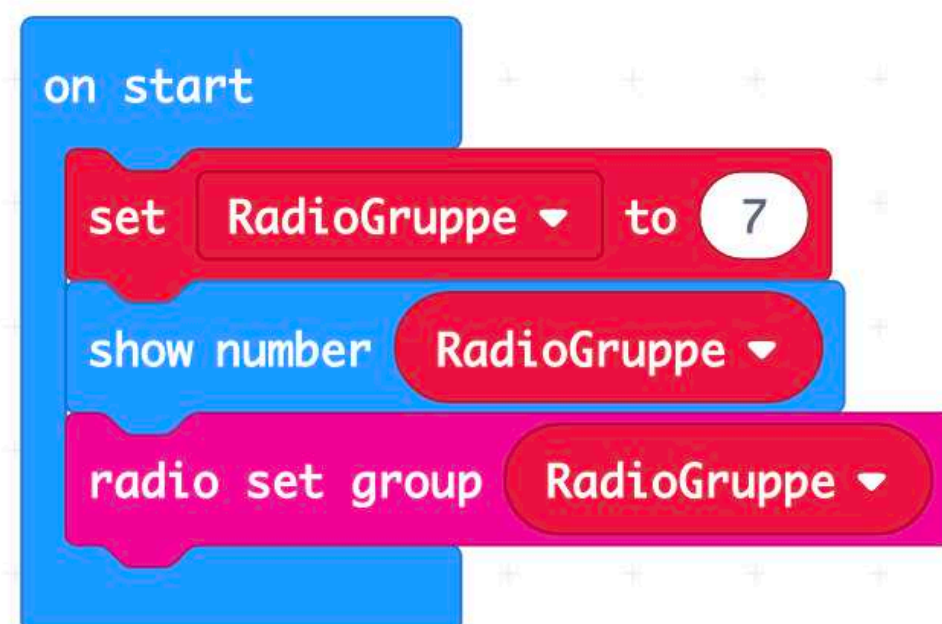
- The center dot follows the mouse arrow's movement
- The throttle-pixel on the left is rising when pressing B multiple times
- $A + B$  results in a lit ARM-pixel and throttle will go down to lowest position.

# Radio

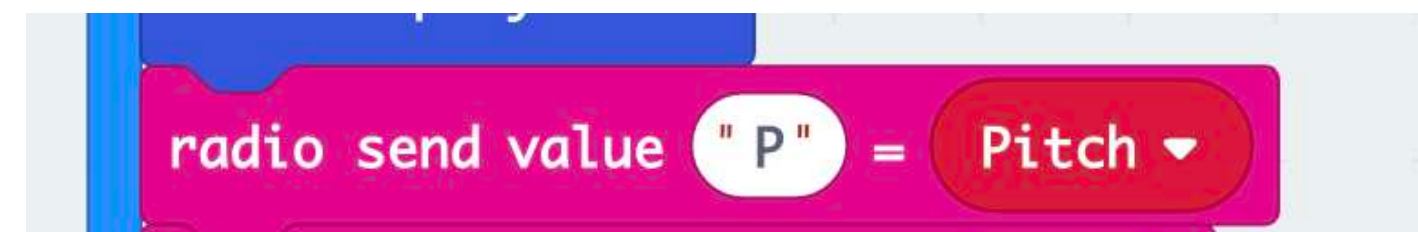


## Task:

- In the On Start block:
  - Make a value called radioGroup or similar and set it to your radio channel
  - Show the number
  - Use radio set group to make the channel take effect
- We need to transmit a letter "string" in pair with the number for the receiver to know the number.
- Use a capital letter for each of the 5 PARTY-values.
- Yaw can be skipped

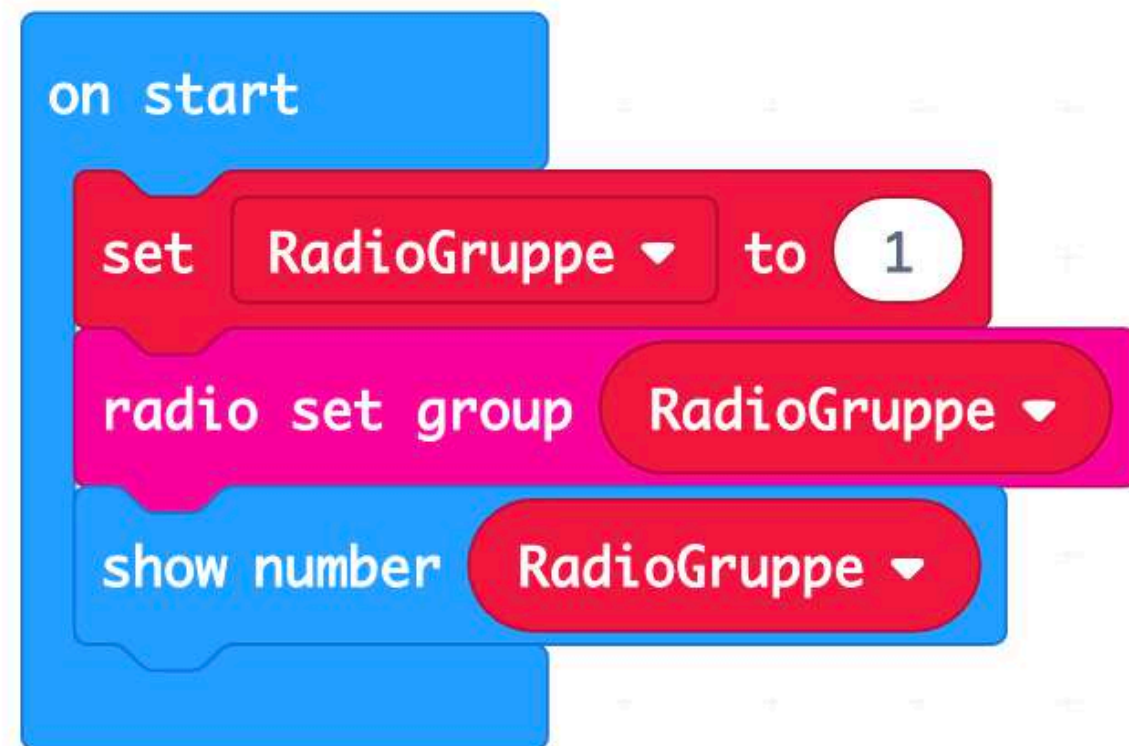


Example:



# Solution

Your unique radio group



Bottom of forever block

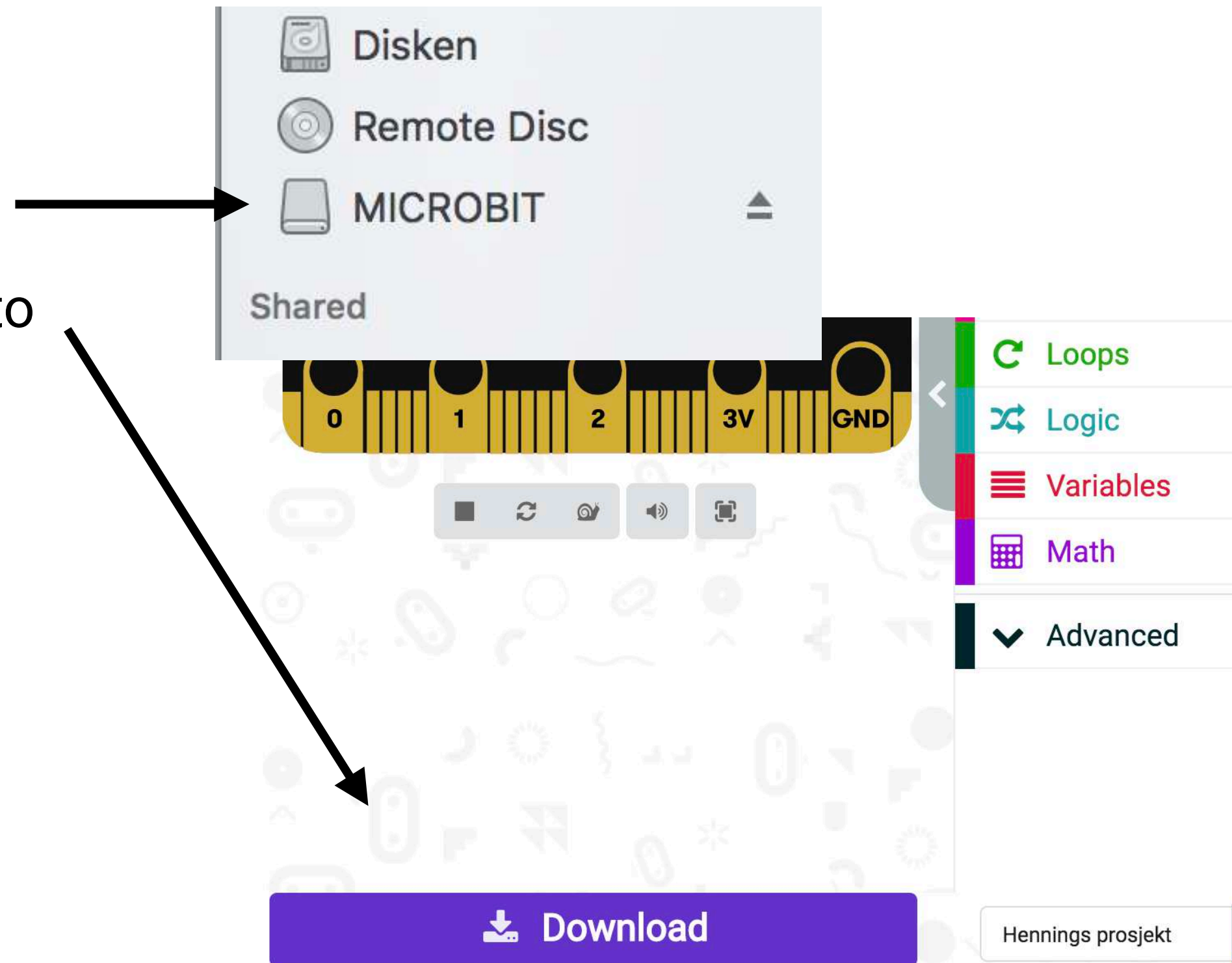


Remember: capital letters  
can be skipped



# Download the code

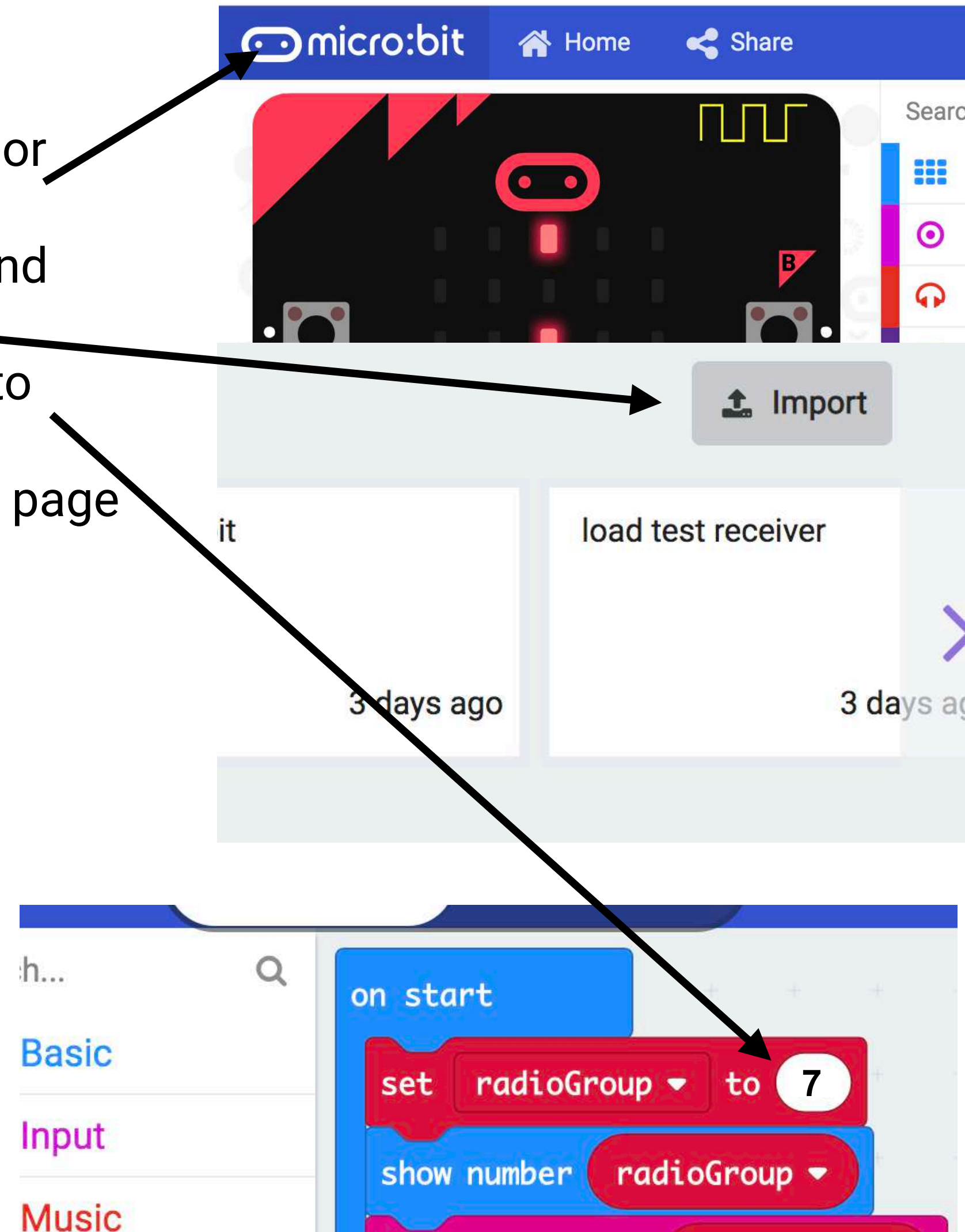
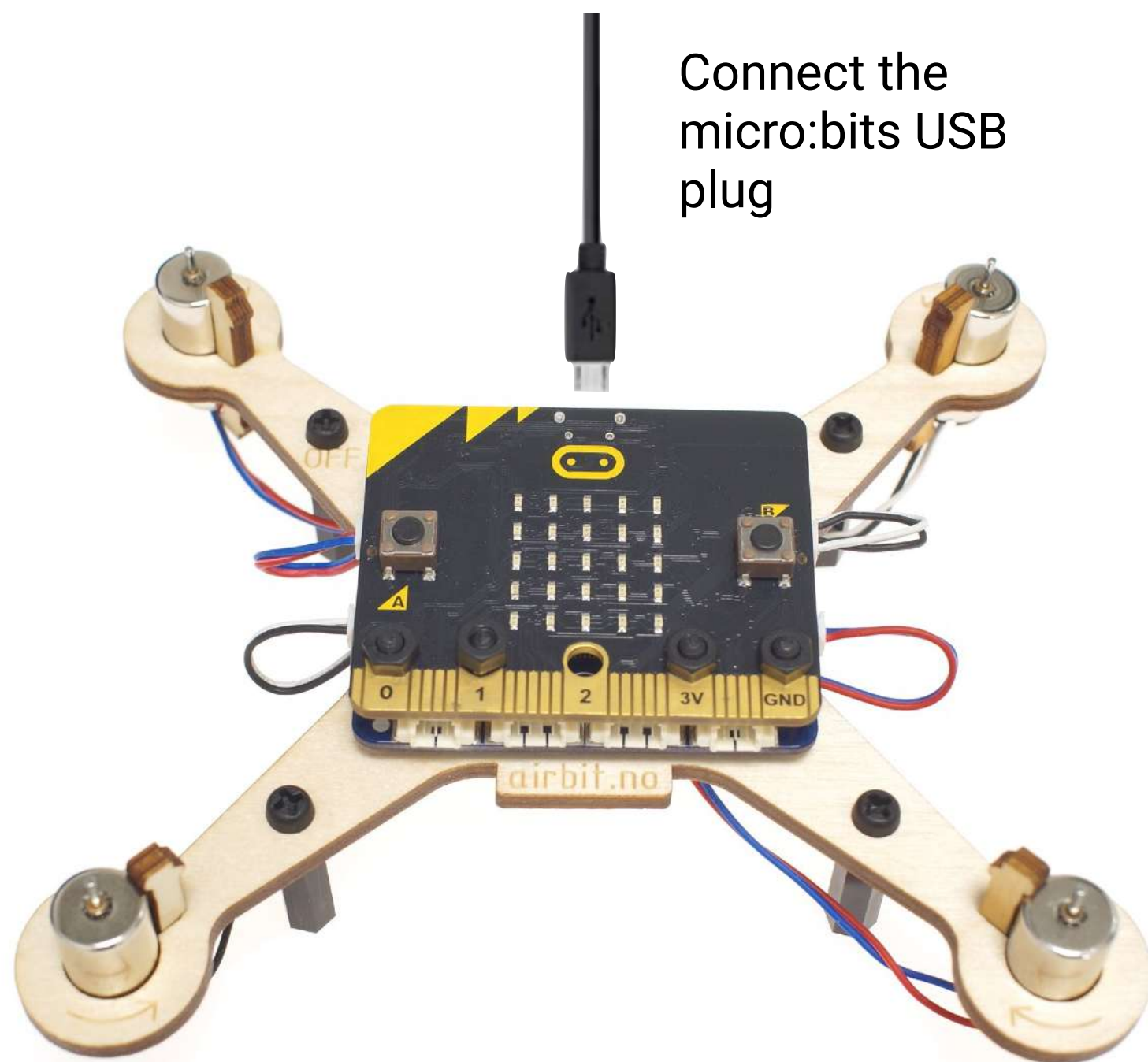
1. Connect your micro:bit to the micro usb
2. Microbit appears as a drive
3. Press "Download" and copy the file to the MICROBIT drive.
4. Watch the orange light on the back and make sure it flashes when code is downloaded



Read more about connecting the micro:bit See the micro:bit introduction at [makekit.no/](https://makekit.no/)

# The drone code

1. Download the drone code called "Airbit 2" or "Wonderbit Airbit"
2. Open the file in the editor. Click "import" and select the file
3. Make sure you have the right radio group to match your transmitter
4. Download the code as seen on the former page





# Startup and Calibration

The drone needs to calibrate before flight.

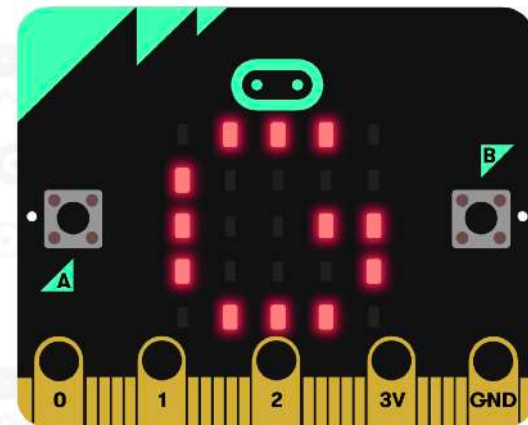
Disconnect any USB power or battery.

Place the drone on a still, level surface and connect the battery. The drone will check for sensors and calibrates.

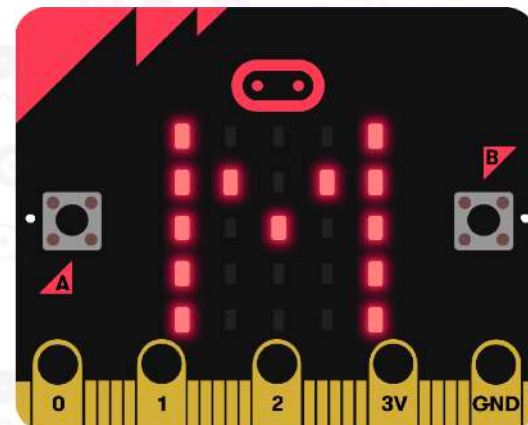
If you get “no gyro”, check the 5 top screws and reconnect power.

If you get “no pca”, disconnect USB and battery, and connect only battery.

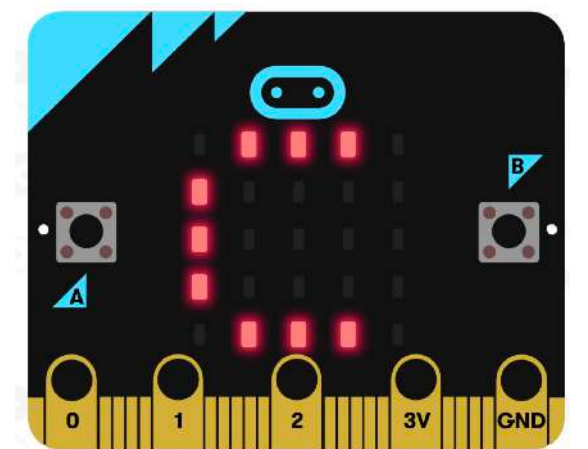
Gyroscope present



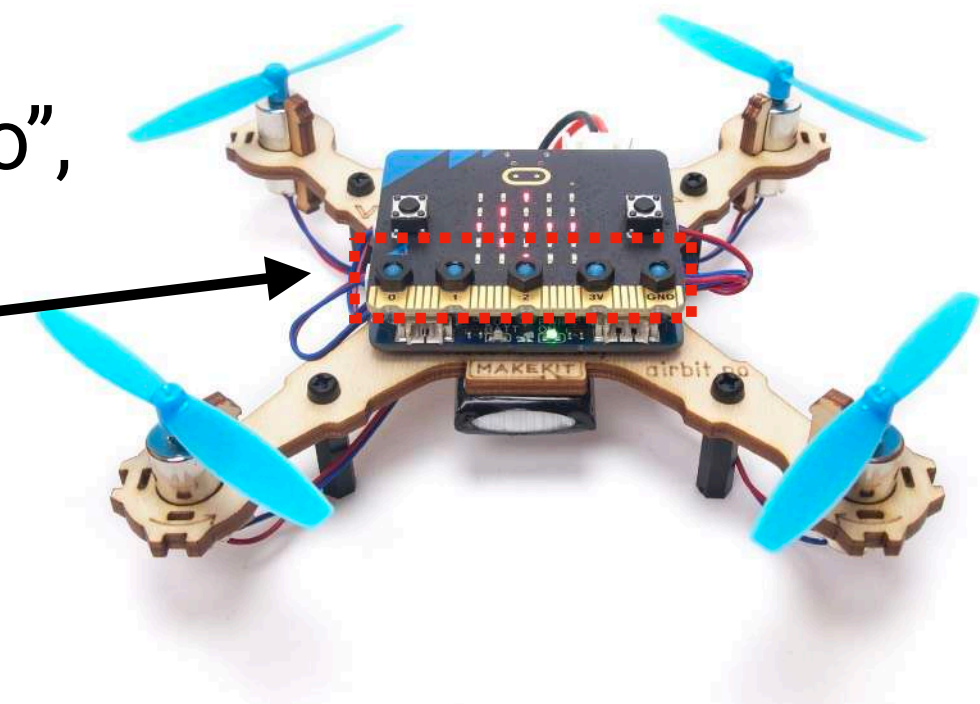
Motor driver present



Calibrating:  
When calibrating,  
the drone needs to  
be still, on a level  
surface.



If you get “no gyro”,  
check /tighten  
screws

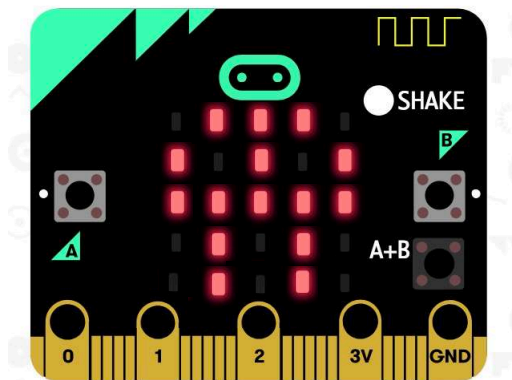




# Flying

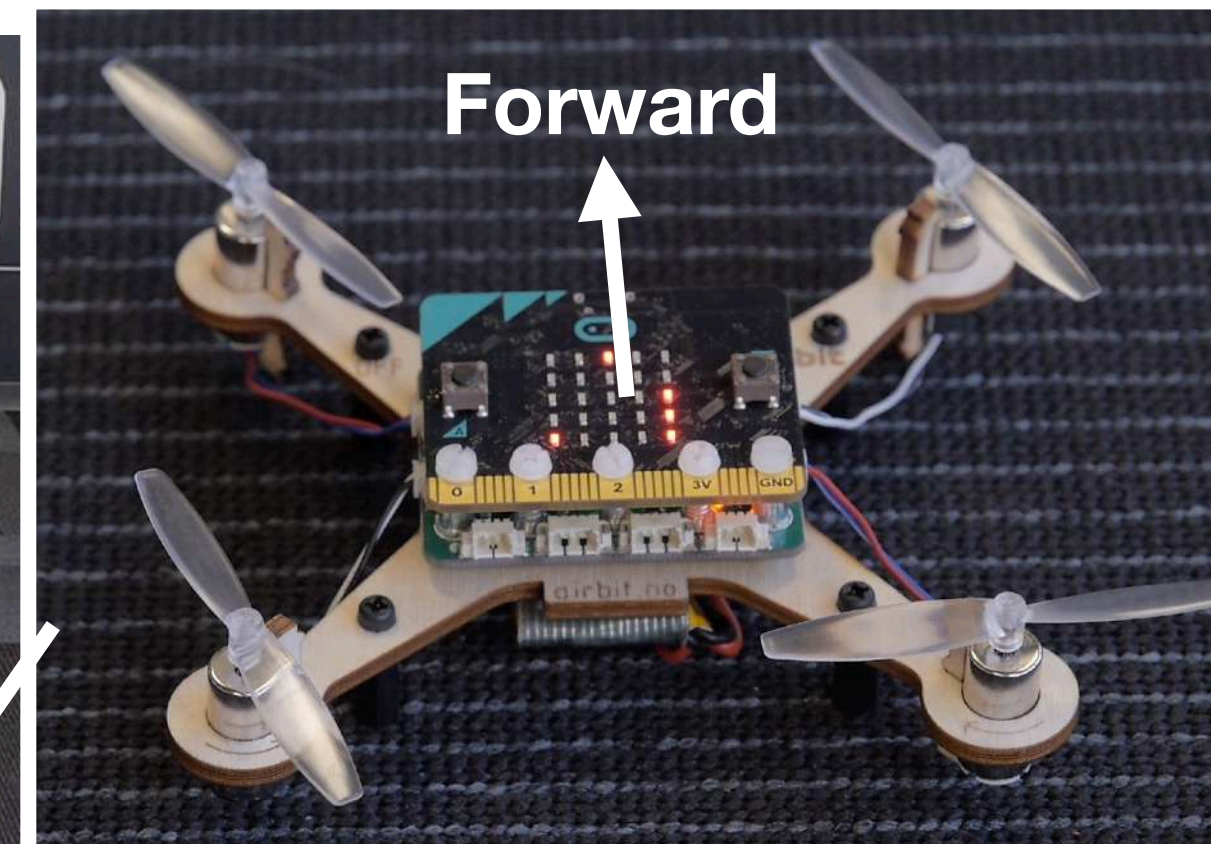
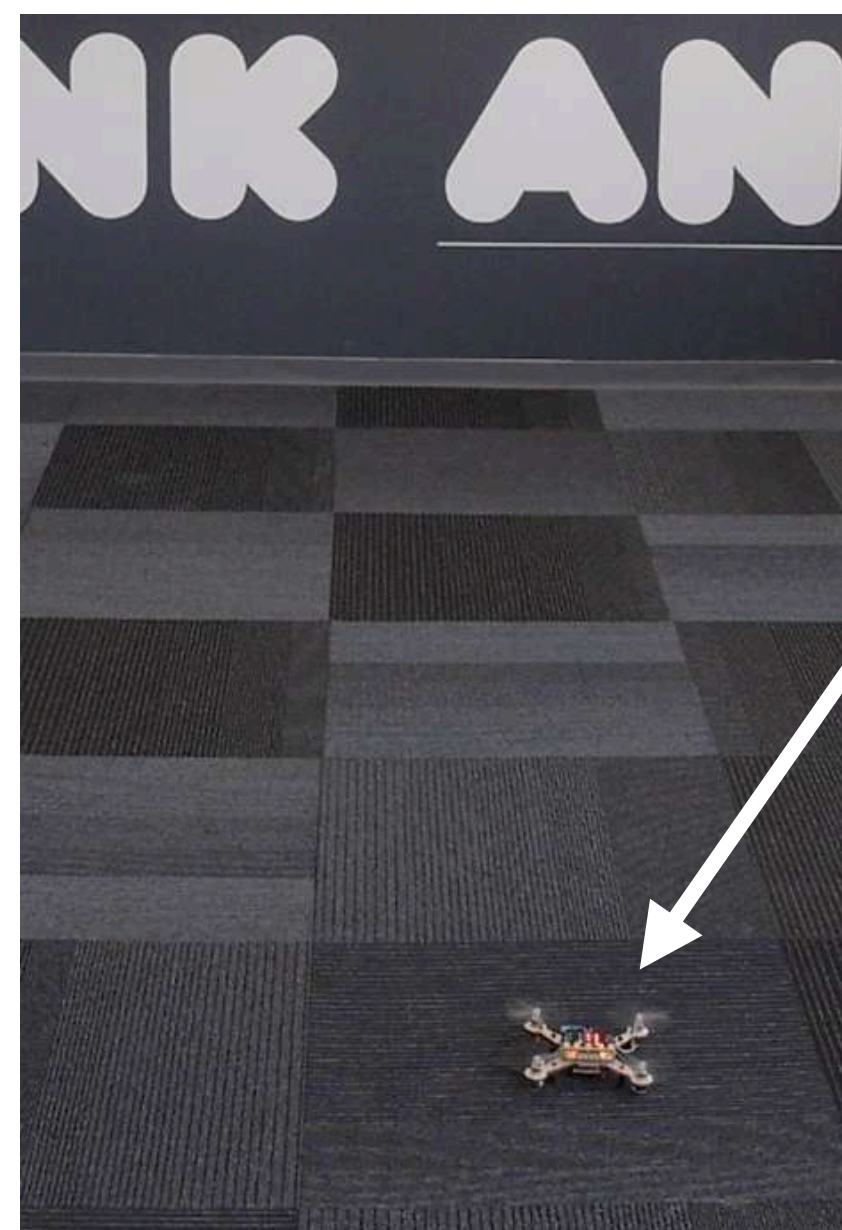
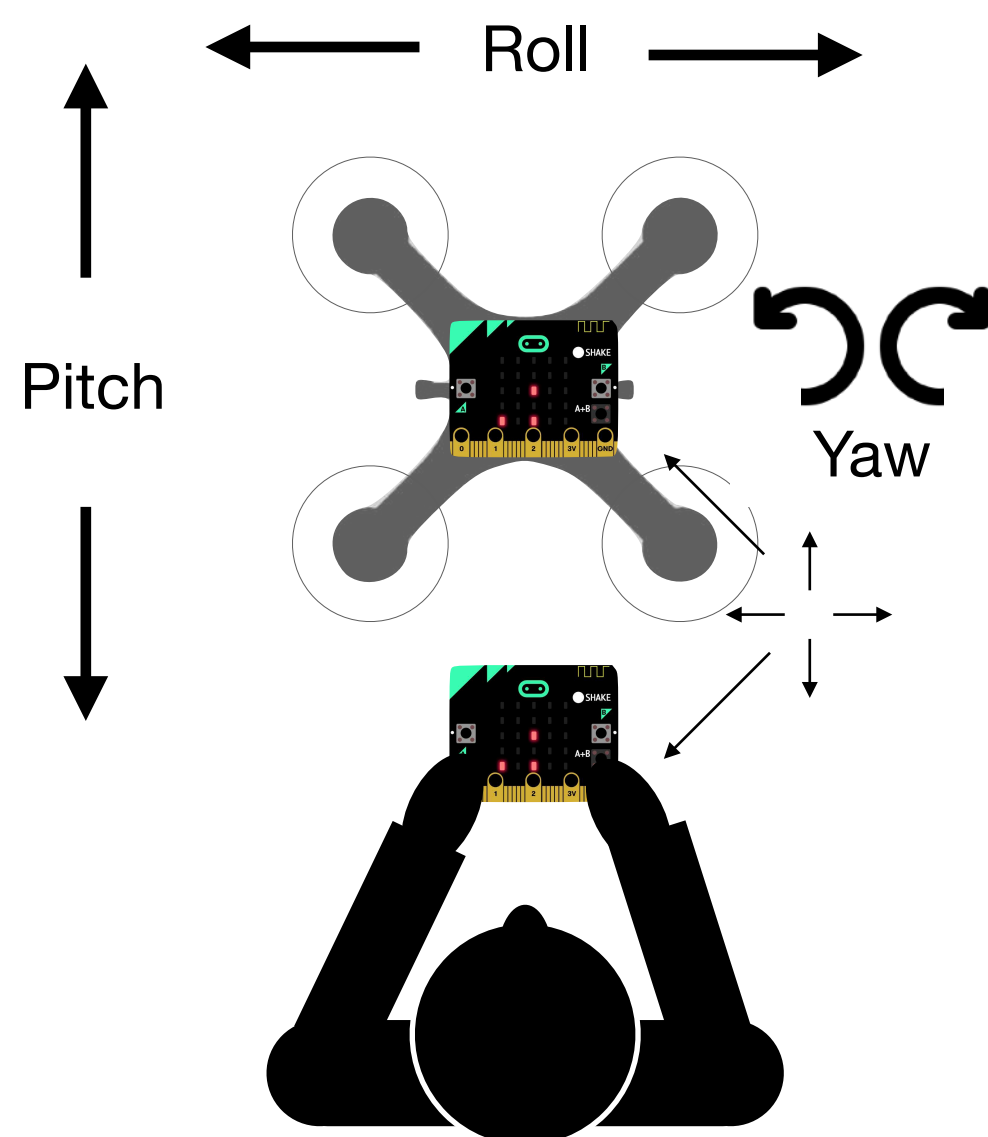


Video: <https://youtu.be/VMF9uehLfg8>



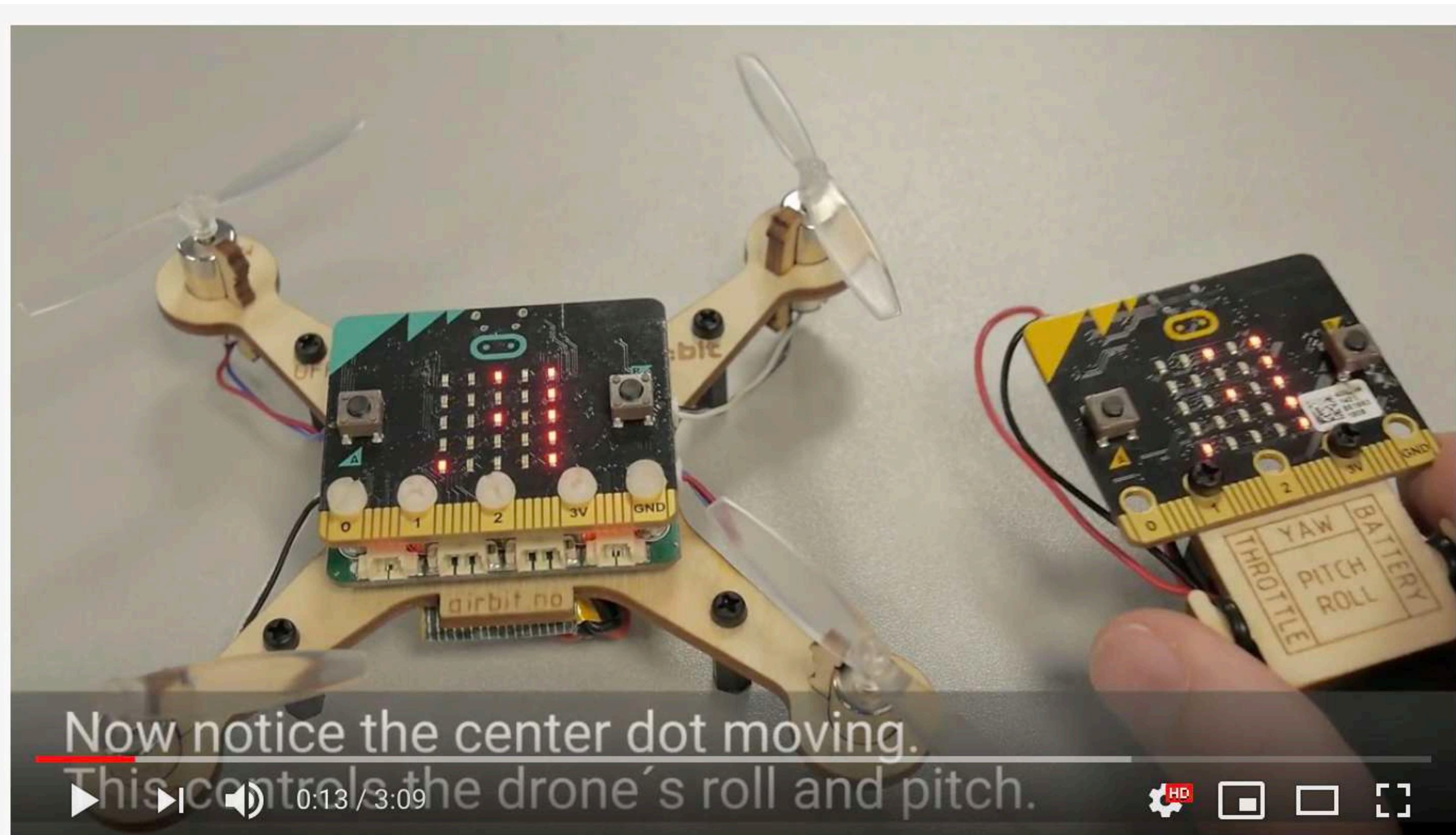
Battery empty

- ☐ Find an open room without obstacles, indoor or outside at a grass field or similar.
- ☐ Carpets and low ceiling lowers the risk of damaging the drone. Fly low and over soft surfaces.
- ☐ Keep children under 8 and animals at a safe distance. Even small propellers can hurt!
- ☐ Place the drone in the middle of the space, with the microbit's "face" towards you.
- ☐ Start the propellers by pressing A + B simultaneously.
- ☐ Increase throttle step by step until the drone is hovering 10-30 cm above the ground.
- ☐ Move the micro:bit transmitter to steer the drones roll and pitch.
- ☐ As soon as drone starts to climb, lower throttle a bit to maintain altitude.
- ☐ If you need to emergency stop: shake the remote.
- ☐ When the battery is empty, the battery icon will blink, then drone will land. Then the battery will be cut off. Do not fly, disconnect battery or charge.
- ☐ Follow local drone regulations.





**How-to fly video: <https://youtu.be/VMF9uehLfg8>**



# Problems?

Place drone flat on ground, connect battery, wait until “C” (calibrating) is done. Press A+B on remote and release quickly. Press a couple of times and watch the propellers closely to see if they starts and stops to spin.

Can't start the motors/propellers



- Is the micro:bit on the drone receiving the radio signal? The dots on the screen should move when you move the remote. Check and re-upload code for transmitter and receiver. Make sure they are on the same radio channel. You can try with ready made code for both transmitter and receiver.
- On the drone, is the control board and the microbit connected? If the text say “No Gyro” or “No Motor controller”, the connection is not stable: tighten the nuts. Disconnect / reconnect battery. Look for the letters G and M during startup.

Motor starts, but can't lift off

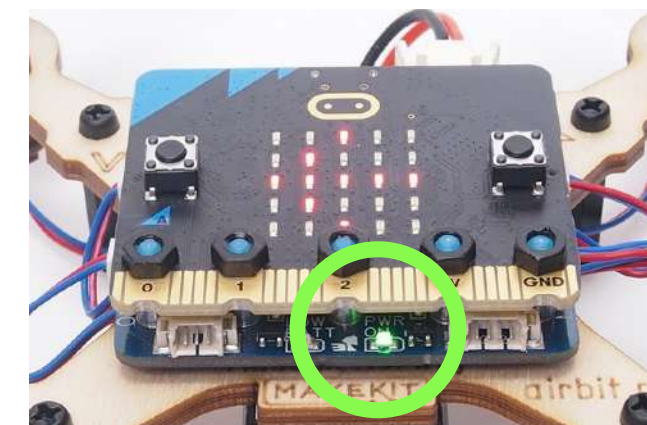


Have you placed the correct propeller on the right motor? (Page 23)

Are all the motors working and spinning?  
Can there be a damaged motor?

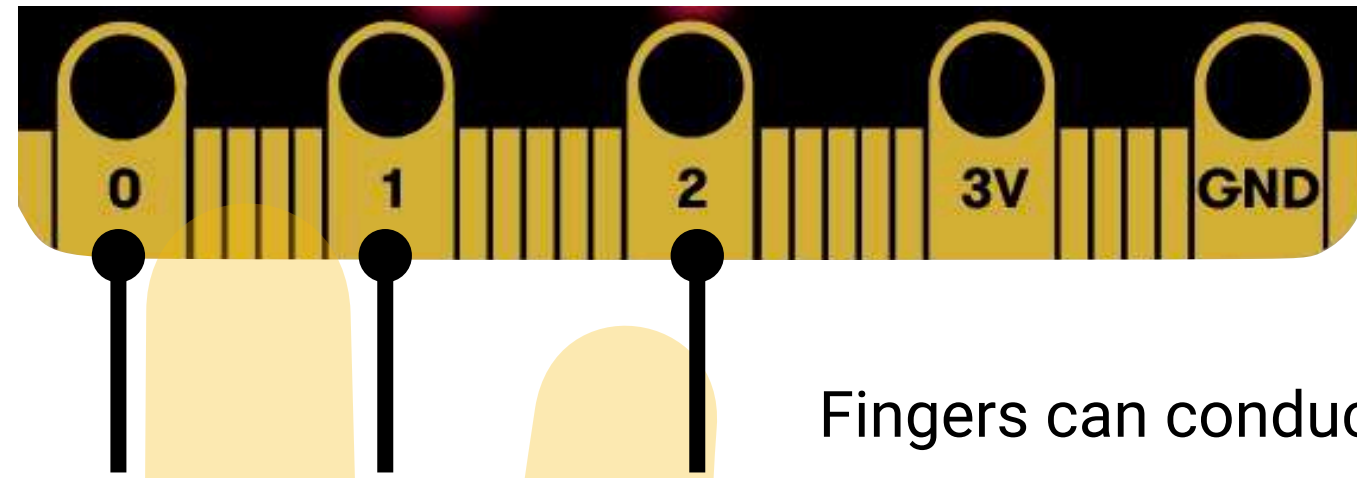
Have you placed the drone the proper way and given enough throttle? You need about 50-60% to lift off. See page 52.

Is the battery charged and is there a green light?

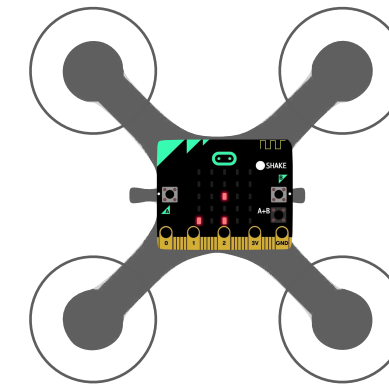




# Extra task: Yaw



Fingers can conduct electric current



Yaw is a sideways rotation

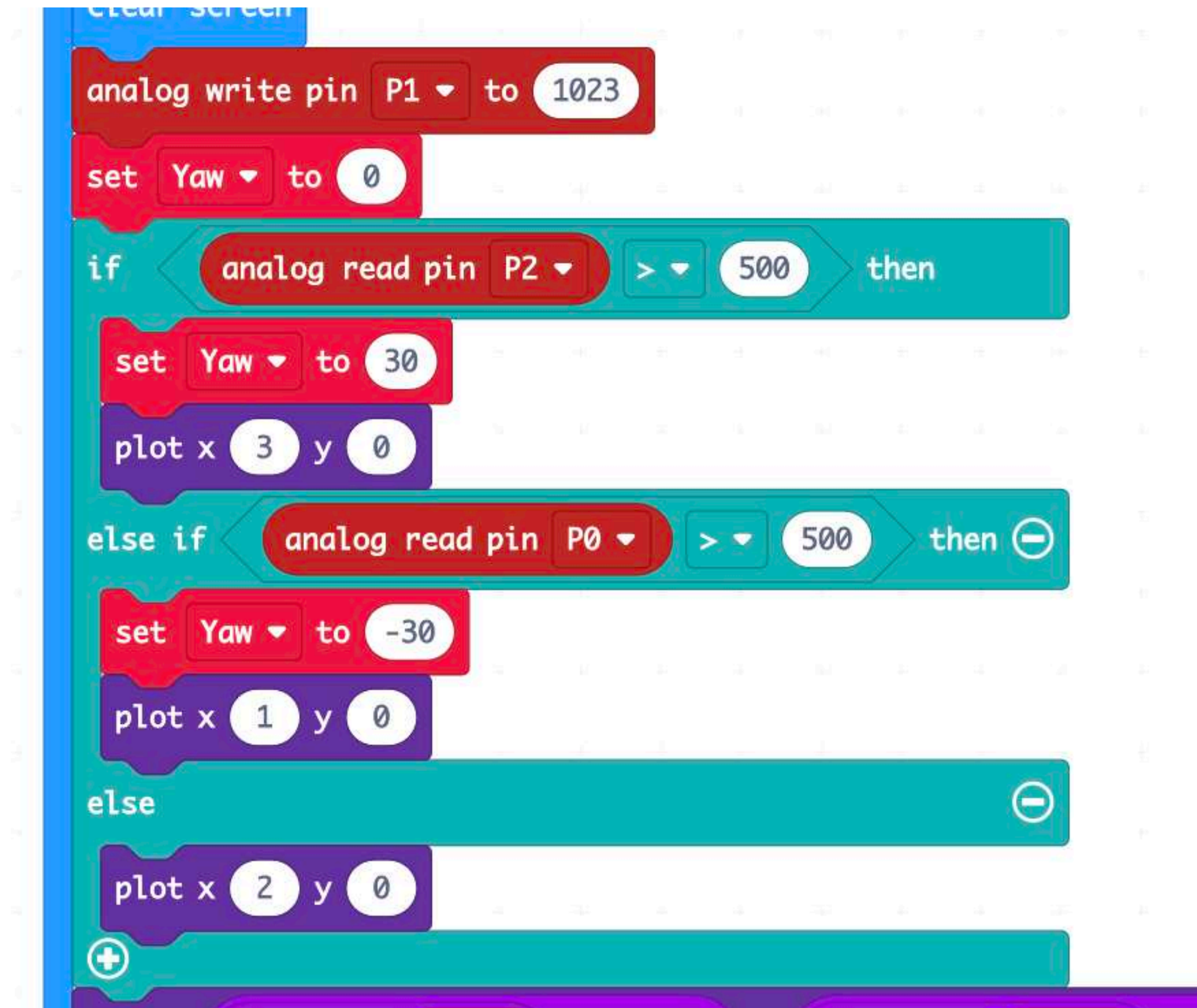
## Yaw

Yaw makes the drone rotate sideways around its center. You can make artificial button and control the yaw.

- Use analog write to make P1 send out power
- Use analog read to make P0 and P2 read power that is lead trough your fingers
- When nothing is touching P0 and P2, the analog read will be low (around 200)
- When conducting electricity from your fingers, the analog read can be about 800
- Using a treshhold on about 500, you can detect a finger press.
- When a press is detected, change the yaw to a negative number for left rotation, and a positive number for right rotation
- If no button is pressed, the yaw should be 0. The easiest way is to set it to zero before checking if a button is pressed.
- The micro:bit V2 is slightly different and might require different values.

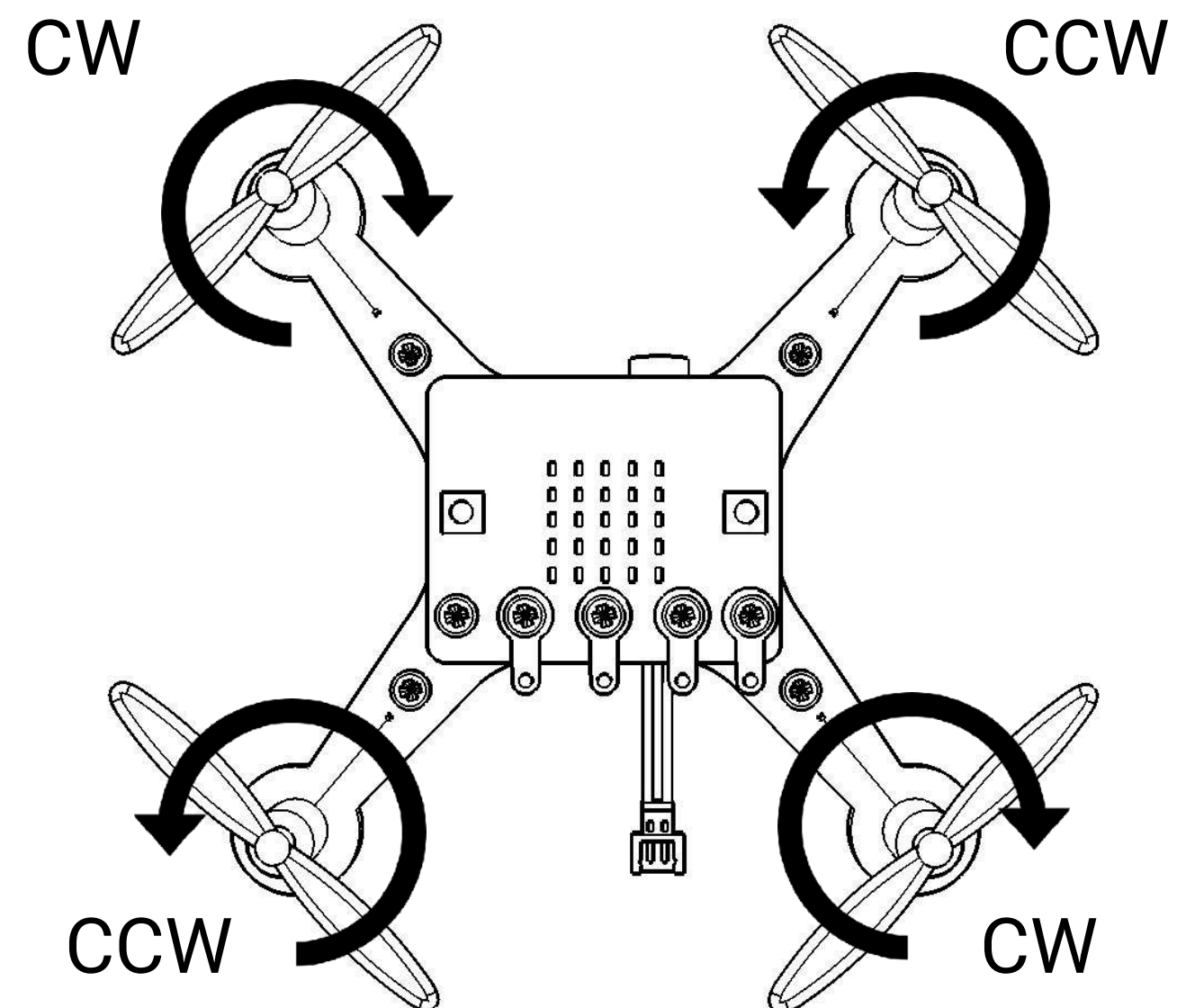
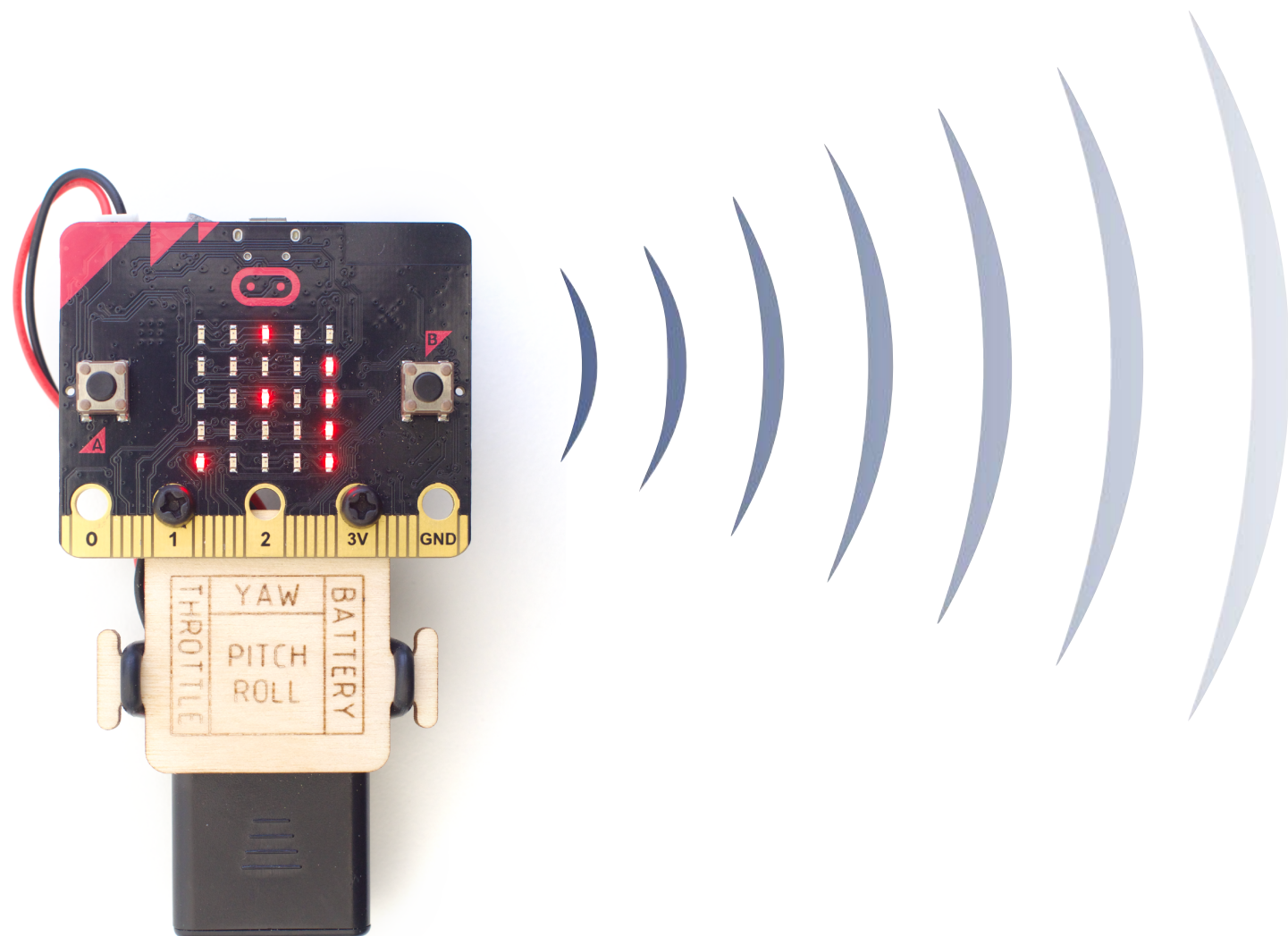


# Yaw: Solution





# How the drone controls its path



- Two motors rotate with the clock (CW) and two rotate against the clock (CCW)
- The counter-rotating setup prevents the drone from spinning around the yaw-axis.
- The drone ascends (**Throttle**) by making all the motors go equally faster
- The drone moves forwards (**Pitch**) by increasing power at the two back motors, while decreasing power to the front motors. This tilts the drone forward.
- Sideways movement is done by speeding up two motors at one side (eg. left) then slowing down two motors at the opposite side (eg. right)
- Drone can rotate around the **Yaw**-axis by speeding up every other motor (eg. every CW motor) while slowing down every CCW motor.
- The flight control board takes all the flight directions (PARTY) and mixes the info onto the four motors.

# Contact:

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